

Orthogonality

Geometric Algorithms

Objectives

Discuss analytic geometry in $\mathbb{R}^n$ in general

Try to understand why it's useful to work with orthogonal vectors

Start to connect orthogonality to matrices and linear transformations

Keywords

orthogonal

orthogonal set

orthogonal basis

orthogonal projection

orthogonal component

orthonormal

orthonormal set

orthonormal basis

orthonormal matrix

orthogonal matrix

Analytic Geometry

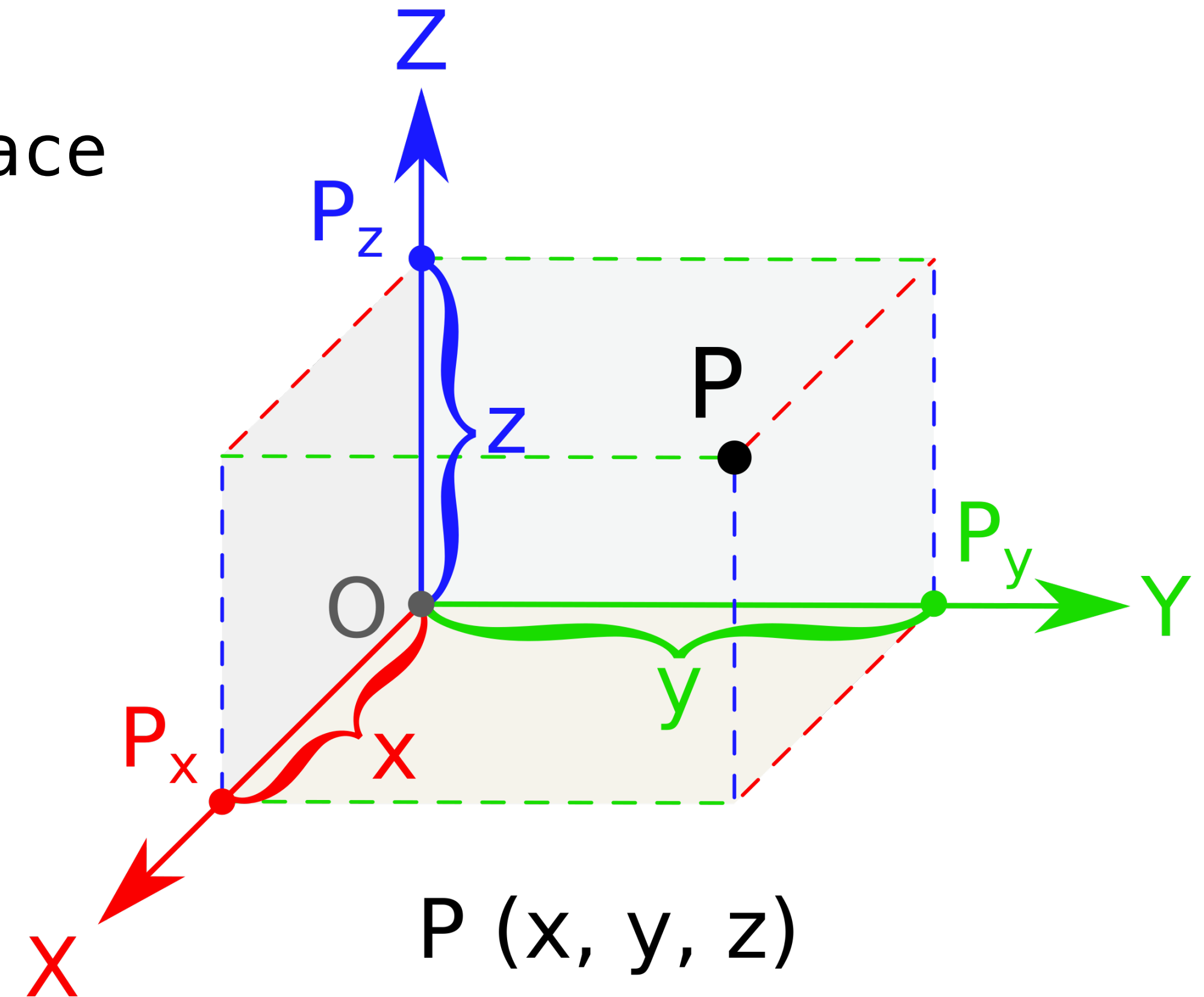
What is Analytic Geometry?

Analytic geometry is the study of space using a *coordinate system*

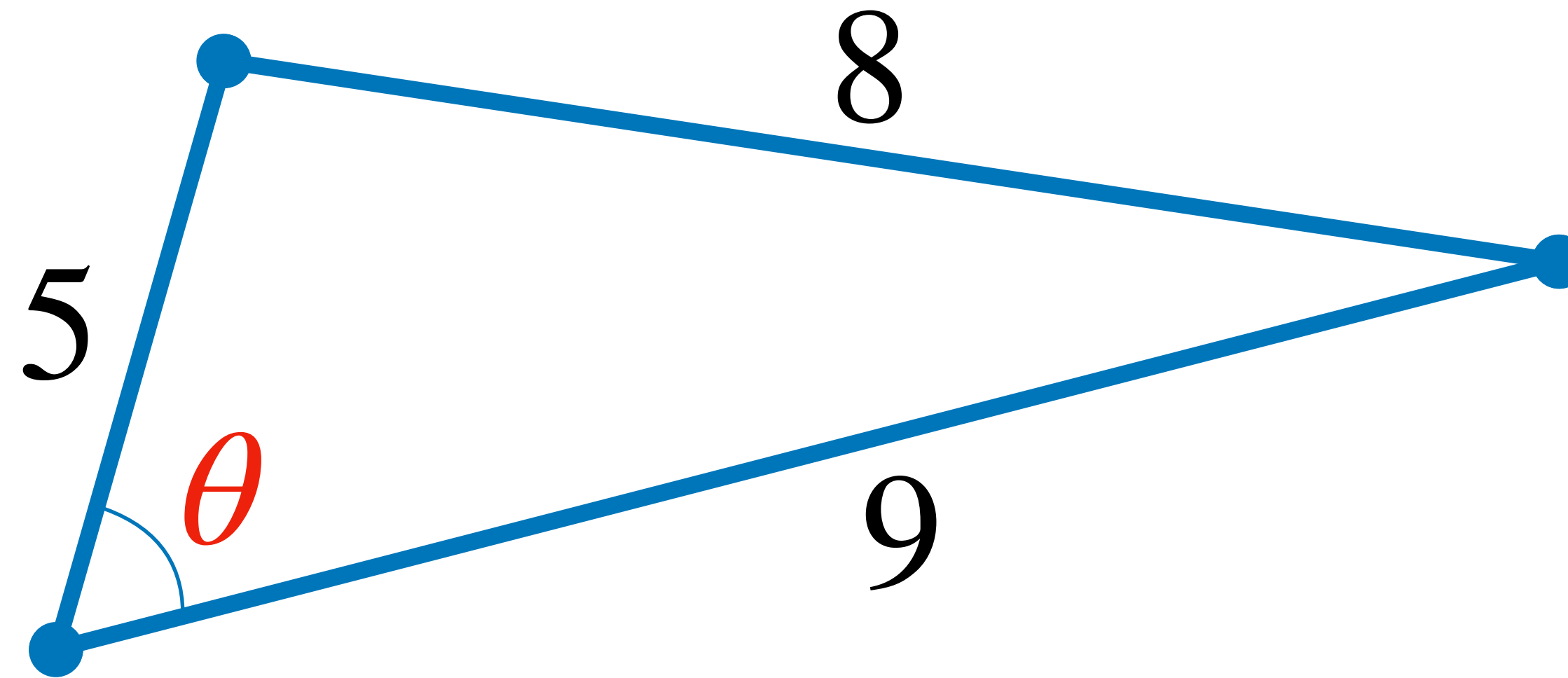
e.g., equations of lines, curves, shapes, angles, etc.

The fundamental concepts:

- » distance
- » position
- » area
- » angle

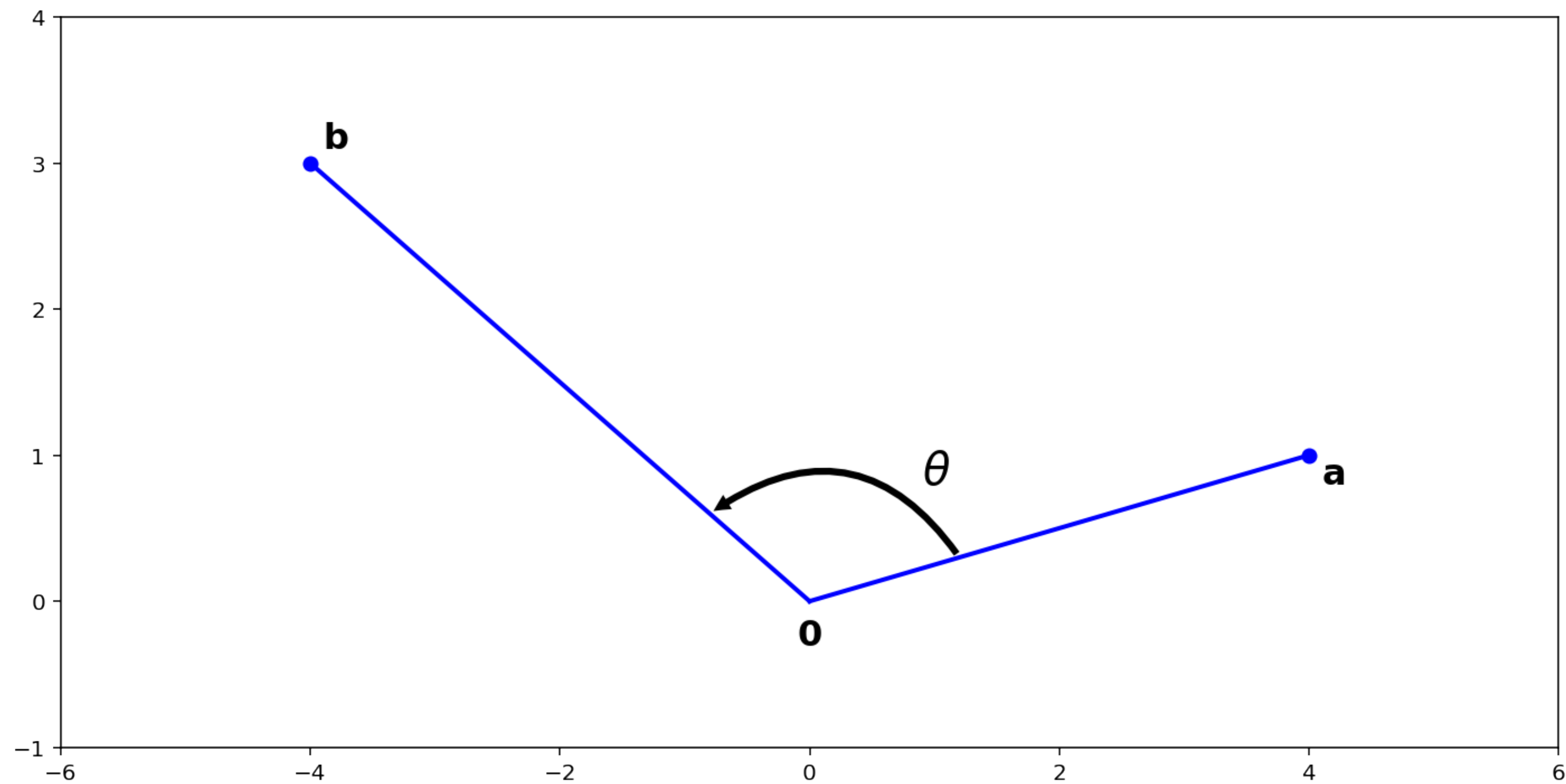


A Potentially Familiar Example



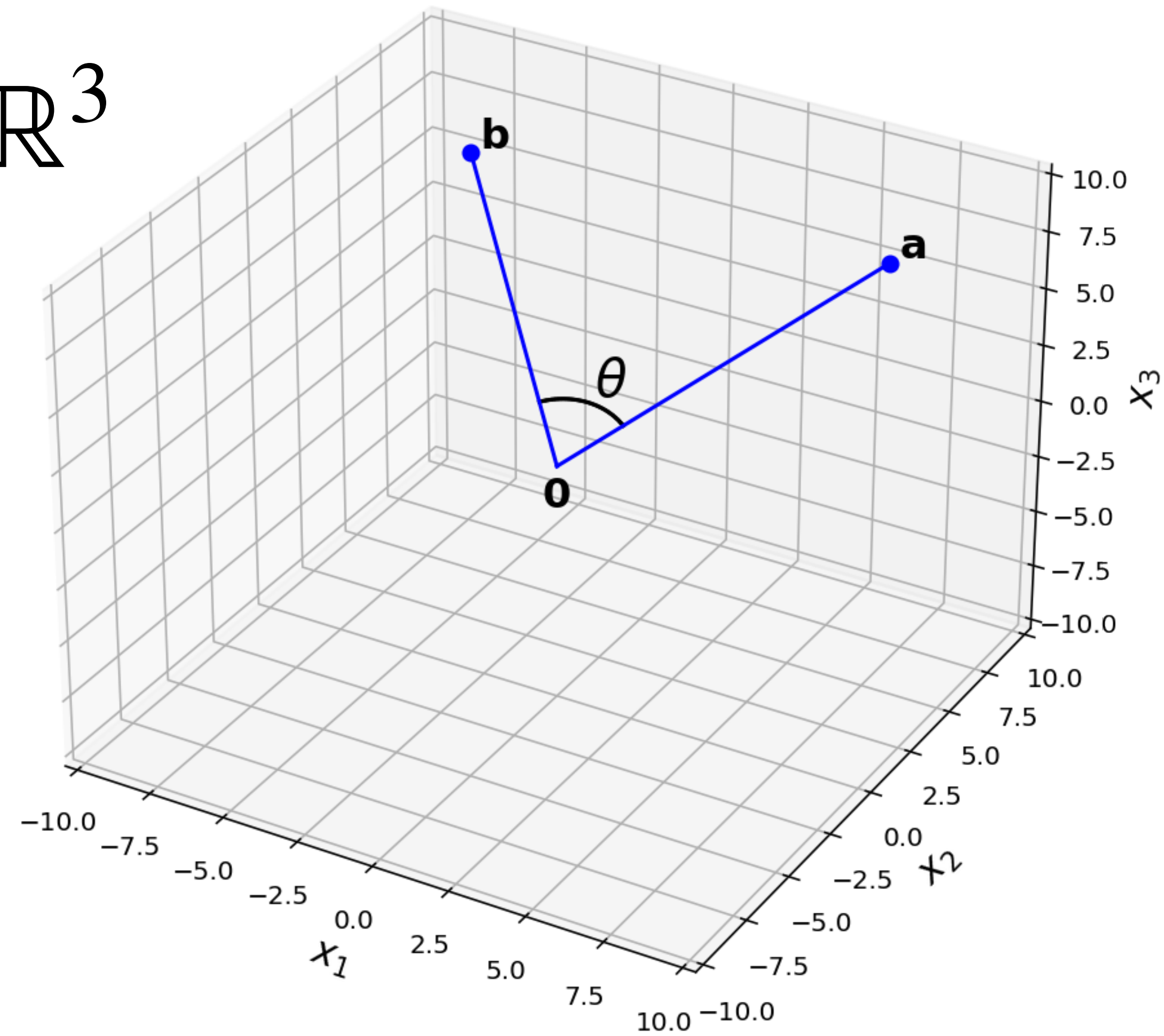
What is the value of θ ?

Angles in $\mathbb{R}^2$



What is the value of θ ?

Angles in $\mathbb{R}^3$



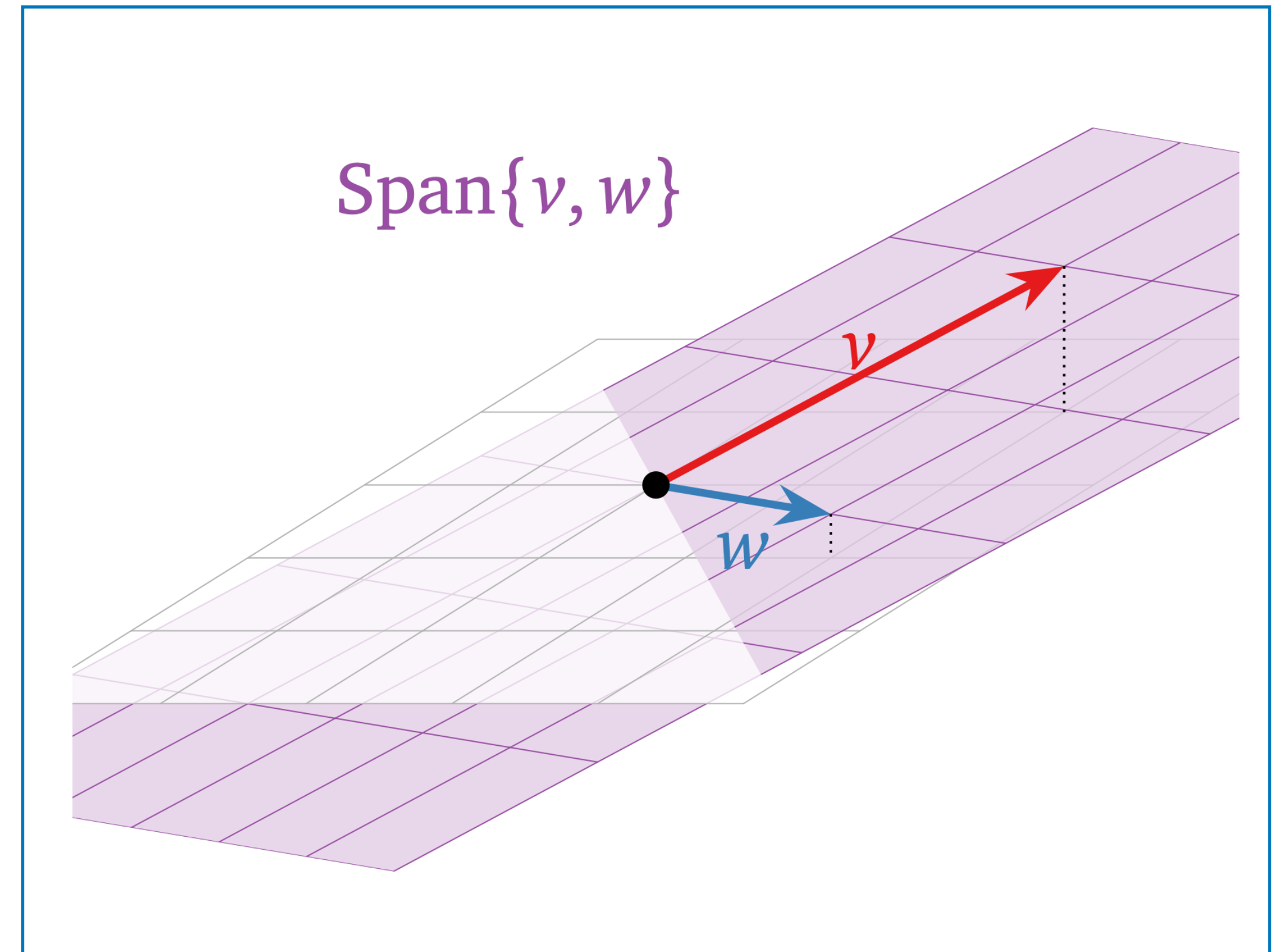
What is the value of θ ?

The First Key Idea

Angles make sense in *any* dimension

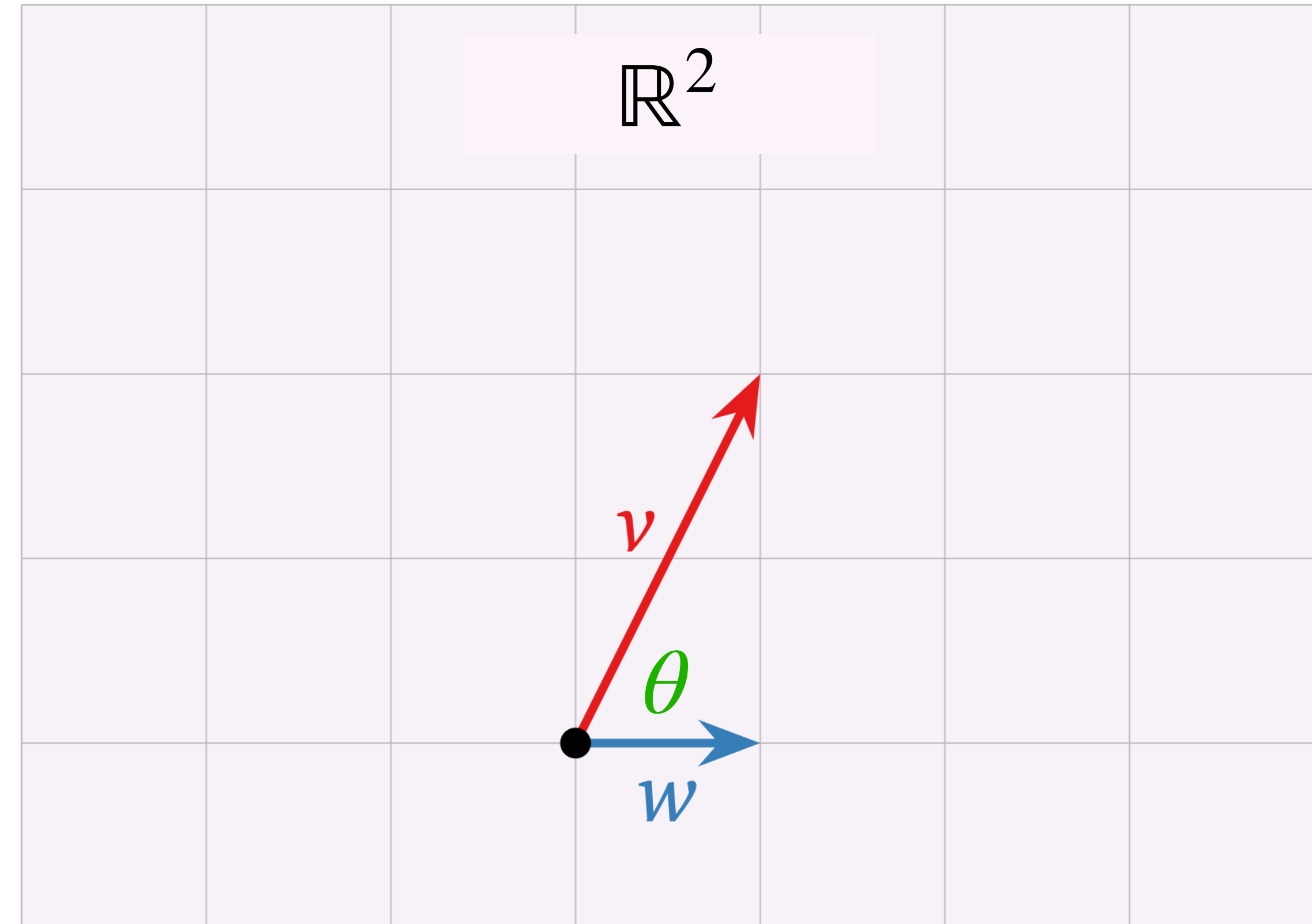
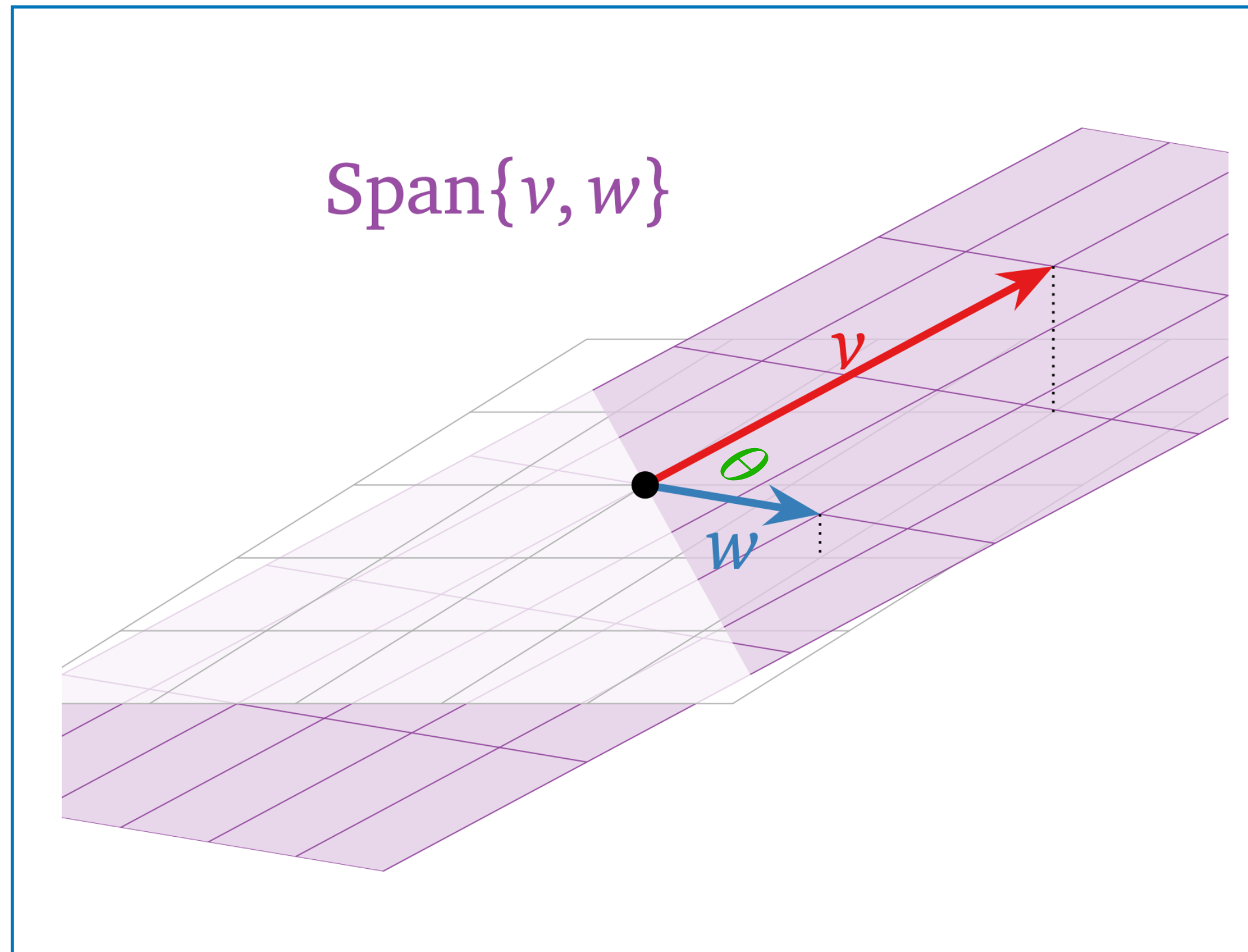
A pairs of L.I. vectors span a plane

(We can formalize this via change of bases)



The Picture

We can do 2D analytic geometry here



change of basis from $\text{span}\{v, w\}$ to $\mathbb{R}^2$

Recall: Inner Products

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$$[u_1 \quad u_2 \quad u_3 \quad u_4] \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = u_1 v_1 + u_2 v_2 + u_3 v_3 + u_4 v_4$$

Recall: Inner Products

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Def. The inner product of $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is

$$\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u} \cdot \mathbf{v} = \mathbf{u}^T \mathbf{v}$$

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dot product

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All of the basic concepts of analytic geometry
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Def (Adv). An **inner product space** is a vector
space with an inner product

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All of the basic concepts of analytic geometry can be defined *in terms of inner products*

Def (Adv). An **inner product space** is a vector space with an inner product

Inner product spaces (like $\mathbb{R}^n$) are places where you can do analytic geometry

Inner Products

Algebraic Properties of Inner Products

» $\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$ (symmetry)

» $(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (\mathbf{u} \cdot \mathbf{w}) + (\mathbf{v} \cdot \mathbf{w})$
» $(\alpha \mathbf{u}) \cdot \mathbf{v} = \alpha(\mathbf{u} \cdot \mathbf{v})$ } (linearity)

» $\mathbf{u} \cdot \mathbf{u} \geq 0$ (nonnegativity)

» $\mathbf{u} \cdot \mathbf{u} = 0$ if and only if $\mathbf{u} = 0$

Verifying Left Additivity

$$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (\mathbf{u} \cdot \mathbf{w}) + (\mathbf{v} \cdot \mathbf{w})$$

Verifying Right Homogeneity

$$\langle \mathbf{v}, c\mathbf{u} \rangle = c\langle \mathbf{v}, \mathbf{u} \rangle$$

Question

$$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$$

$$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = (\mathbf{u} \cdot \mathbf{w}) + (\mathbf{v} \cdot \mathbf{w})$$

Simplify the expression $\langle \mathbf{u} + \mathbf{v}, \mathbf{u} - \mathbf{v} \rangle$ using the properties of inner products

Answer: $\langle \mathbf{u}, \mathbf{u} \rangle - \langle \mathbf{v}, \mathbf{v} \rangle$

Distance and Length

Algebraic Properties of Inner Products

$$\gg \mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u} \quad (\text{symmetry})$$

$$\left. \begin{aligned} \gg (\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} &= (\mathbf{u} \cdot \mathbf{w}) + (\mathbf{v} \cdot \mathbf{w}) \\ \gg (\alpha \mathbf{u}) \cdot \mathbf{v} &= \alpha(\mathbf{u} \cdot \mathbf{v}) \end{aligned} \right\} (\text{linearity})$$

$$\gg \mathbf{u} \cdot \mathbf{u} \geq 0 \quad (\text{nonnegativity})$$

$$\gg \mathbf{u} \cdot \mathbf{u} = 0 \quad \text{if and only if } \mathbf{u} = \mathbf{0}$$

Nonnegativity and Scaling

$$\langle c\mathbf{v}, c\mathbf{v} \rangle = c^2 \langle \mathbf{v}, \mathbf{v} \rangle = c^2 \sum_{i=1}^n v_i^2$$

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Increasing the length of a vector increases its inner product with itself

Nonnegativity and Scaling

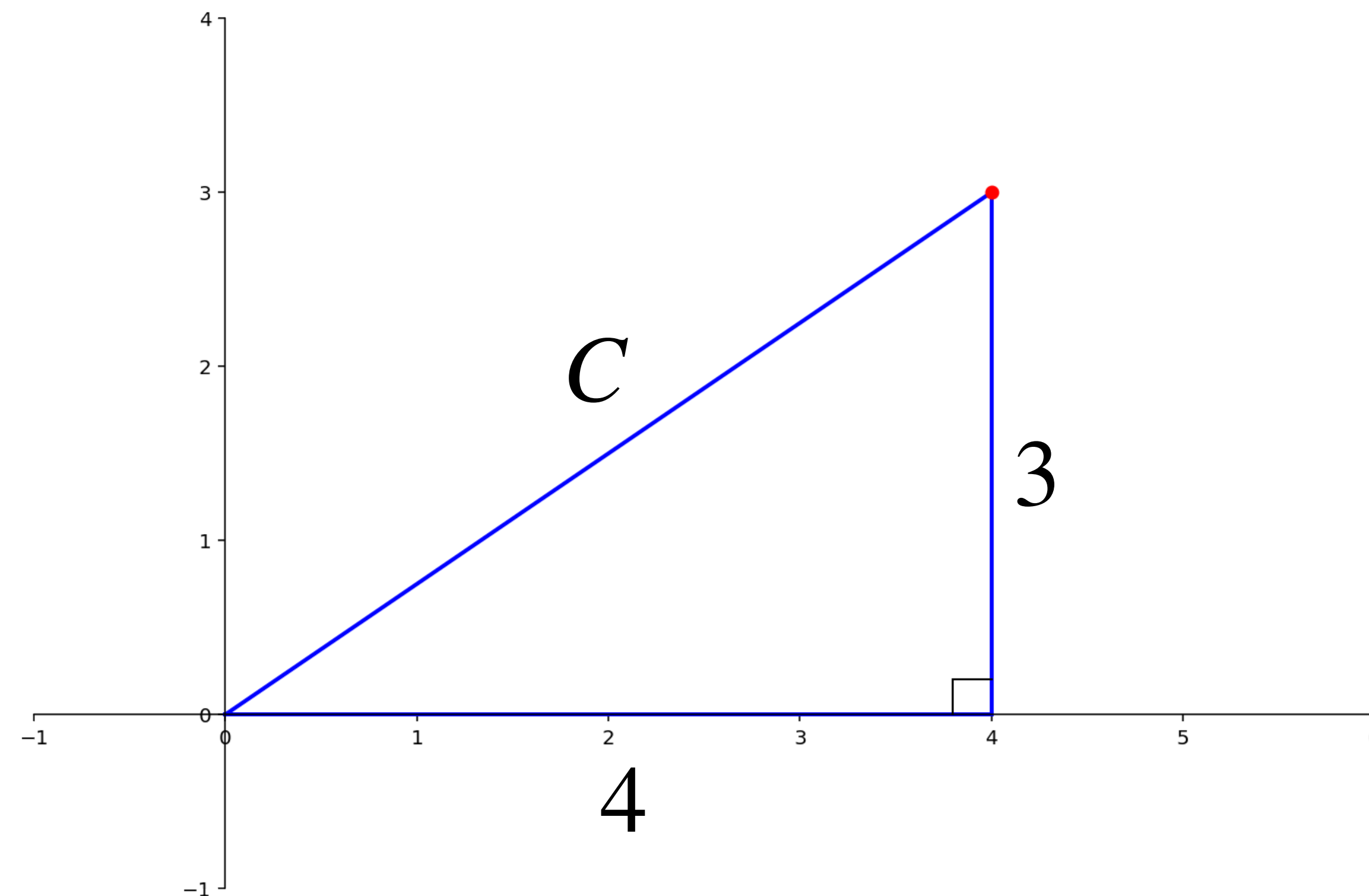
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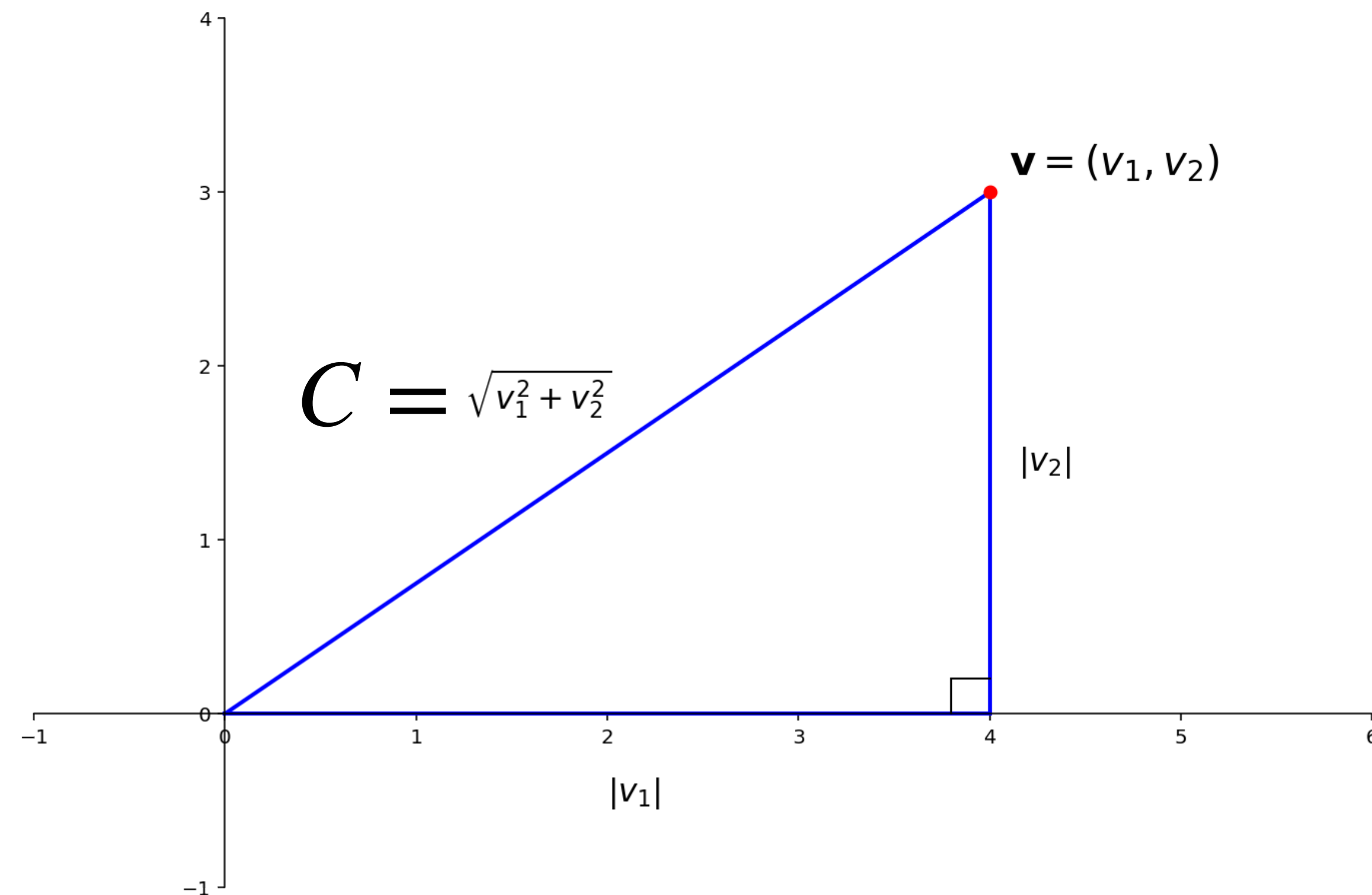
This means $\langle \mathbf{v}, \mathbf{v} \rangle$ is capturing some notion of magnitude

Another Potentially Familiar Question



How long is the side C ?

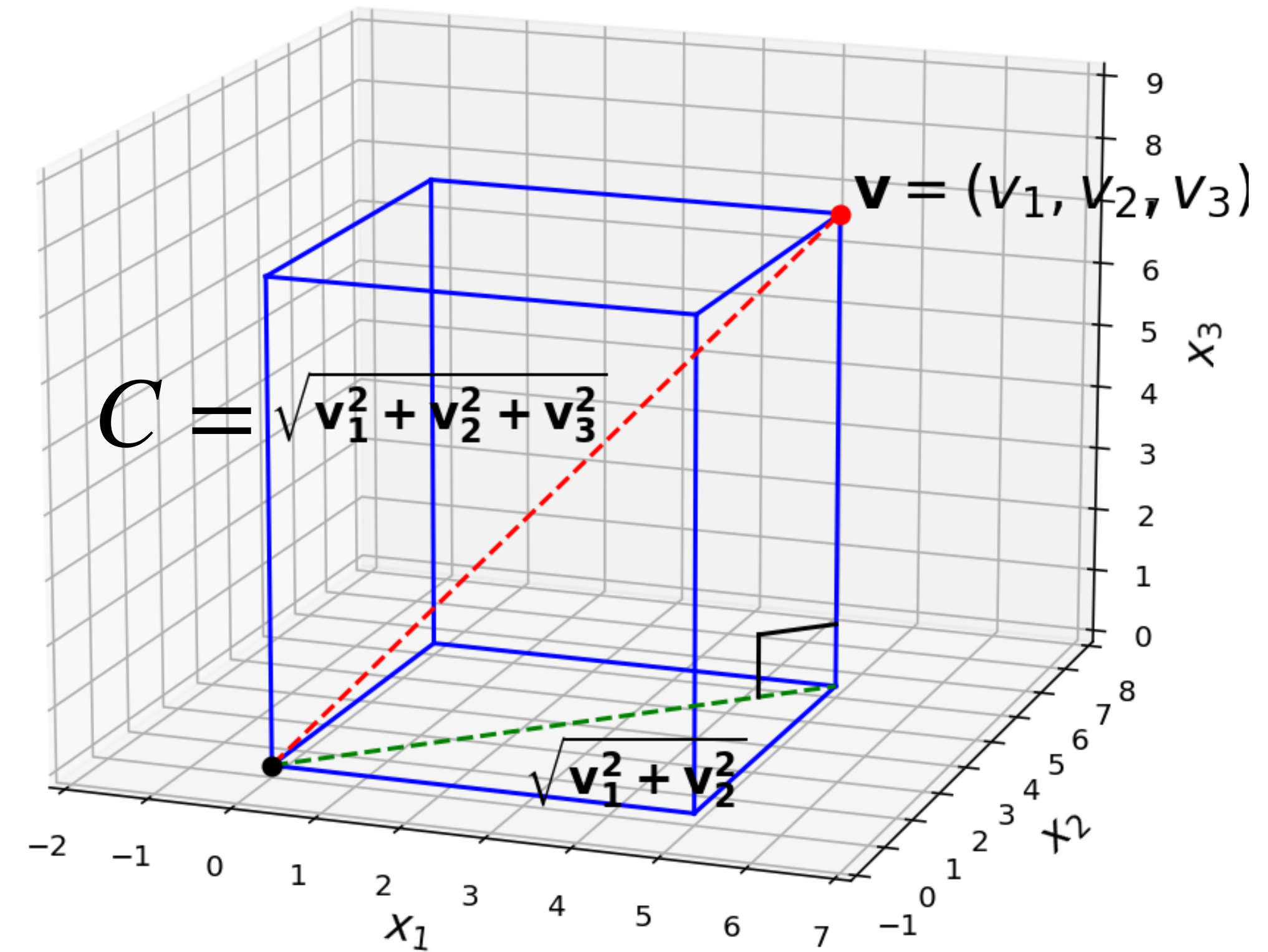
Pythagorean Theorem



Thm (Pythagoras). *For a right triangle, the square of the length of the hypotenuse is the sum of the squares of the lengths of the remaining two sides*

This still works in $\mathbb{R}^3$

Thm (Pythagoras). $C = \sqrt{v_1^2 + v_2^2 + v_3^2}$



Norms and Inner Products

Def. The ℓ^2 norm of a vector $\mathbf{v}$ in $\mathbb{R}^n$ is

$$\|\mathbf{v}\| = \sqrt{\mathbf{v} \cdot \mathbf{v}}$$

The norm of a vector is the square root of the inner product with itself

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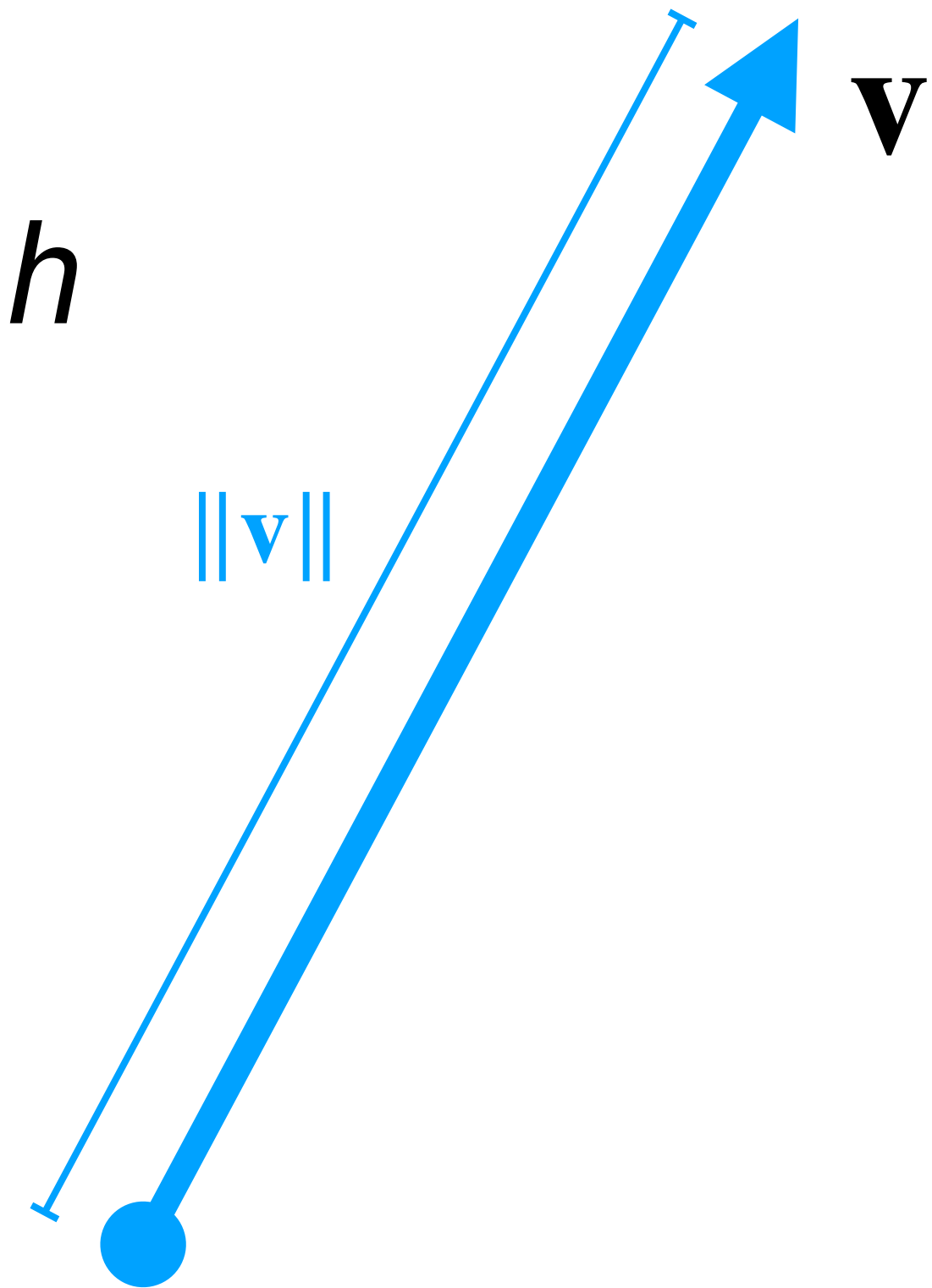
The norm of a vector is the square root of the inner product with itself

It's important that $\mathbf{v}^T \mathbf{v}$ is nonnegative

Norms and Distance

Norms give us a notion of *length*

In $\mathbb{R}^2$ and $\mathbb{R}^3$ this is the usual notion of length

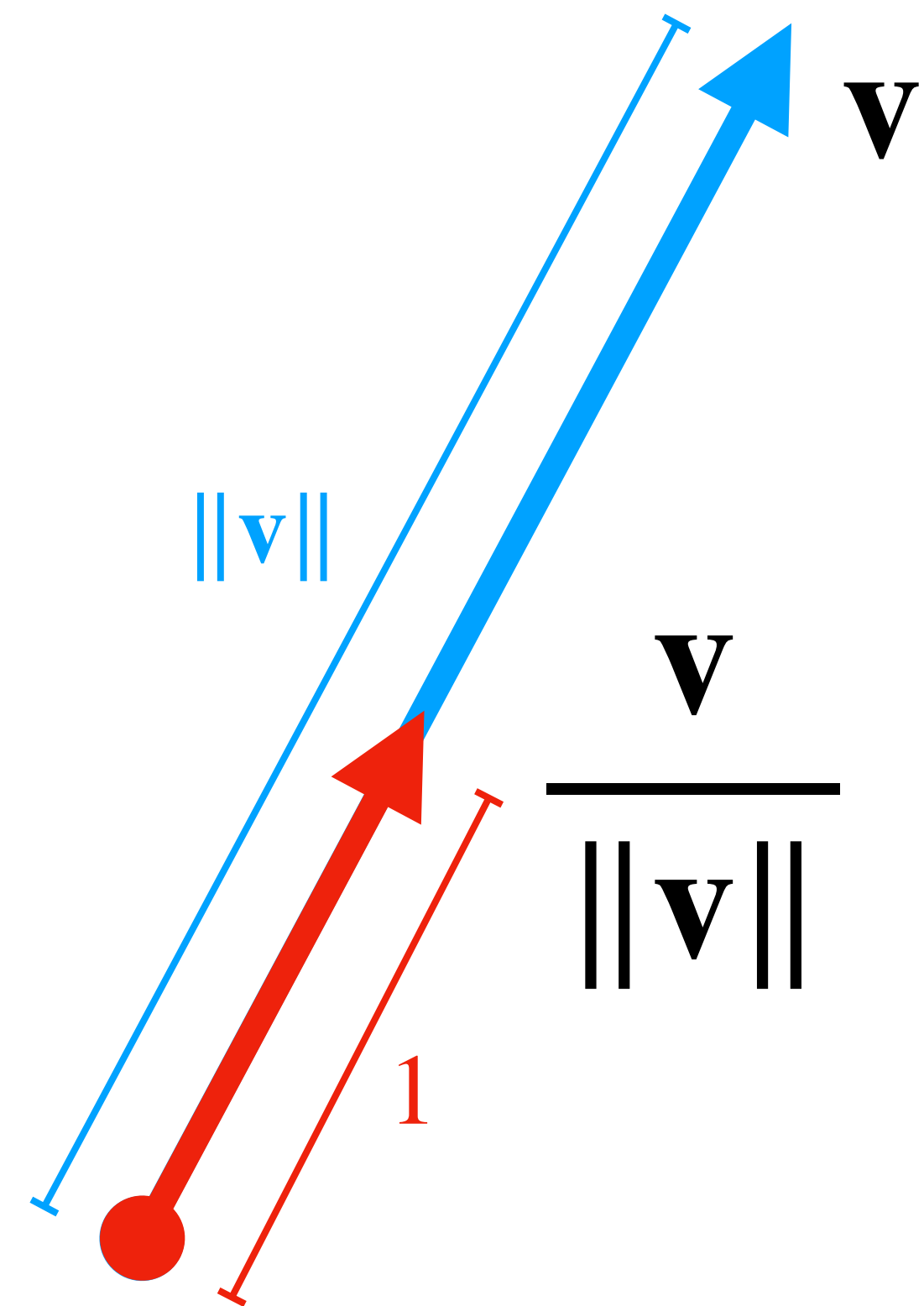


ℓ^2 Normalization

Def. A **unit vector** is a vector $\mathbf{v}$ such that $\|\mathbf{v}\| = 1$

Normalize vectors give us a notion of direction:

$$\mathbf{v} \mapsto \frac{\mathbf{v}}{\|\mathbf{v}\|}$$



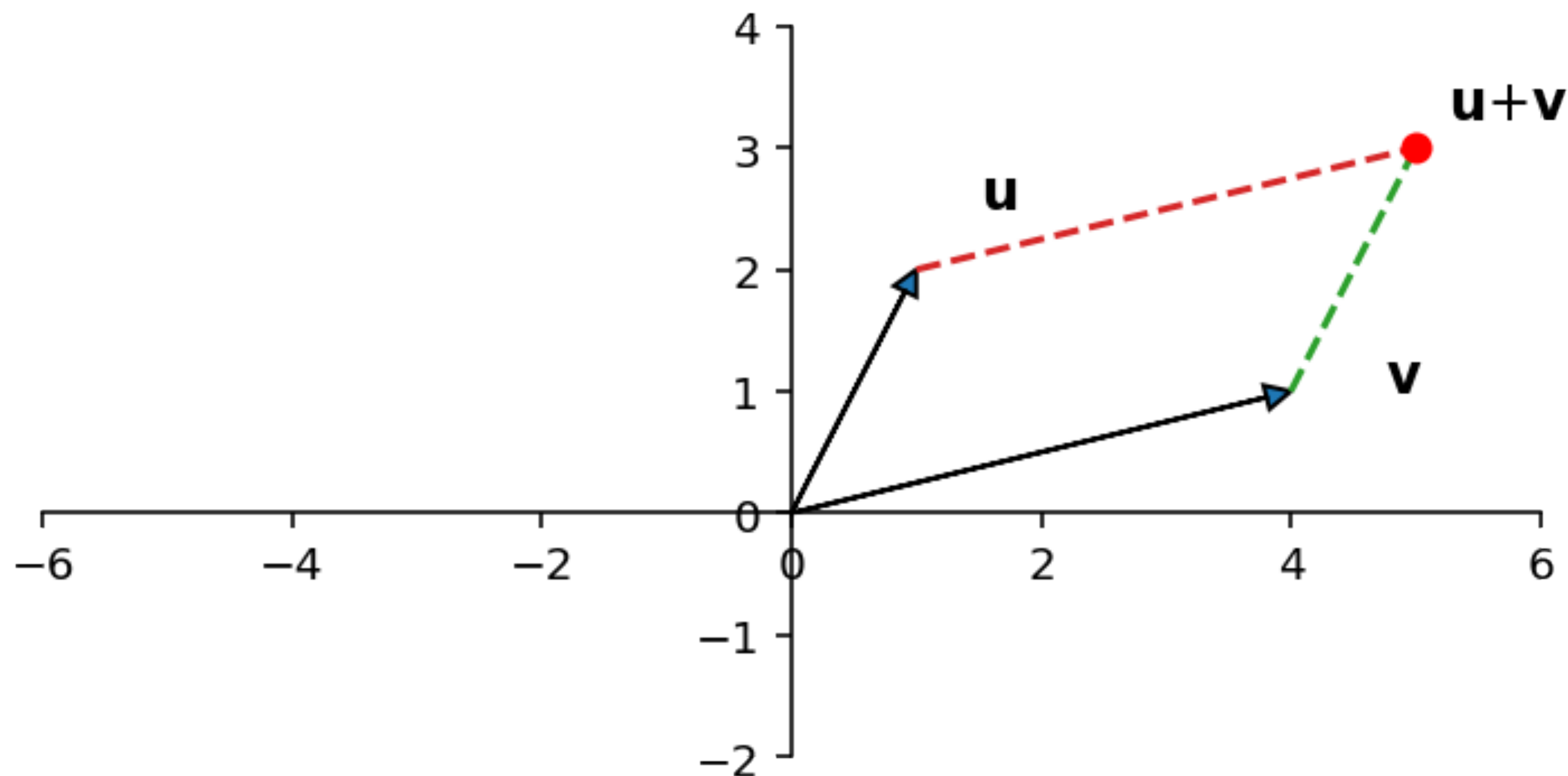
Example

Find the unit vector in the same direction as $\begin{bmatrix} 1 \\ -2 \\ 2 \\ 0 \end{bmatrix}$

If we know how to calculate
lengths of vectors, we know how
to calculate distances

Recall: Vector Addition

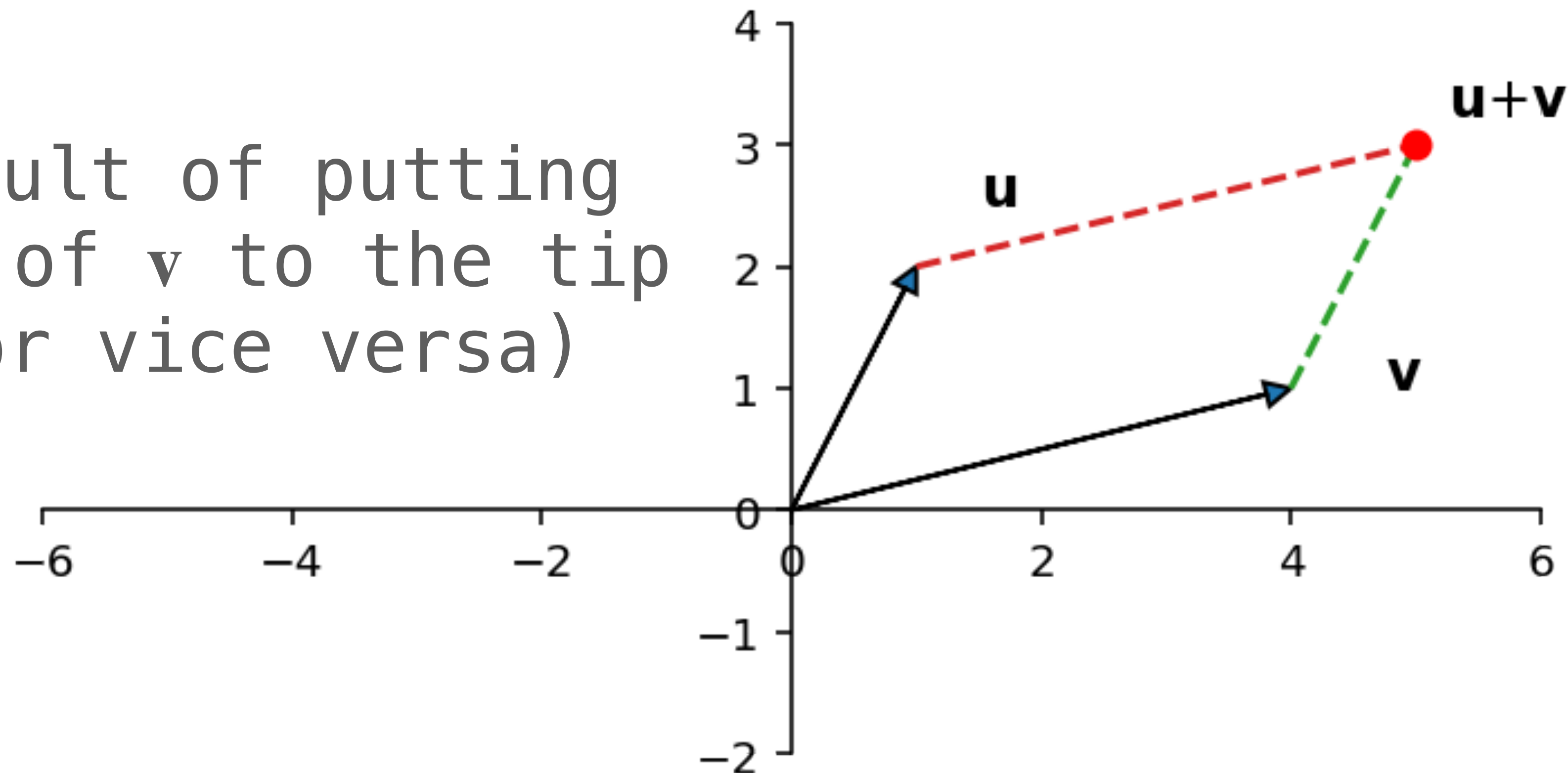
tip-to-tail rule:



Recall: Vector Addition

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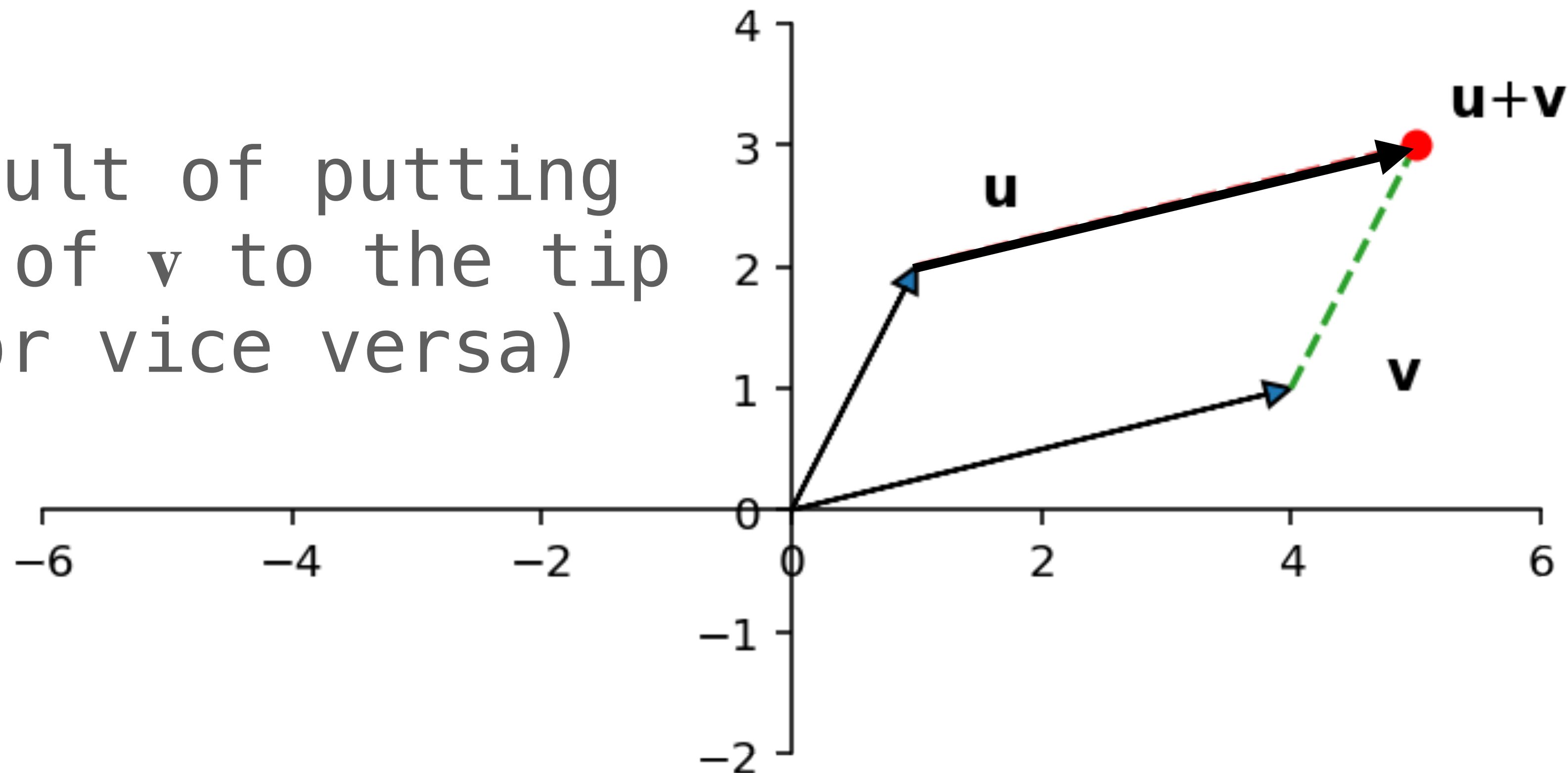
$\mathbf{u} + \mathbf{v}$ result of putting
the tail of $\mathbf{v}$ to the tip
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Recall: Vector Addition

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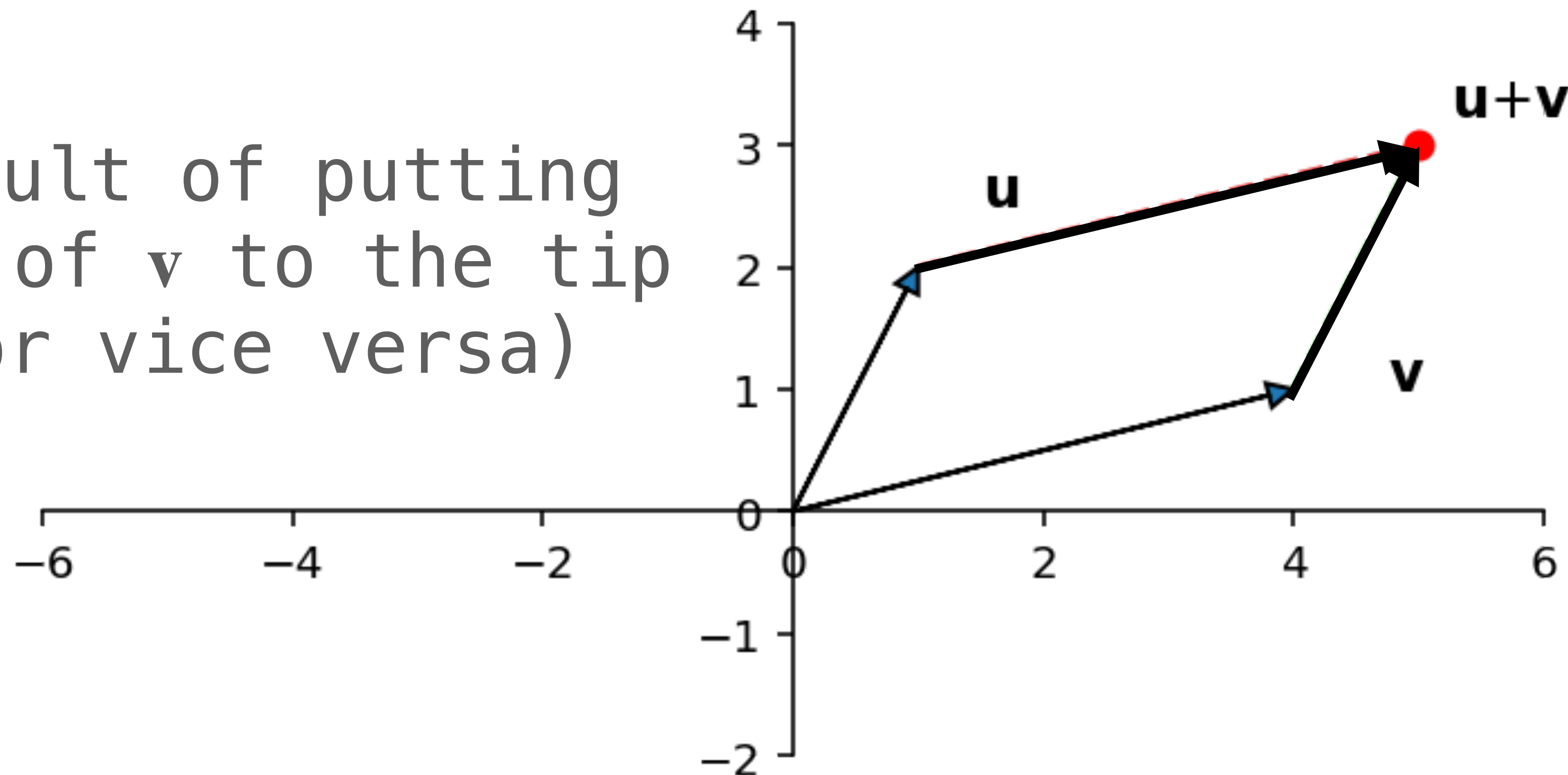
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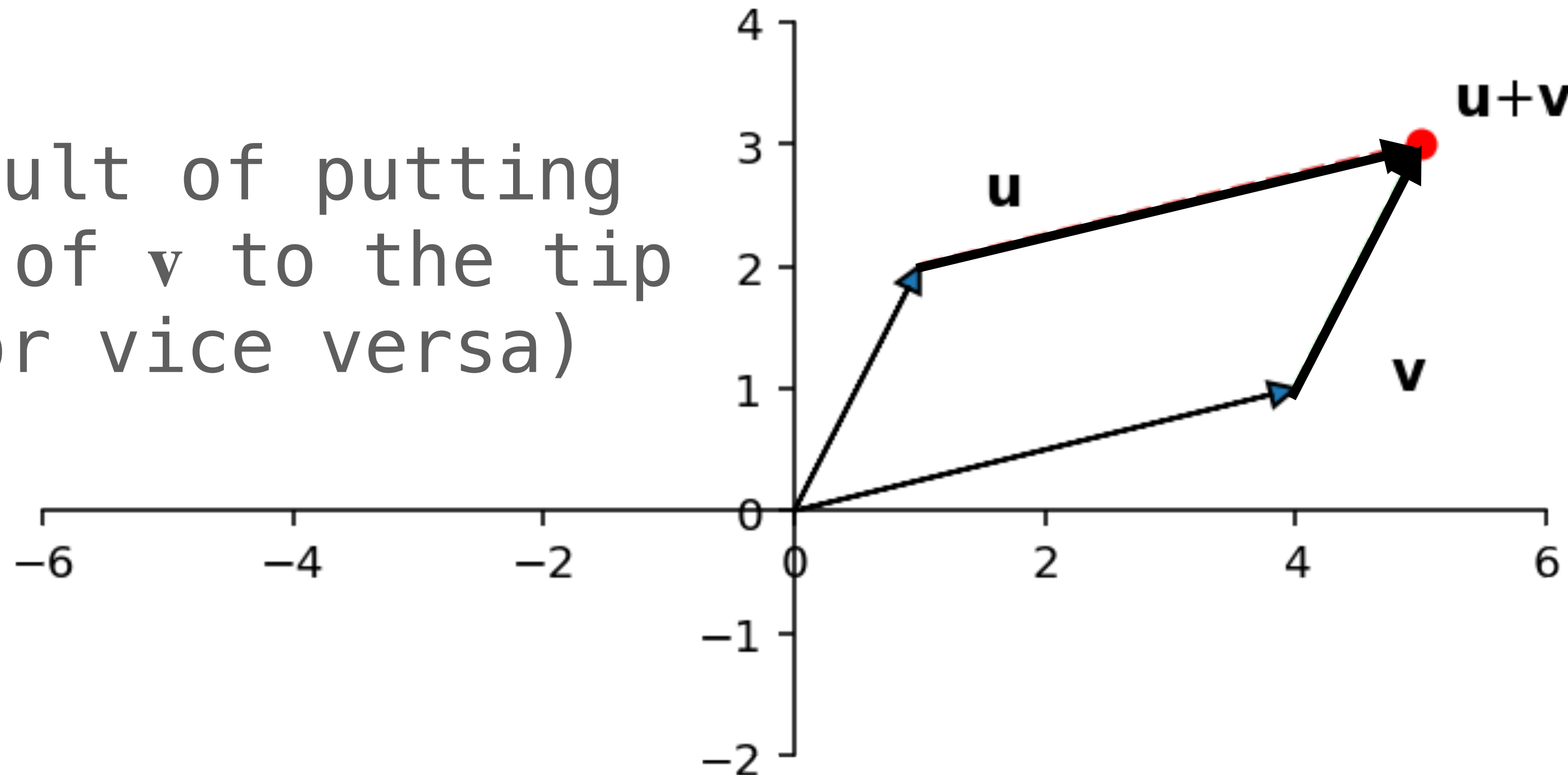


Recall: Vector Addition

The distance between u and $u+v$ is the length of v

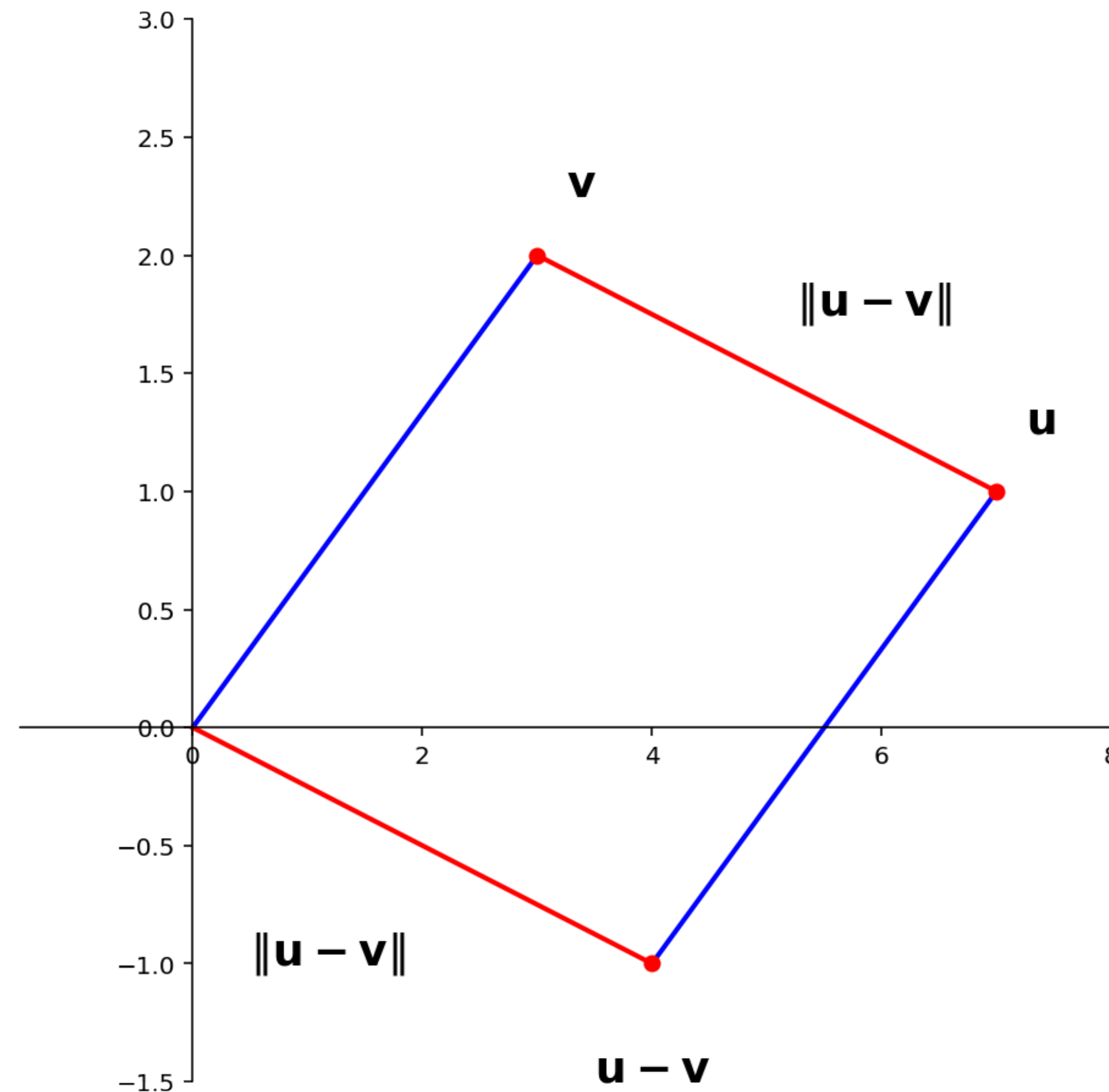
tip-to-tail rule:

$u+v$ result of putting the tail of v to the tip of u (or vice versa)



Distance (Pictorially)

If we can calculate lengths then we can calculate distances

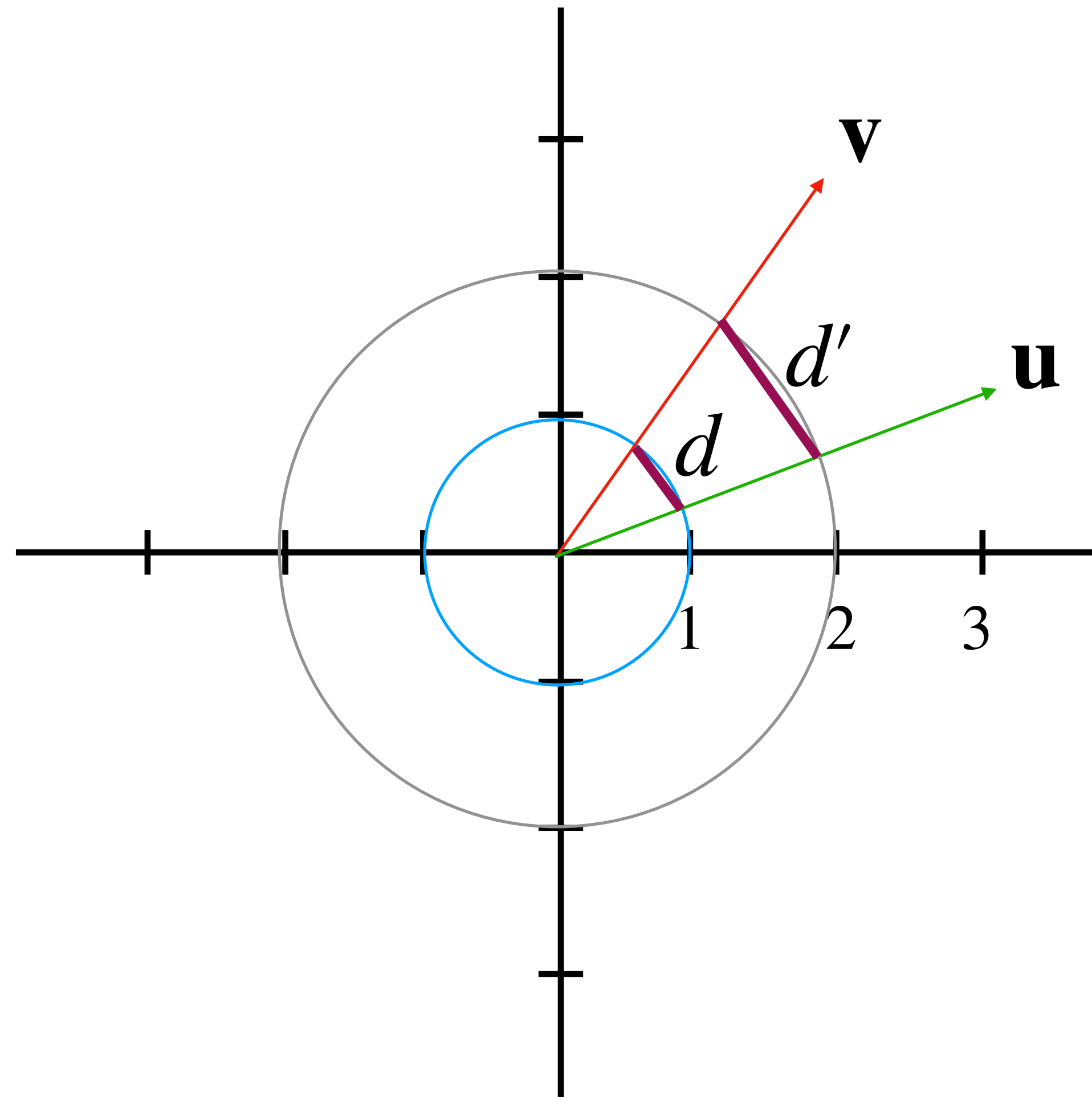


Distance (Algebraically)

Def. The distance between $\mathbf{u}$ and $\mathbf{v}$ in $\mathbb{R}^n$ is given by $\text{dist}(\mathbf{u}, \mathbf{v}) = \|\mathbf{u} - \mathbf{v}\|$

e.g., $\mathbf{u} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 2 \end{bmatrix}$

Question



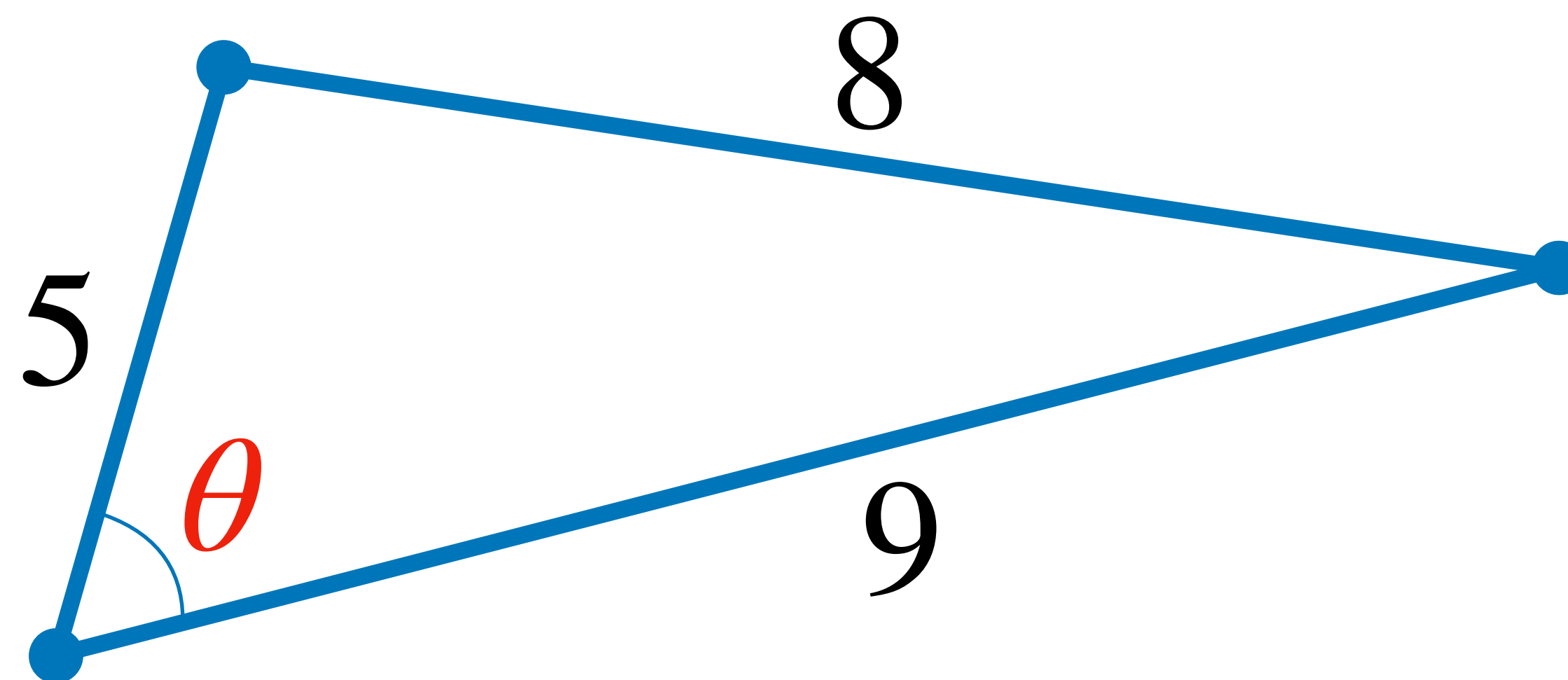
Find an expression for the distance d

Challenge. Find an expression for d'

Answer

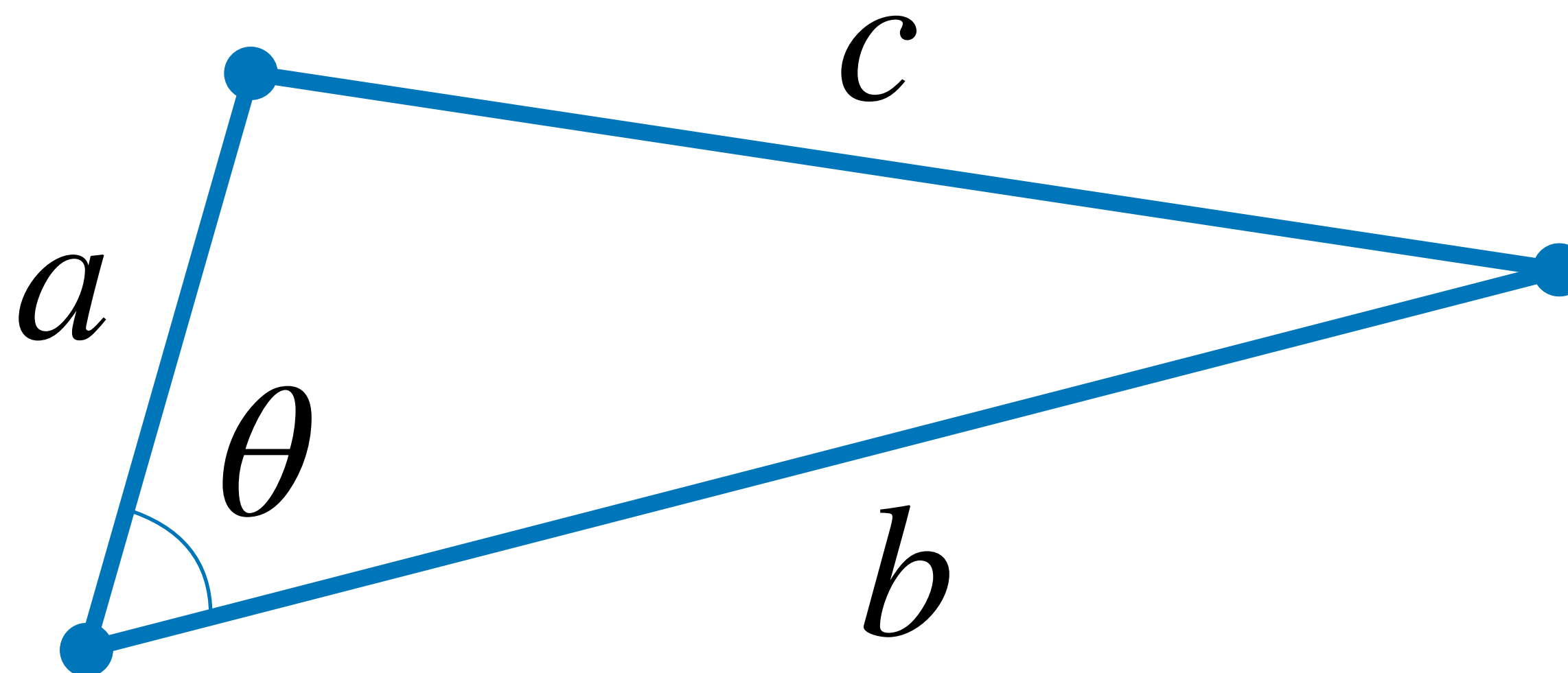
Angles

Recall: A Potentially Familiar Example



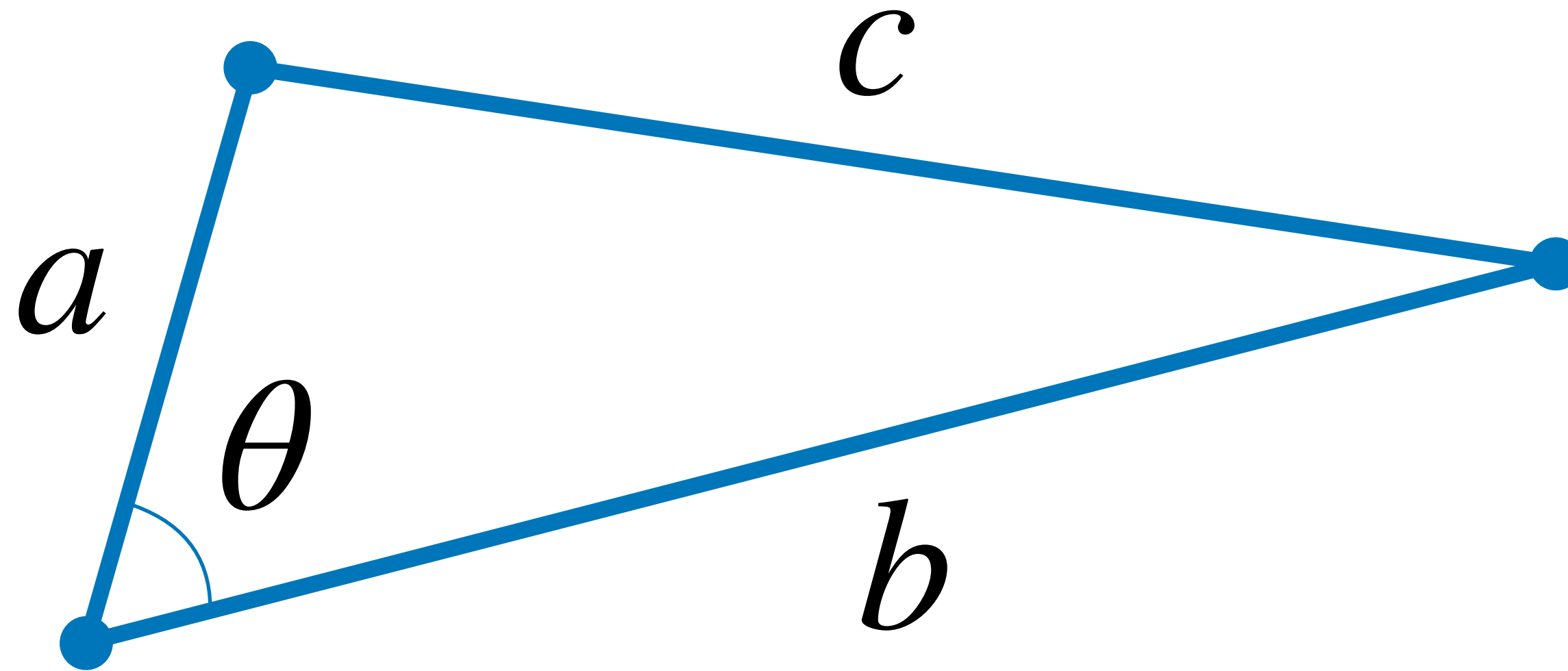
What is the value of θ ?

Law of Cosines



Thm. $c^2 = a^2 + b^2 - 2ab \cos \theta$

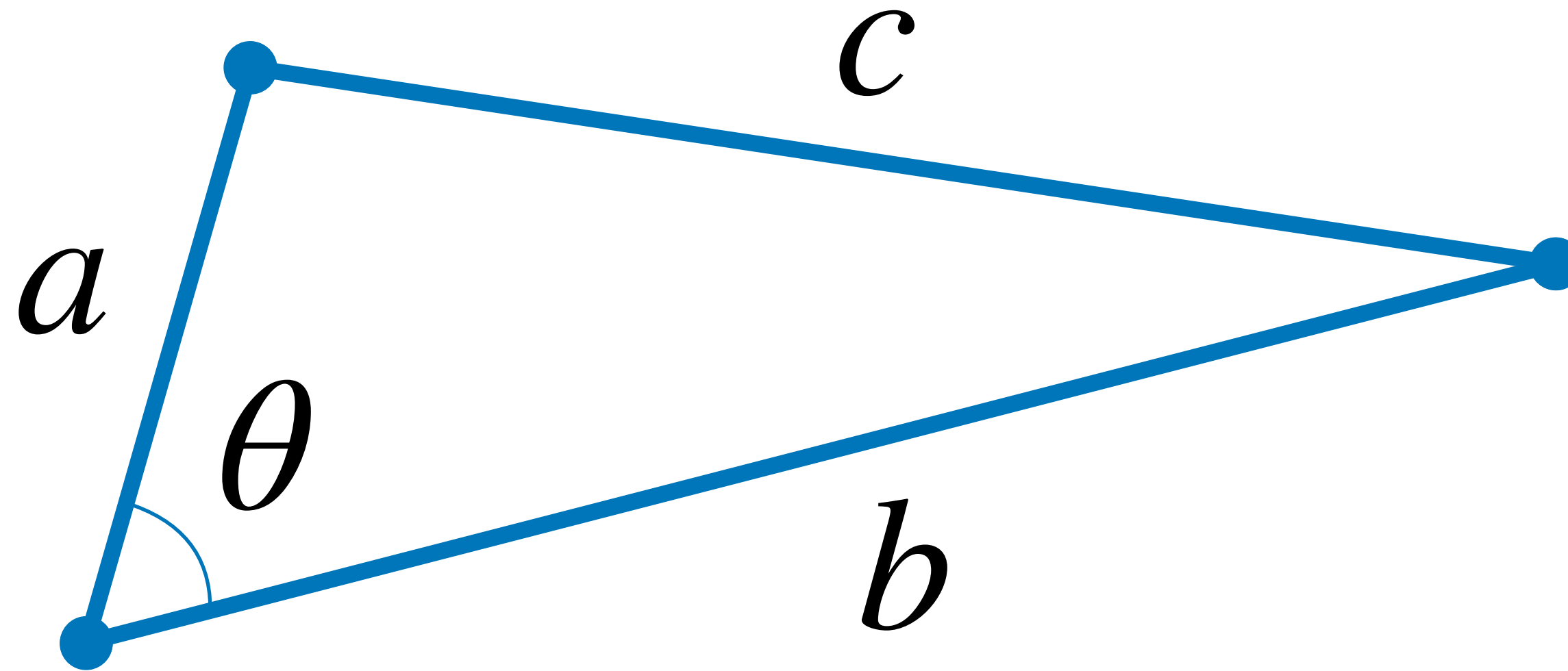
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Generalized the Pythagorean Theorem

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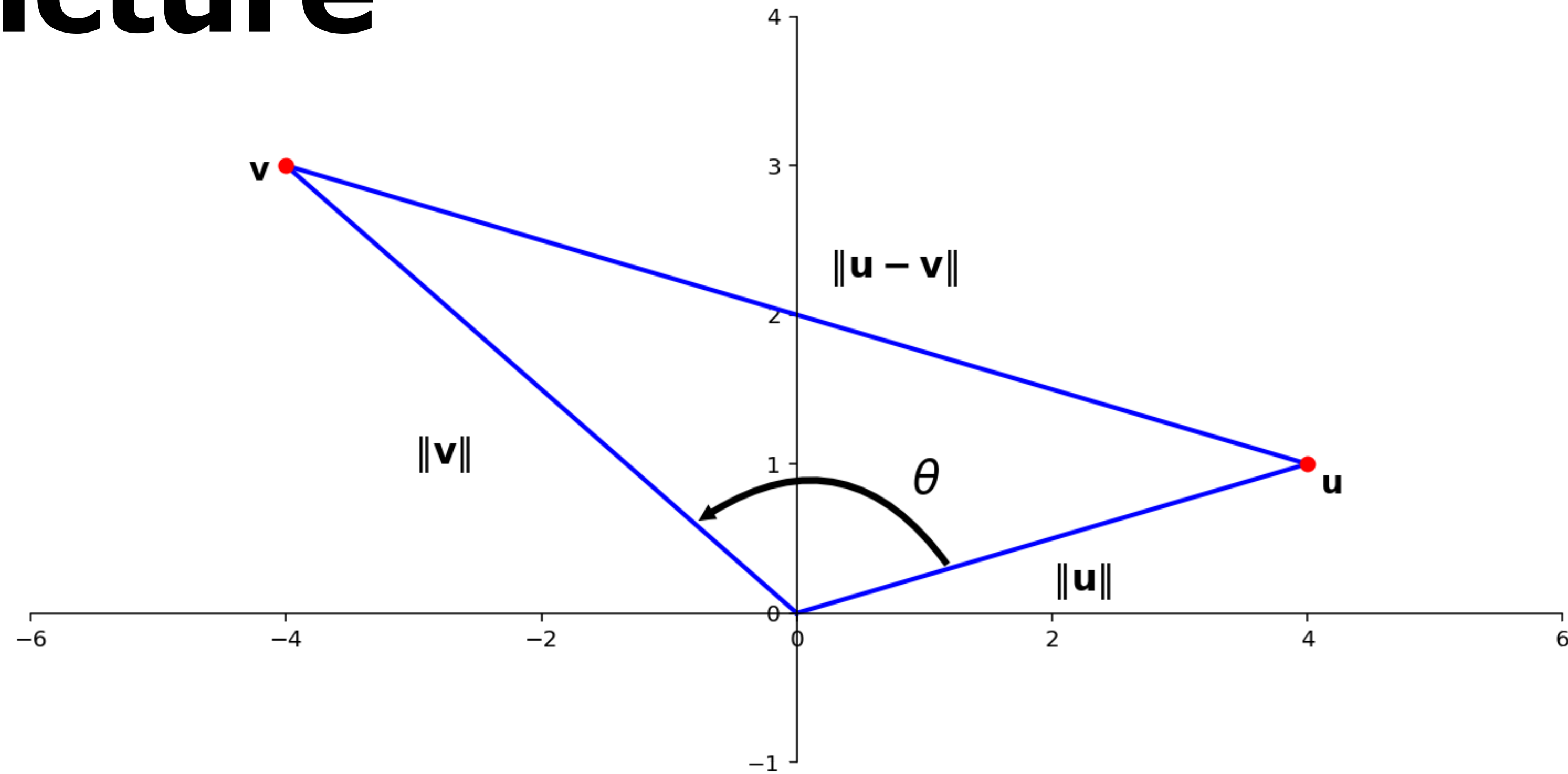


θ exactly when $\theta = 90^\circ$

Thm. $c^2 = a^2 + b^2 - 2ab \cos \theta$

Generalized the Pythagorean Theorem

The Picture



Isolating θ

$$\|\mathbf{u} - \mathbf{v}\|^2 = \|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 - 2\|\mathbf{u}\|\|\mathbf{v}\|\cos\theta$$

Cosines and Unit Vectors

Thm. For $\mathbf{u}, \mathbf{v} \in \mathbb{R}^n$ with an angle θ between them

$$\cos \theta = \left\langle \frac{\mathbf{u}}{\|\mathbf{u}\|}, \frac{\mathbf{v}}{\|\mathbf{v}\|} \right\rangle$$

The cosine of the angle between two vectors is the inner product of their ℓ^2 normalizations

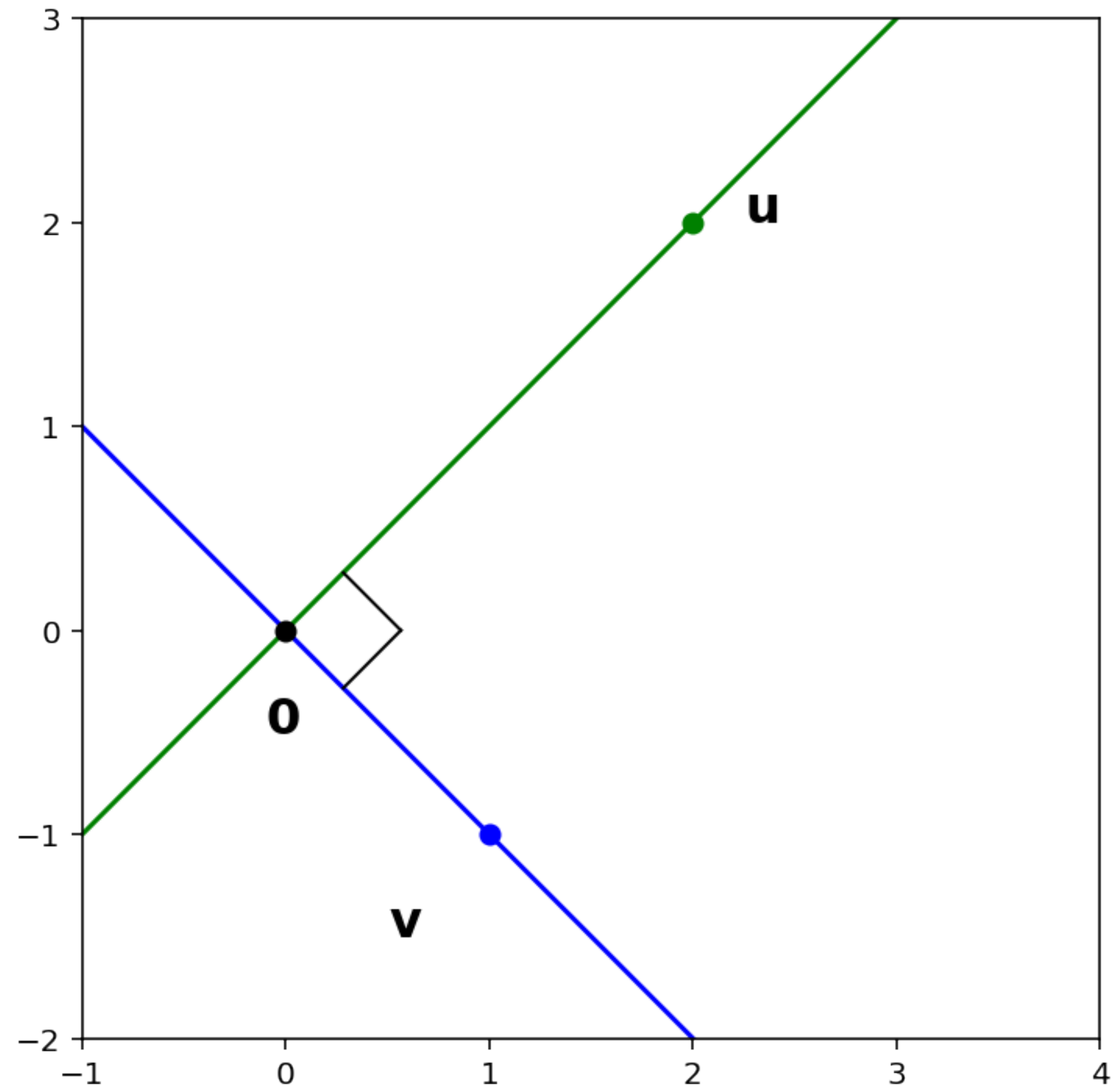
Orthogonality (Perpendicularity)

Orthogonality

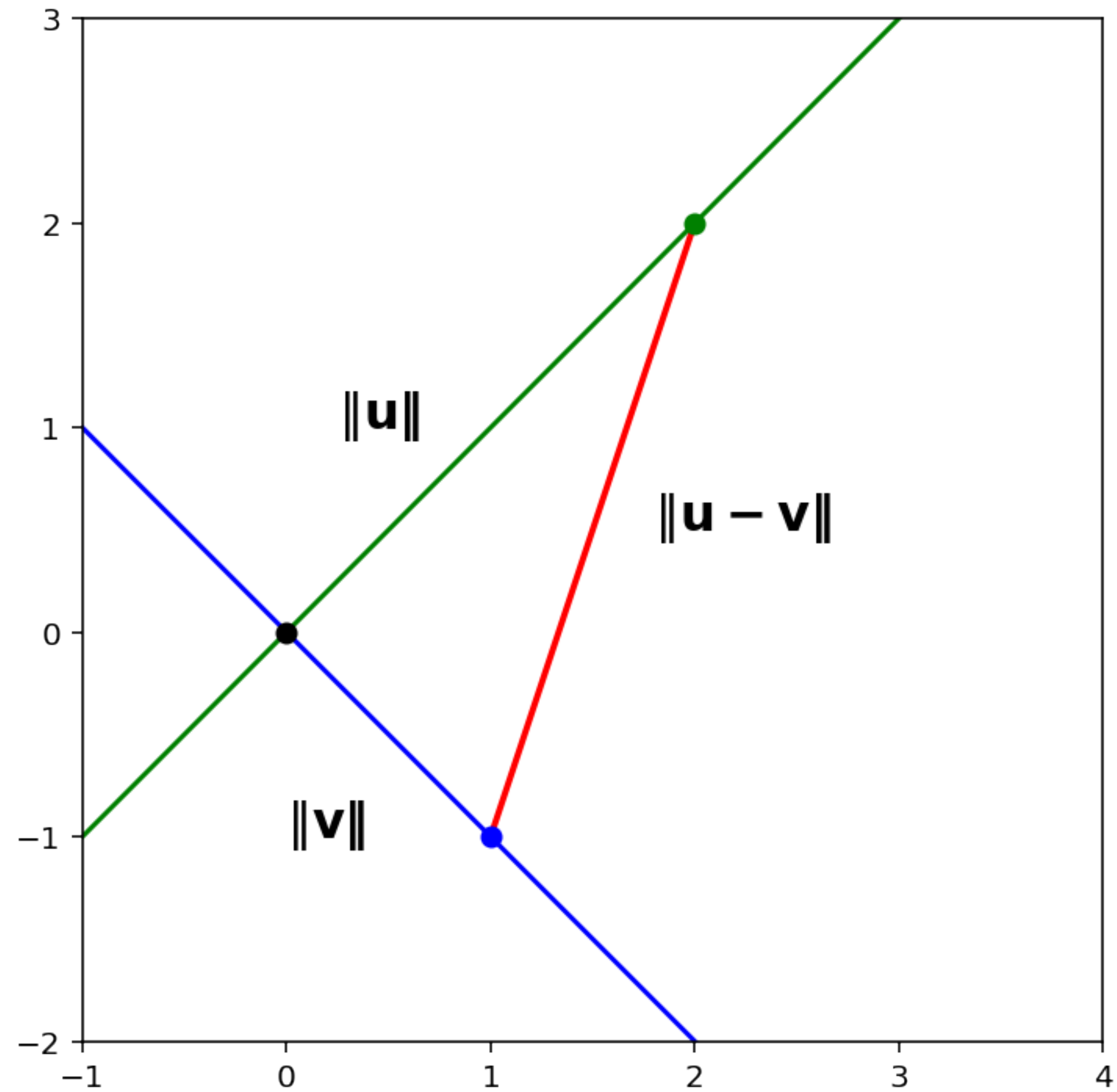
Def. Vectors $\mathbf{u}$ and $\mathbf{v}$ are **orthogonal** if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

This definition gives an easy computational way to determine orthogonality

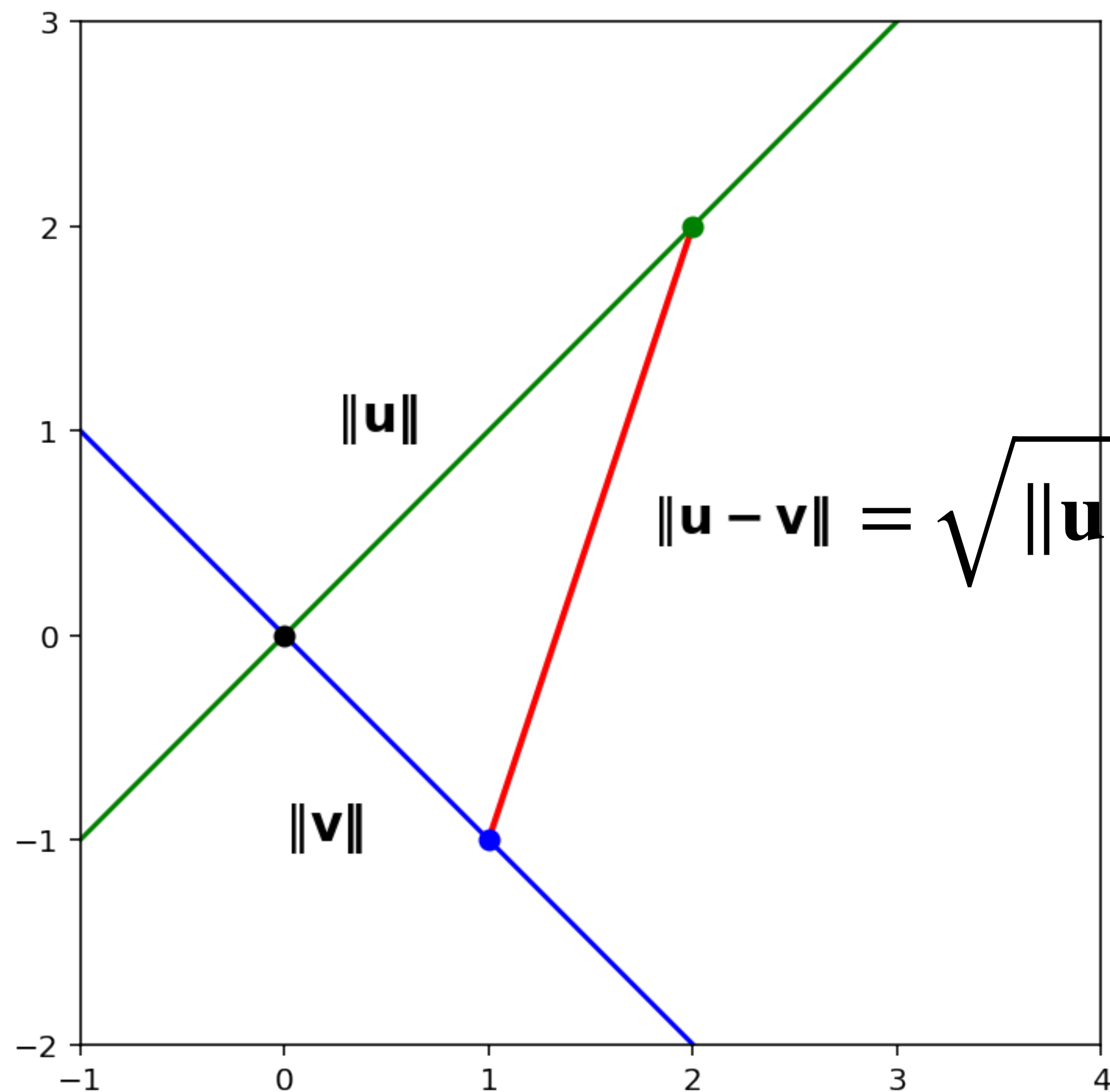
The Picture



The Picture



Derivation by Picture



$$\|\mathbf{u} - \mathbf{v}\| = \sqrt{\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2} \quad (\text{Pythagoras})$$

Isolating $\langle \mathbf{u}, \mathbf{v} \rangle$

$\mathbf{u}$ and $\mathbf{v}$ are orthogonal when $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{u} - \mathbf{v}\|^2$

With inner product we can...

- » Given a vector we can determine its *length*
- » Given two points (vectors) we can determine the *distance* between them
- » Given two vectors we can determine the *angle* between them

Orthogonal Sets

Orthogonal Sets

Def. A set $\{u_1, u_2, \dots, u_p\}$ of vectors from $\mathbb{R}^n$ is an **orthogonal set** if every pair of distinct vectors is orthogonal, i.e., if $i \neq j$ then

$$\langle u_i, u_j \rangle = 0$$

Each vector is pairwise/mutually perpendicular

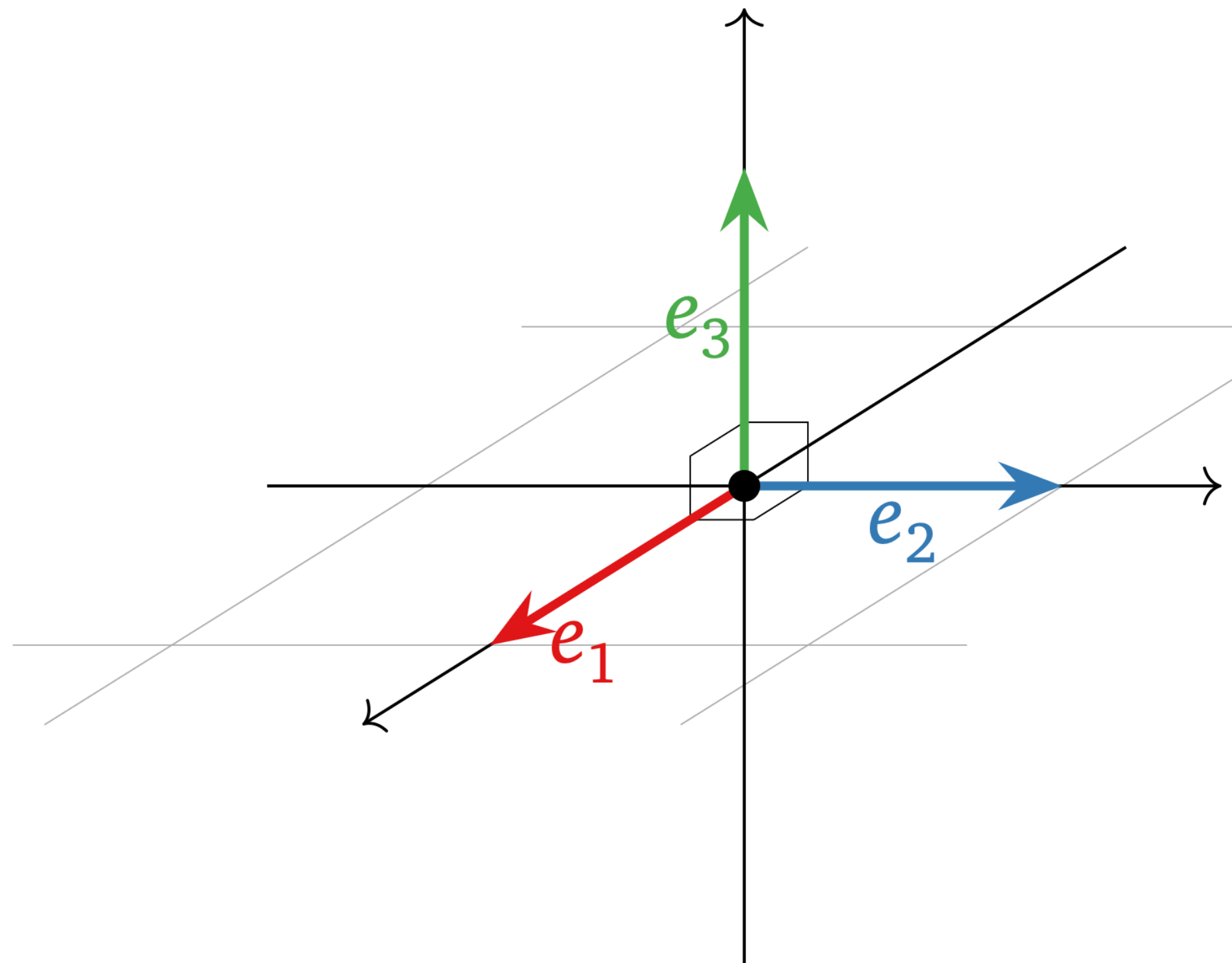
Example

$$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

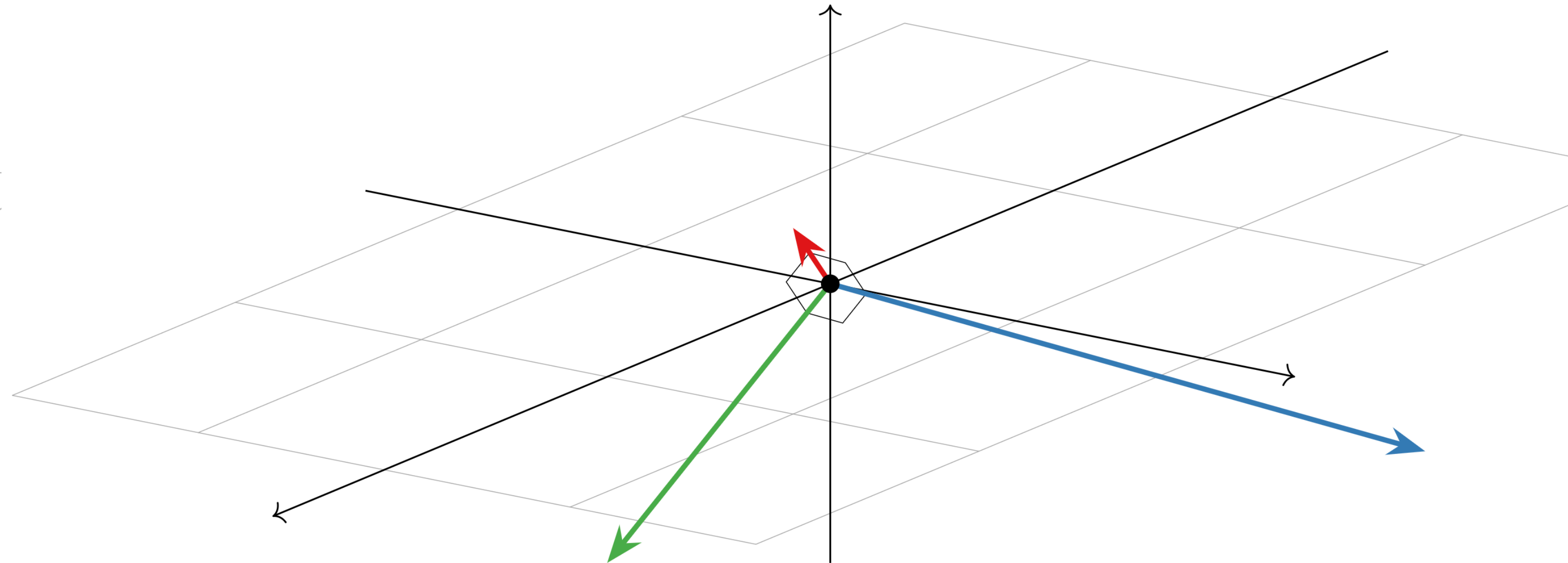
$$u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

$$u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

The Picture



the standard basis forms a
"centered" orthogonal set



an orthogonal set is like
the standard basis *after*
some rotations and scalings

Orthogonal Sets and Independence

Thm. If $\{u_1, u_2, \dots, u_k\}$ is an orthogonal set of *nonzero* vectors from $\mathbb{R}^n$, then it is *L.I.*

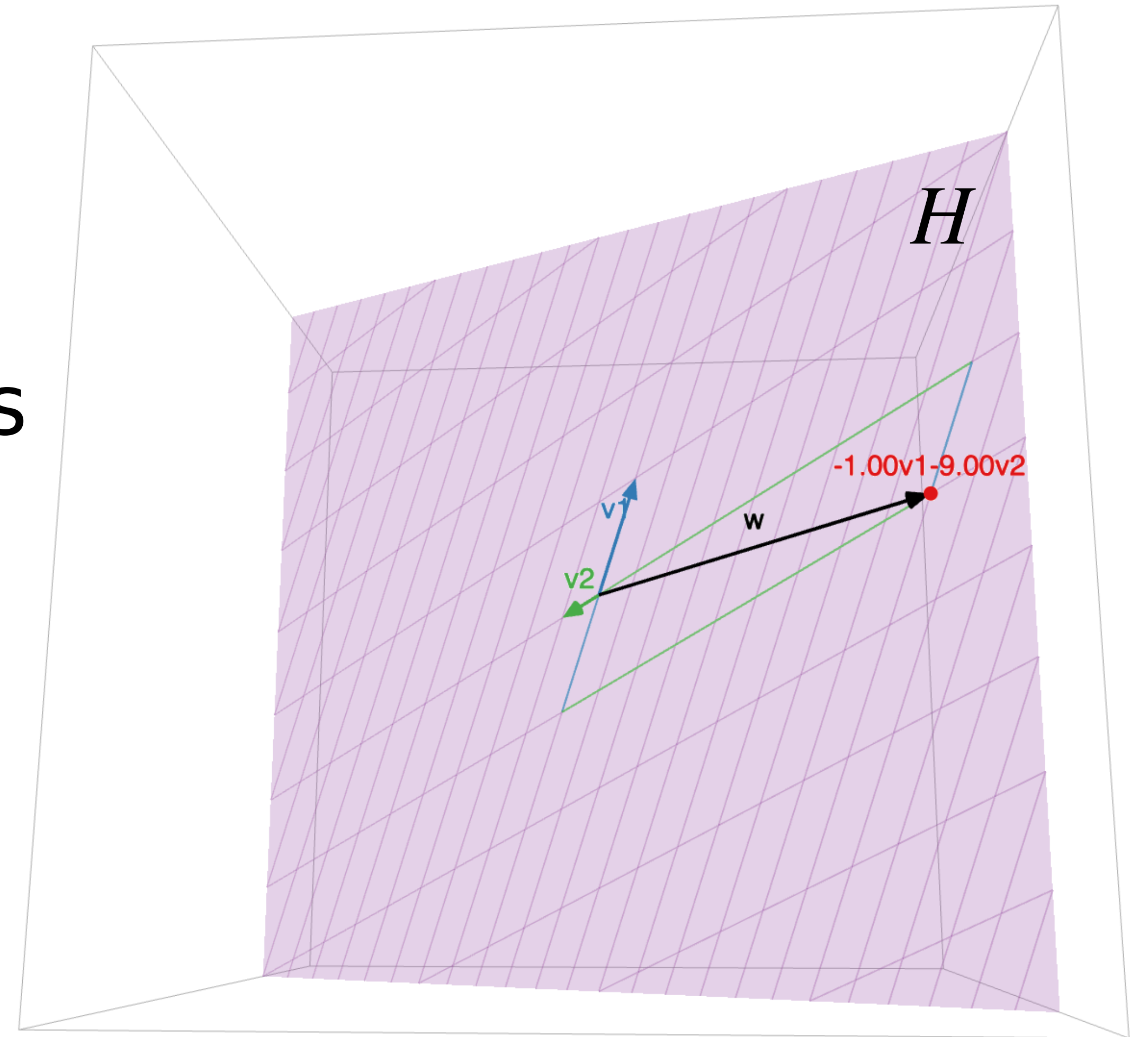
If $\{u_1, u_2, \dots, u_k\}$ is an orthogonal set,
then it is a **basis** for $\text{span}\{u_1, u_2, \dots, u_k\}$

Orthogonal Basis

Def. An orthogonal basis for a subspace W of $\mathbb{R}^n$ is a basis for W which is also an orthogonal set

Orthogonal Basis

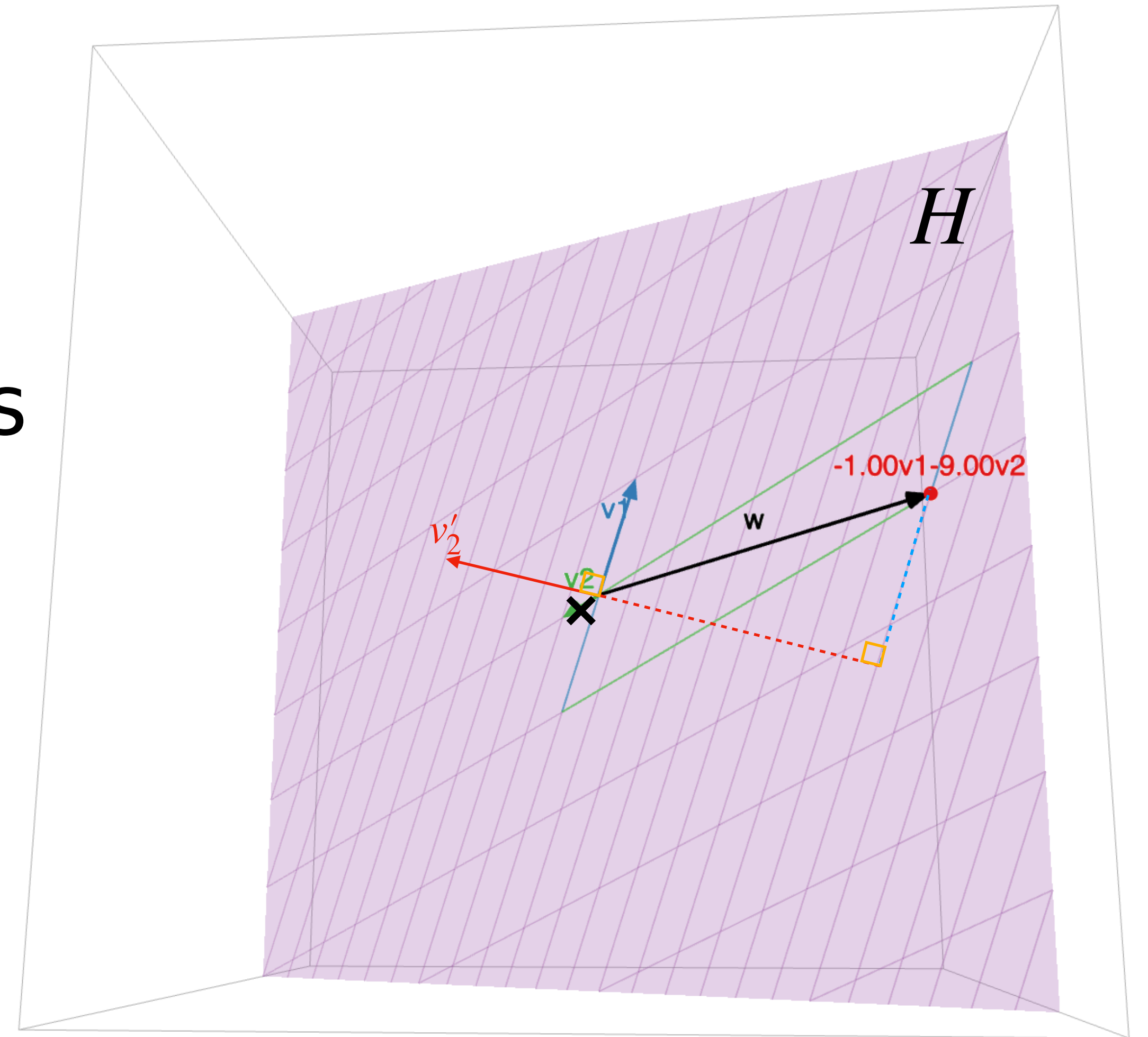
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v_1 and v_2 form a basis of H

Orthogonal Basis

Def. An **orthogonal basis** for a subspace W of $\mathbb{R}^n$ is a basis for W which is also an orthogonal set



v_1 and v_2 form a basis of H

v_1 and v_2' form an **orthogonal** basis of H

What's nice about an
orthogonal basis?

Recall: Bases

Recall: Bases

Q. Given a basis $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$ for a subspace W of R^n and a vector $\mathbf{w}$ in W , find weights $c_1, c_2, \dots, c_p$ such that

$$\mathbf{w} = c_1 \mathbf{u}_1 + c_2 \mathbf{u}_2 + \dots c_p \mathbf{u}_p$$

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$$\mathbf{w} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \dots c_p\mathbf{u}_p$$

A. Solve the vector equation

$$x_1\mathbf{u}_1 + x_2\mathbf{u}_2 + \dots x_p\mathbf{u}_p = \mathbf{w}$$

by Gaussian elimination, matrix inversion, etc.

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by Gaussian elimination, matrix inversion, etc.

This takes work

Orthogonal Bases and Linear Combinations

Thm. For an orthogonal set $\{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_p\}$, if $\mathbf{y} = c_1\mathbf{u}_1 + \dots + c_p\mathbf{u}_p$ then for $j = 1, \dots, p$

$$c_j = \frac{\mathbf{y}^T \mathbf{u}_j}{\mathbf{u}_j^T \mathbf{u}_j}$$

Orthogonal Bases and Linear Combinations

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Much easier to compute

Question

Express $[6 \ 1 \ (-8)]^T$ as a linear combination of vectors in $\{u_1, u_2, u_3\}$ where

$$u_1 = \begin{bmatrix} 3 \\ 1 \\ 1 \end{bmatrix}$$

$$u_2 = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix}$$

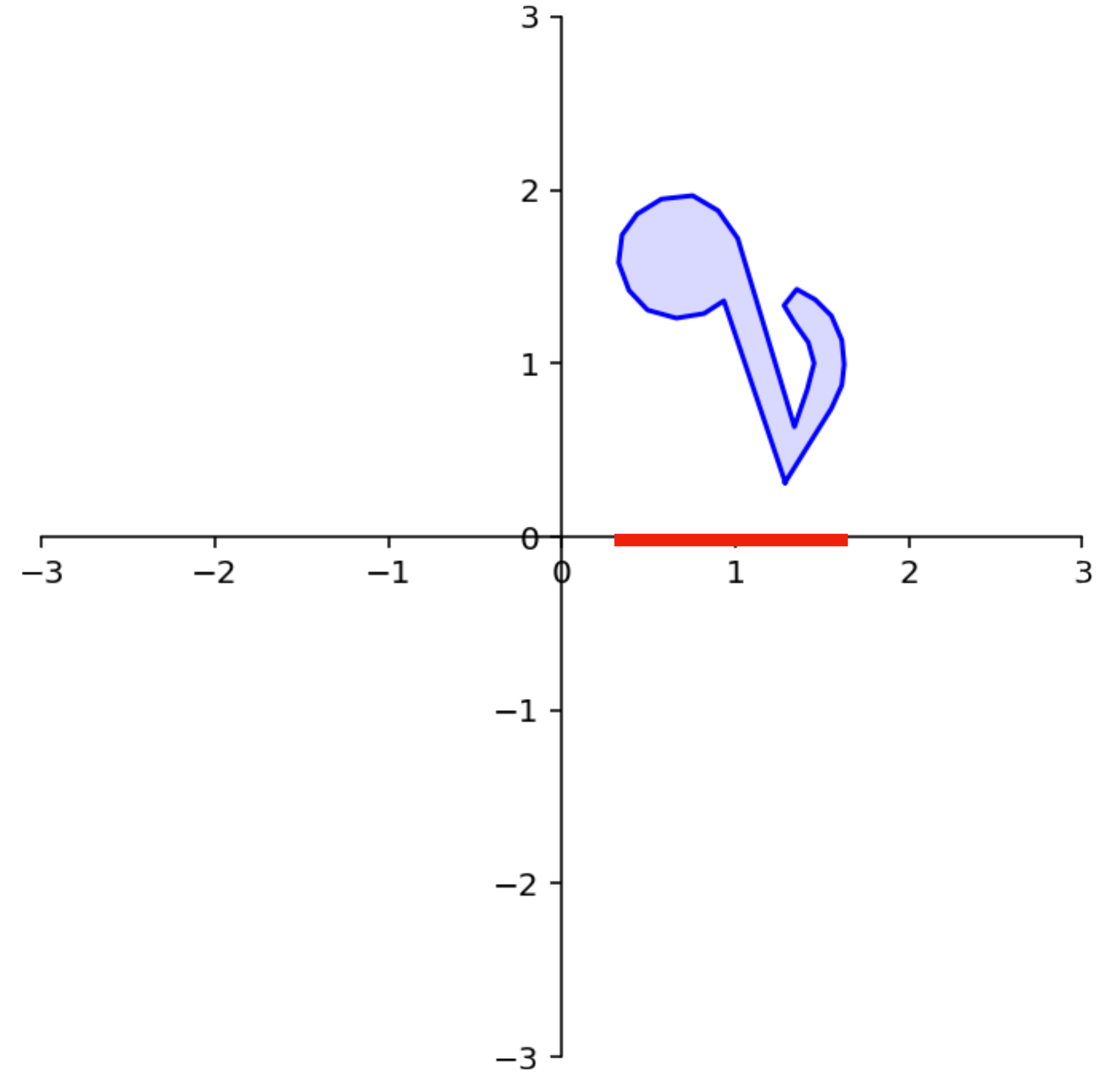
$$u_3 = \begin{bmatrix} -1/2 \\ -2 \\ 7/2 \end{bmatrix}$$

Answer: $u_1 - 2u_2 - 2u_3$

Orthogonal Projection

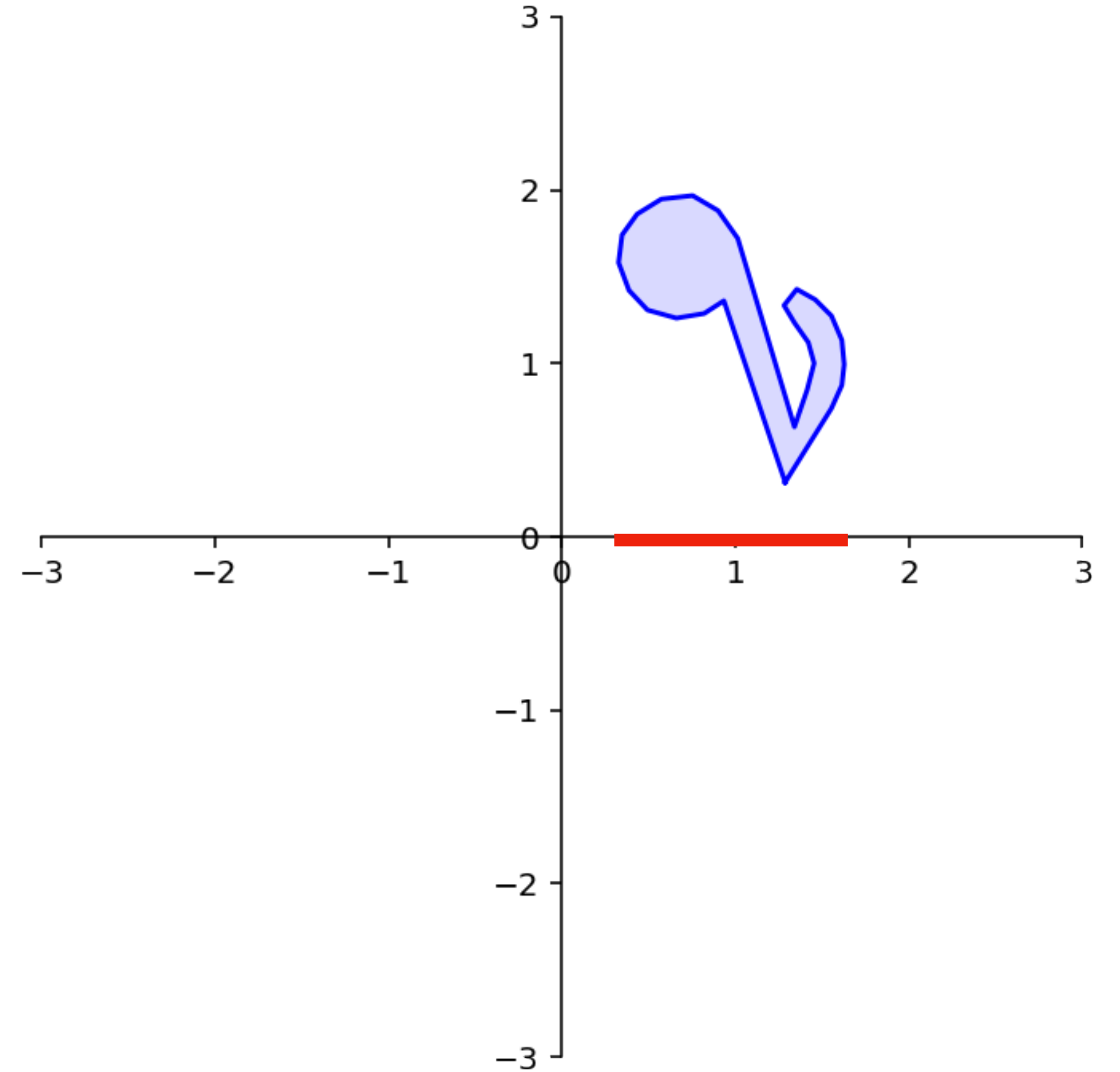
Why does that formula in
the last example work?

Recall: Projection onto the x -axis



Recall: Projection onto the x -axis

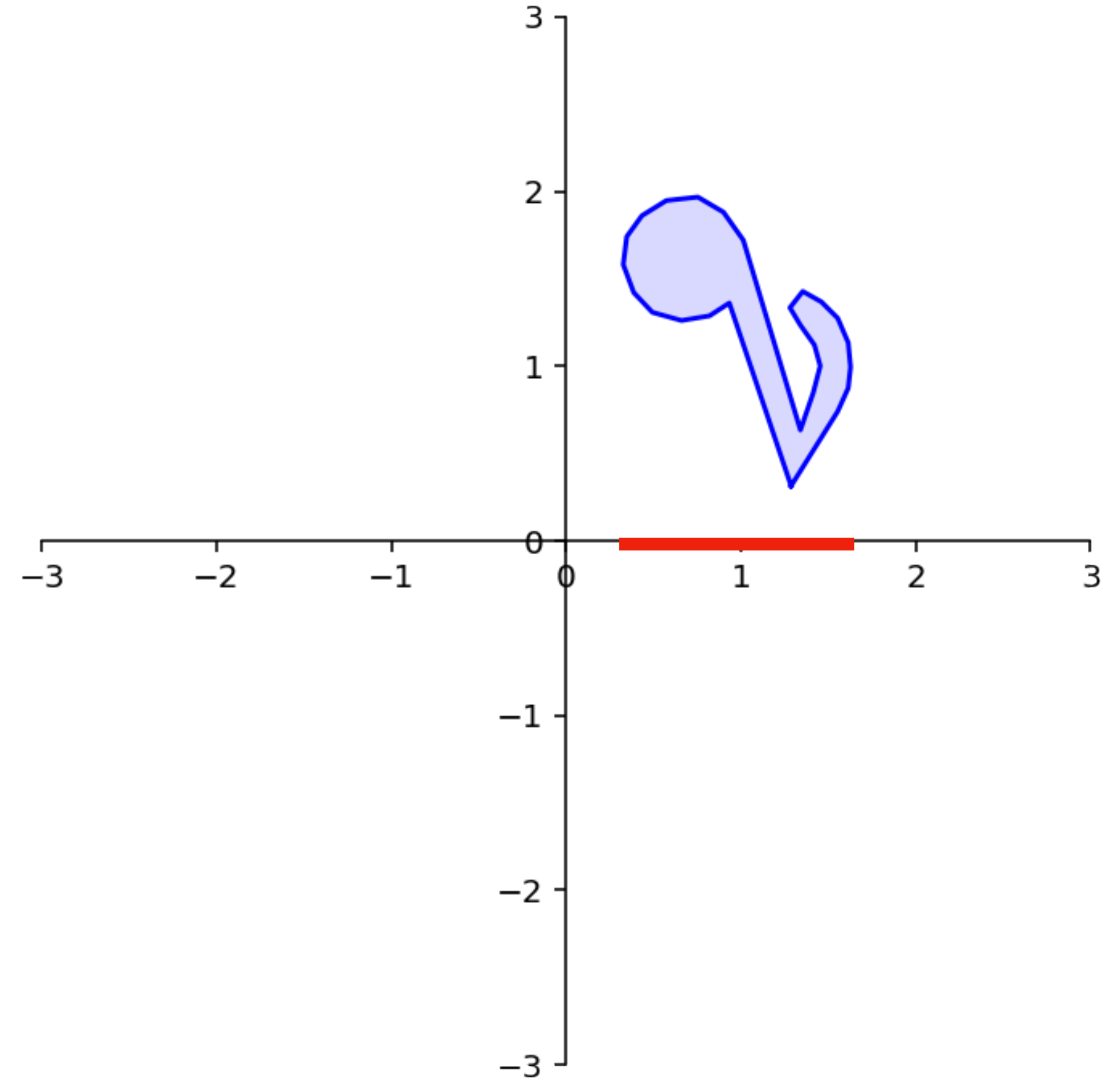
We've seen simple
projections in $\mathbb{R}^2$



Recall: Projection onto the x -axis

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projections in $\mathbb{R}^2$

We're going to generalize
this idea

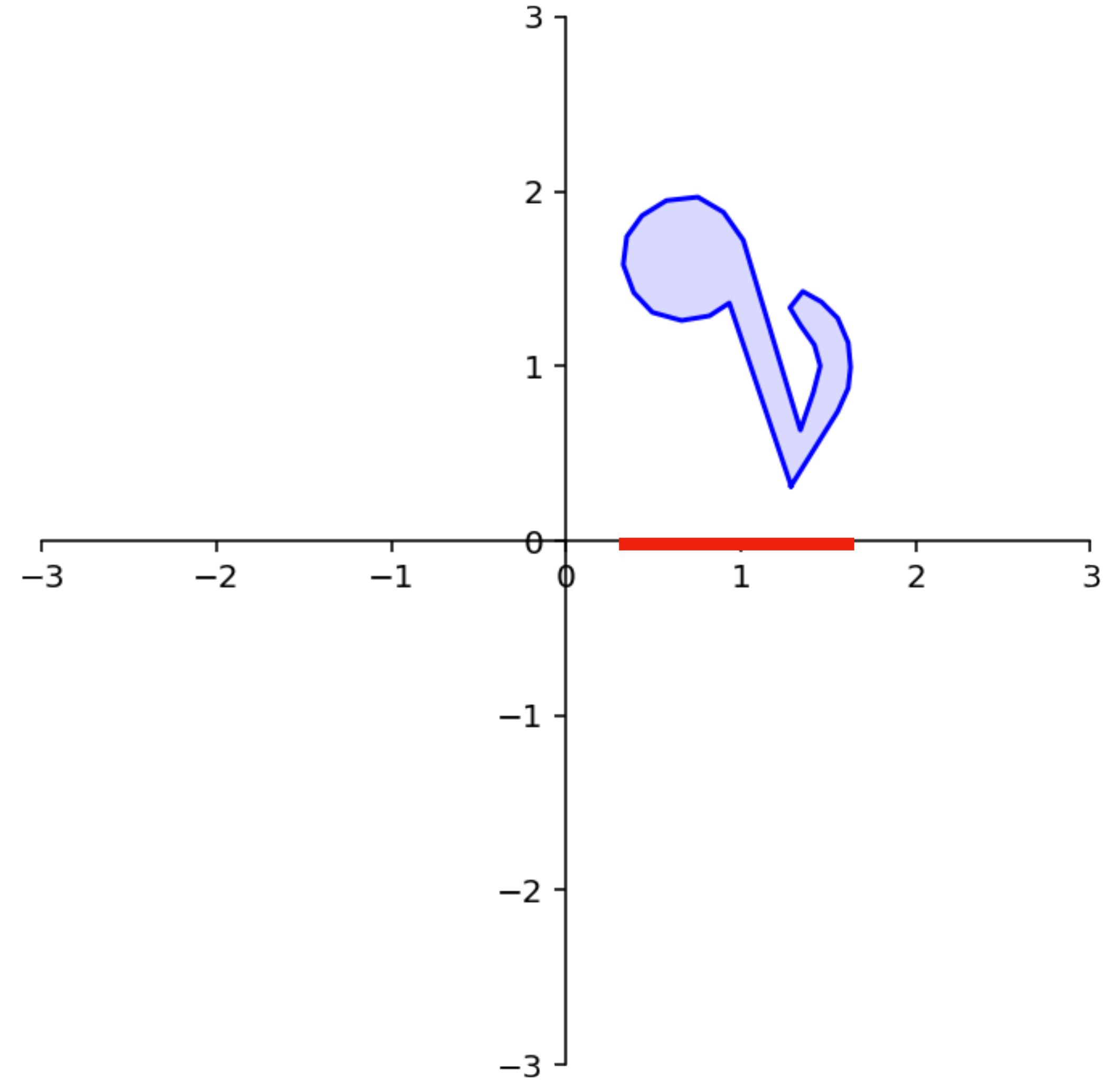


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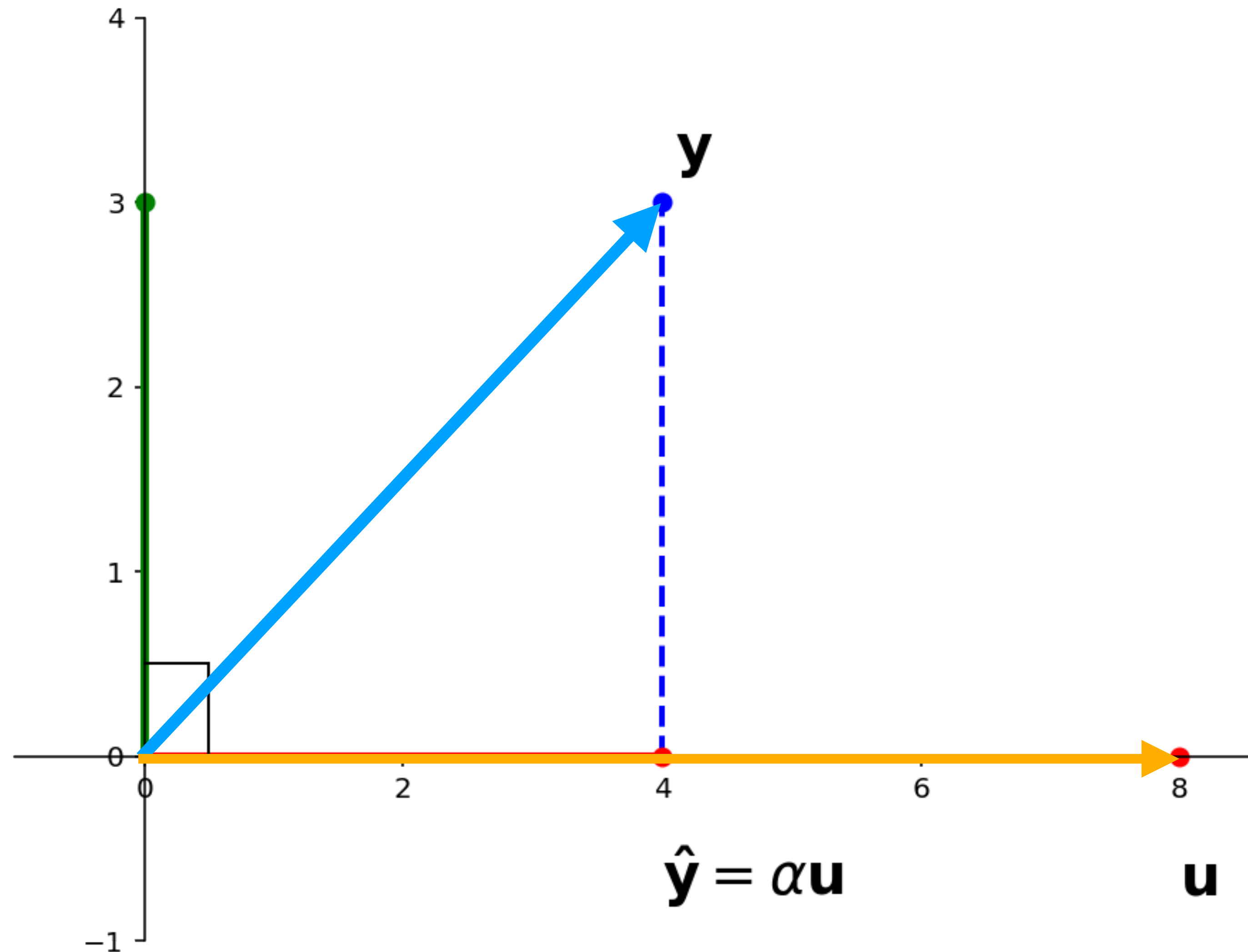
We've seen simple
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We're going to generalize
this idea

**What we really did was a
kind of projection onto
the basis vectors**

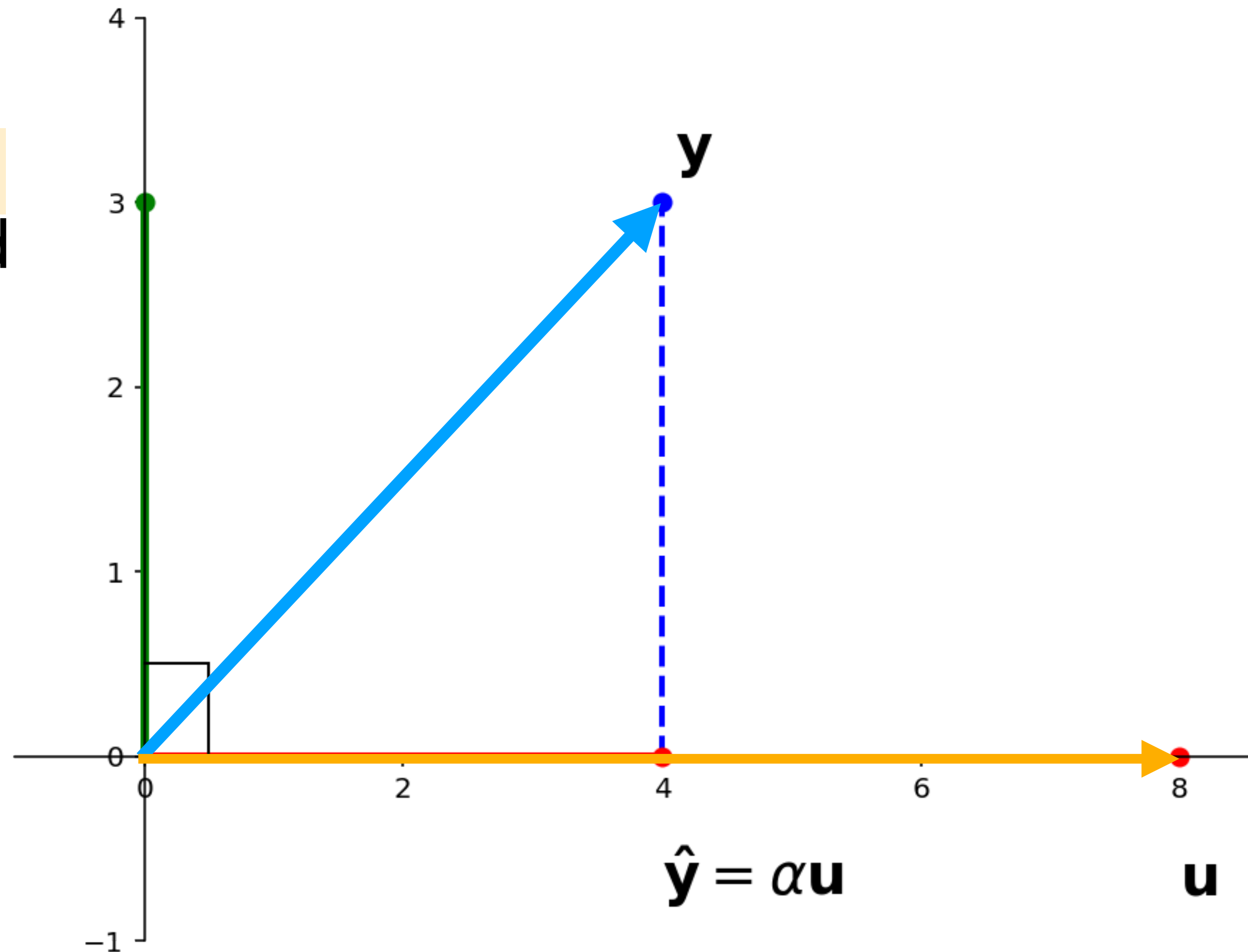


Orthogonal Projection



Orthogonal Projection

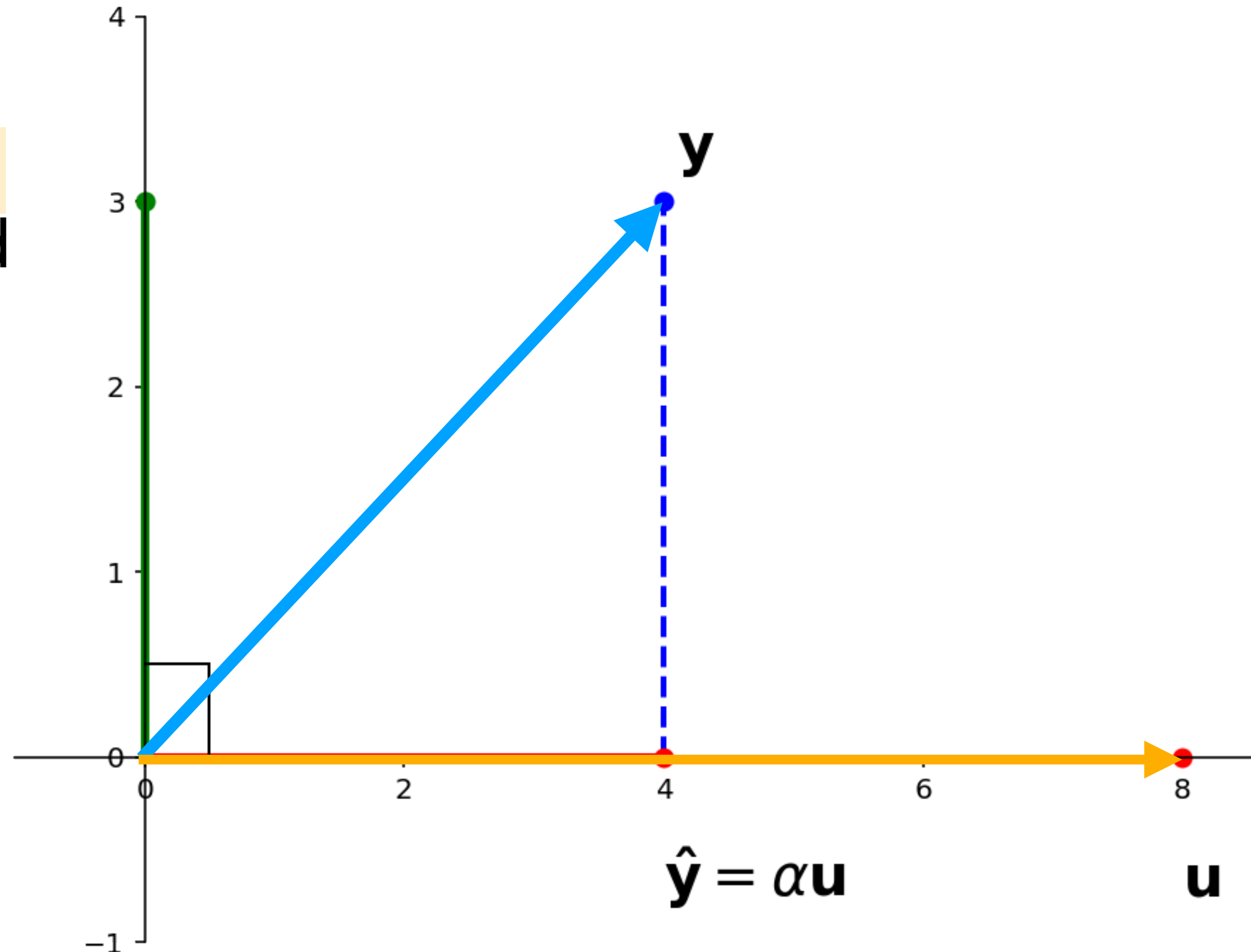
Q. Given vectors $\mathbf{y}$ and $\mathbf{u}$ in $\mathbb{R}^n$, find vectors $\hat{\mathbf{y}}$ and $\mathbf{z}$ such that



Orthogonal Projection

Q. Given vectors **y** and **u** in $\mathbb{R}^n$, find vectors $\hat{y}$ and **z** such that

» **z** is orthogonal to **u**
(i.e., $\mathbf{z} \cdot \mathbf{u} = 0$)

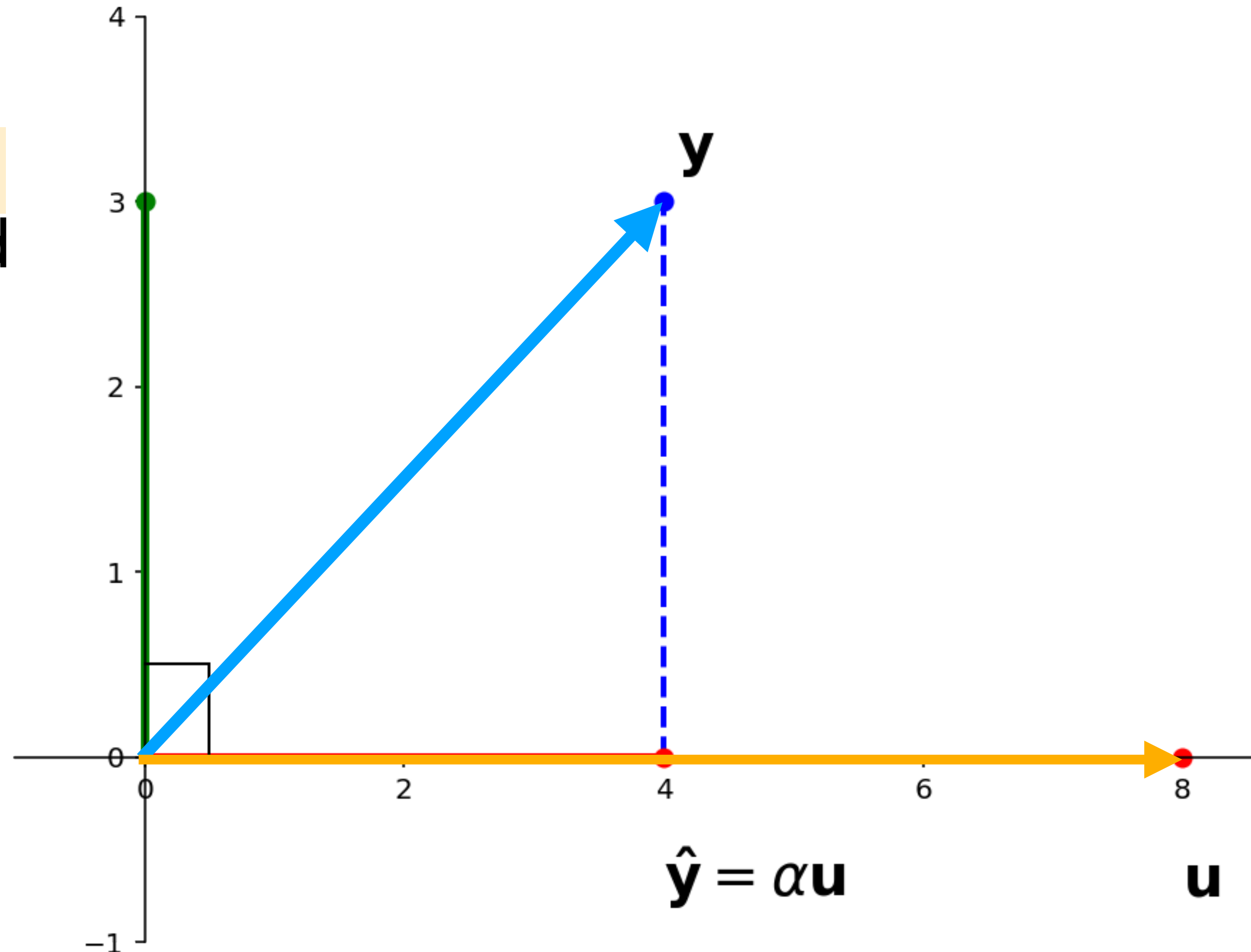


Orthogonal Projection

Q. Given vectors **y** and **u** in $\mathbb{R}^n$, find vectors $\hat{y}$ and **z** such that

» **z** is orthogonal to **u**
(i.e., $\mathbf{z} \cdot \mathbf{u} = 0$)

» $\hat{\mathbf{y}} \in \text{span}\{\mathbf{u}\}$



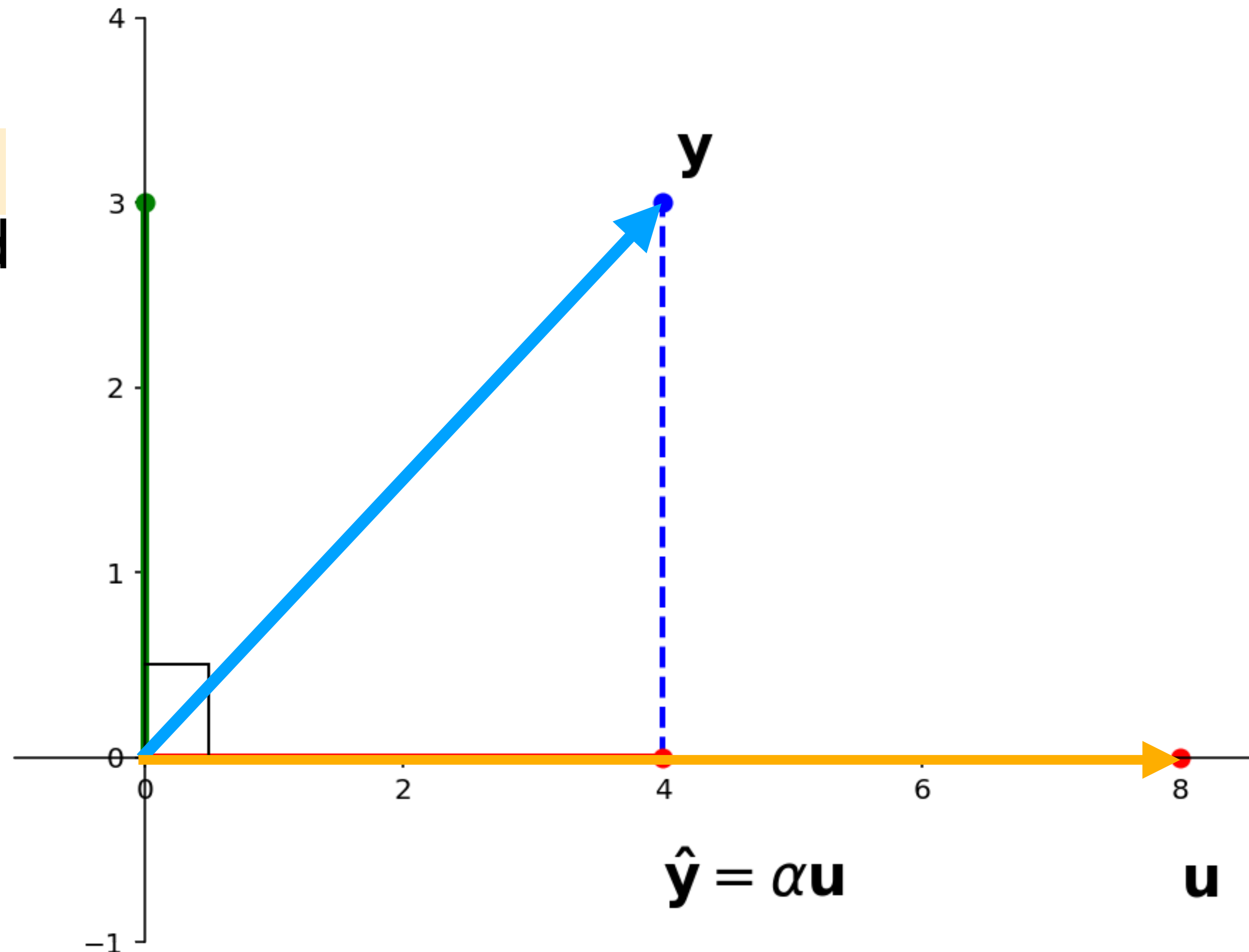
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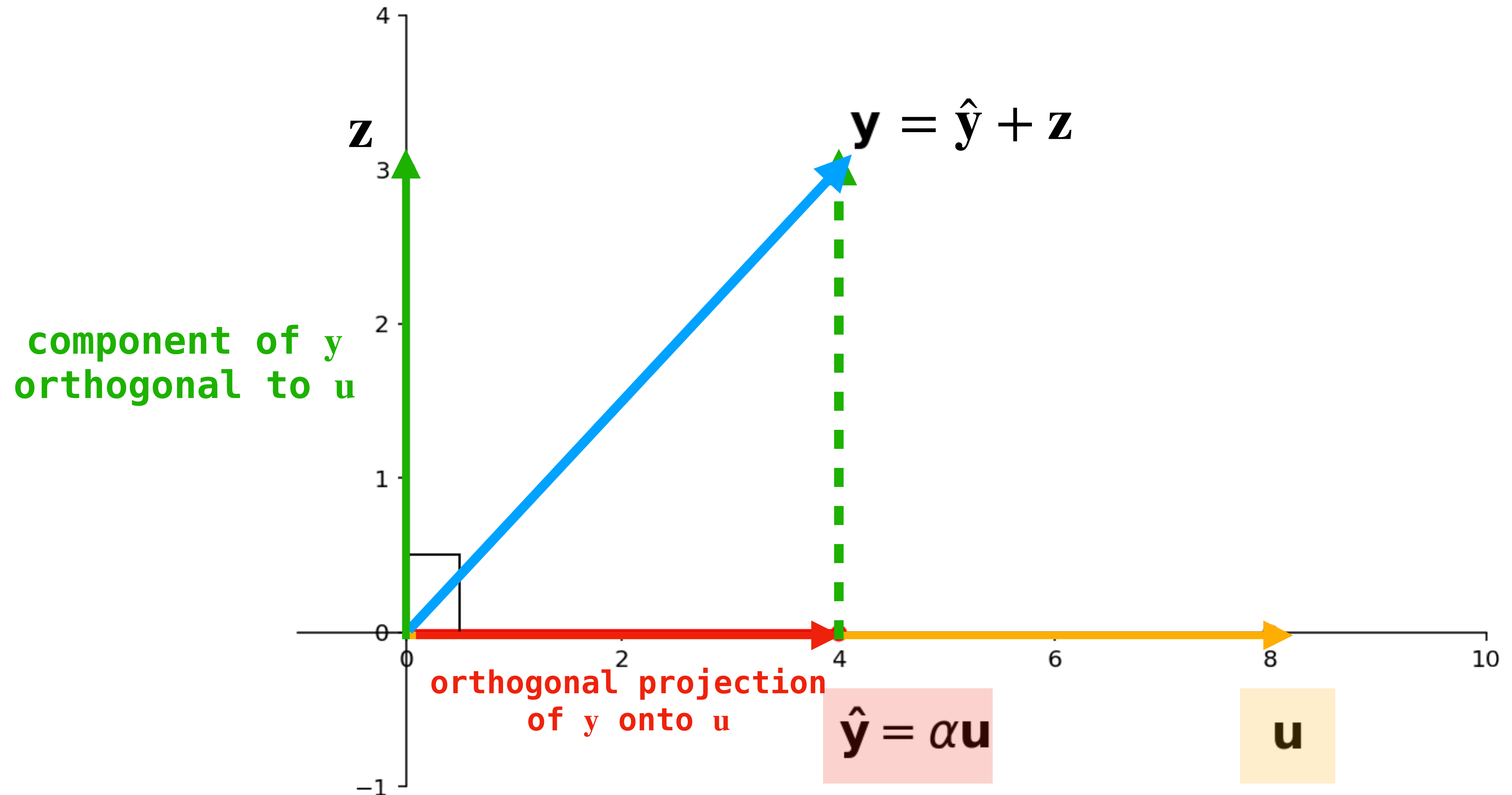
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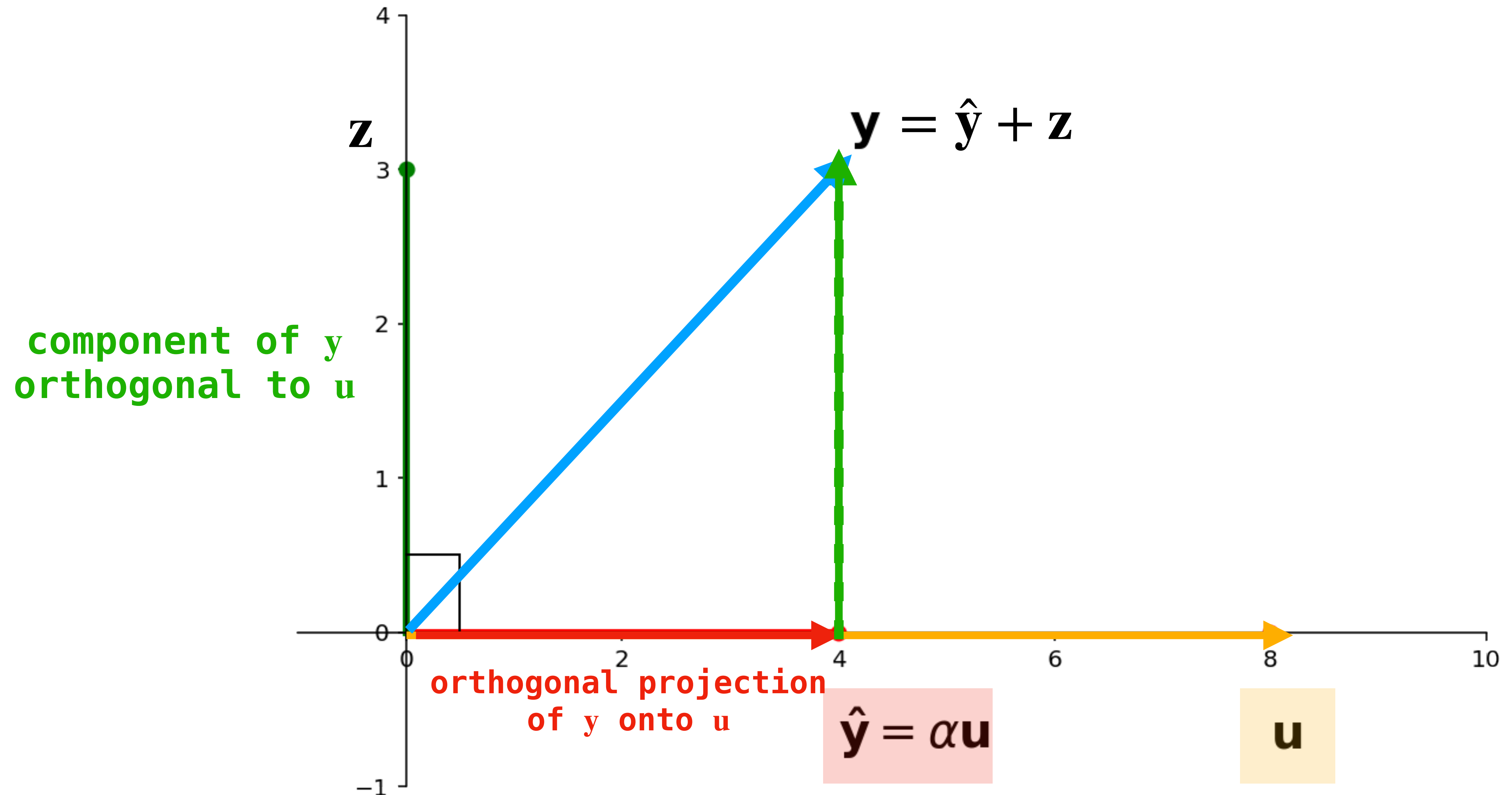
» $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$



Orthogonal Projection

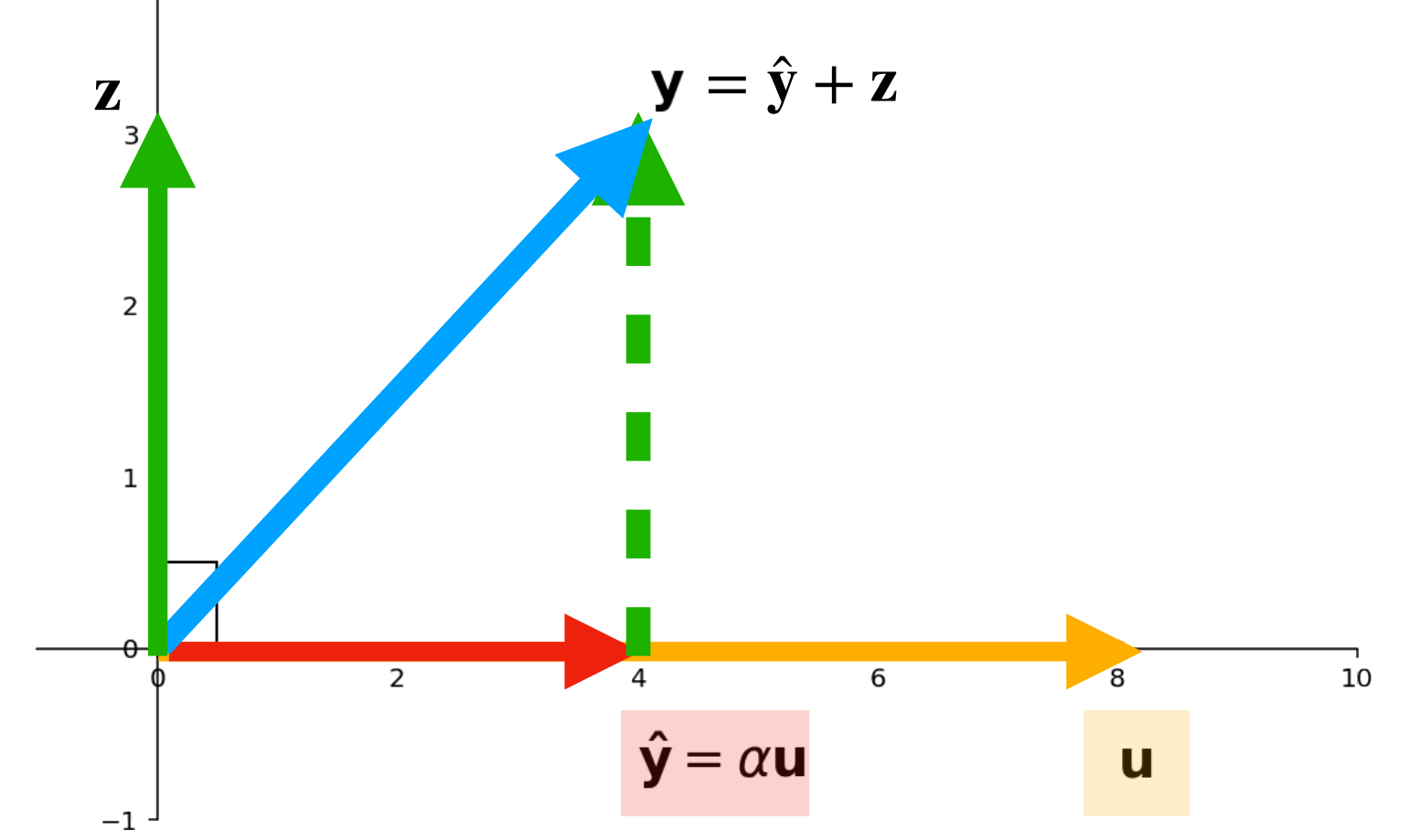


Orthogonal Projection



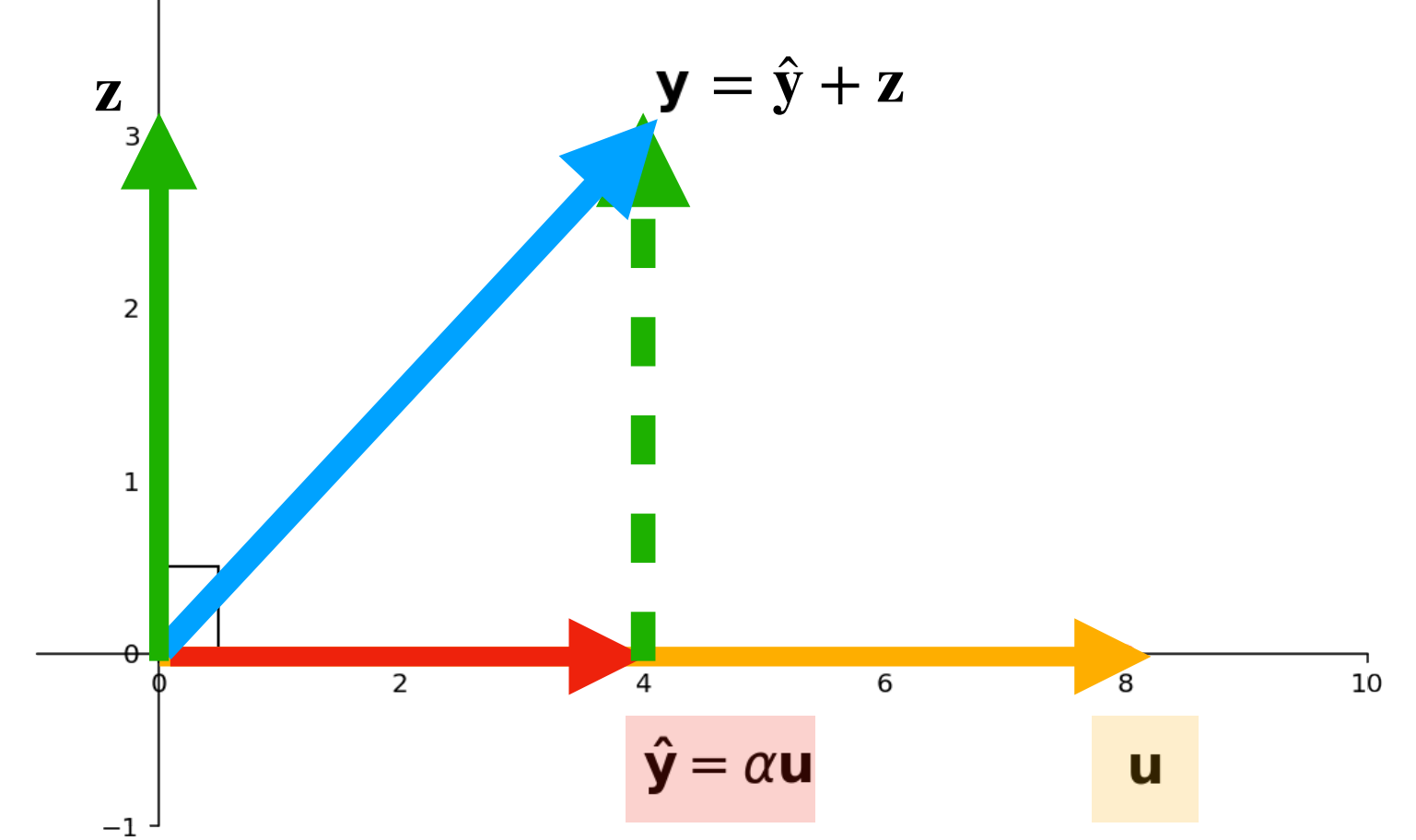
How do we find the orthogonal
projection and orthogonal component?

What we know

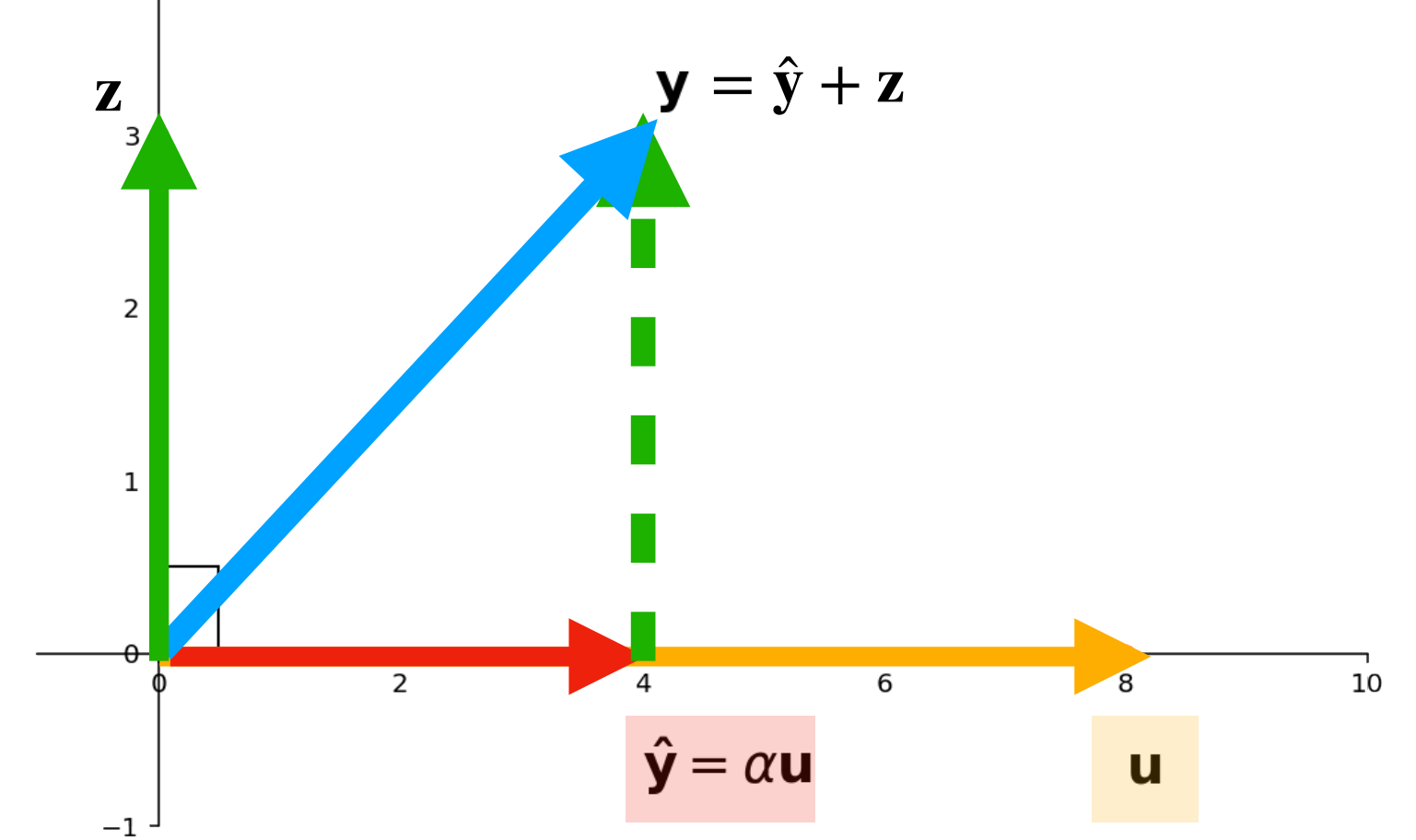


What we know

» $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in \text{span}\{\mathbf{u}\}$)



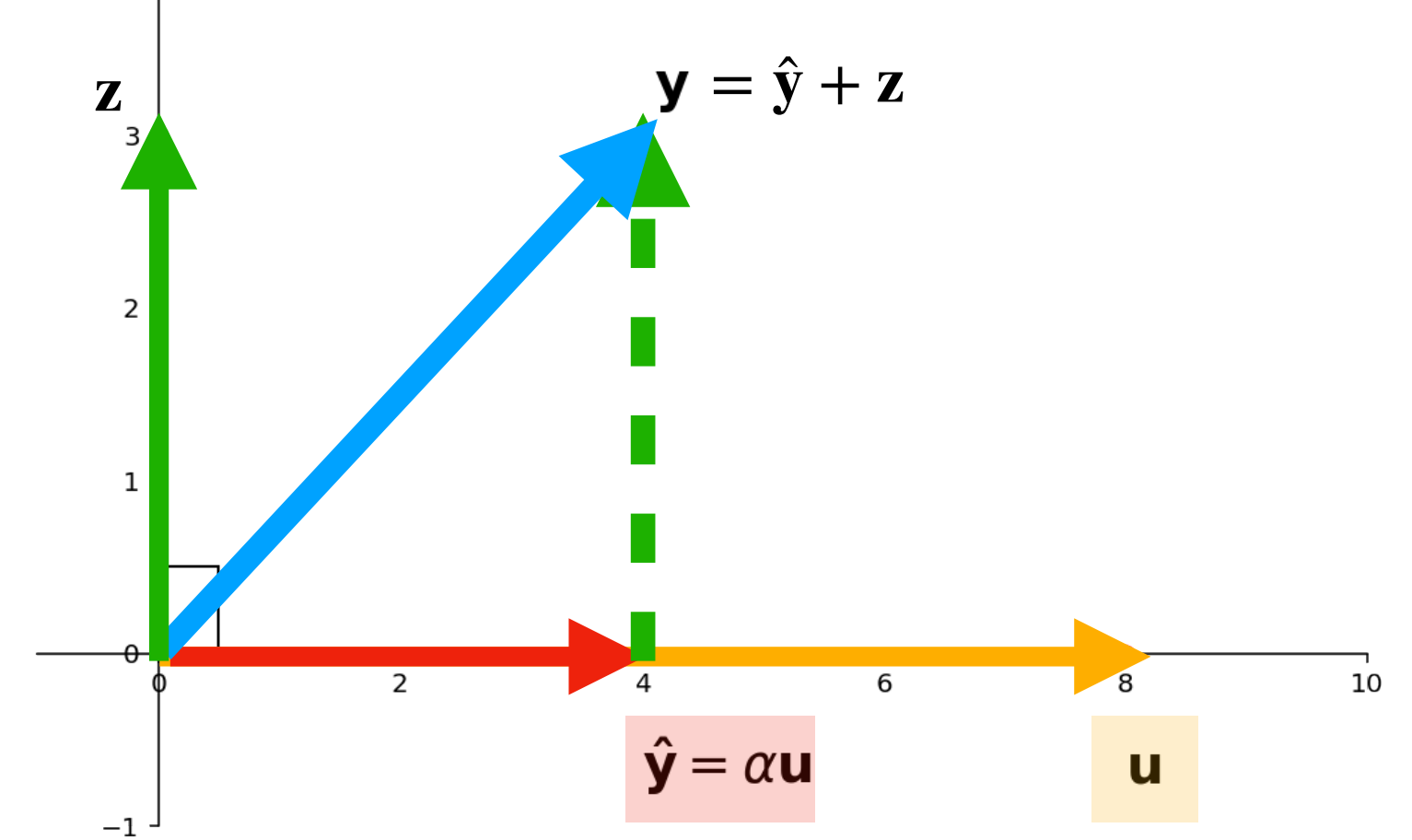
What we know



» $\hat{y} = \alpha u$ for some scalar α (since $\hat{y} \in \text{span}\{u\}$)

» $z = y - \hat{y} = y - \alpha u$ (since $y = \hat{y} + z$)

What we know

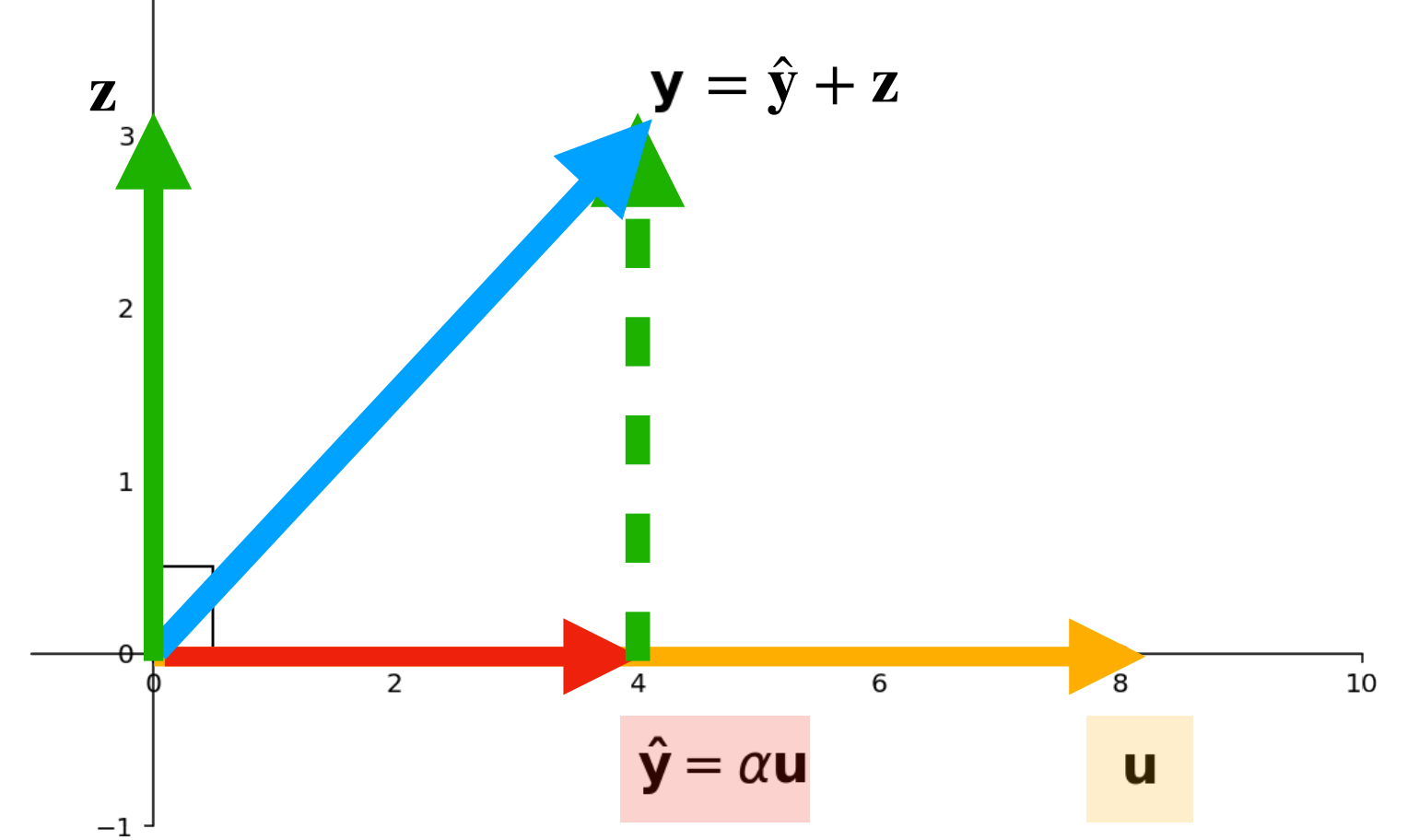


» $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in \text{span}\{\mathbf{u}\}$)

» $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \alpha \mathbf{u}$ (since $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$)

» $\langle \mathbf{z}, \mathbf{u} \rangle = 0$ (since $\mathbf{z}$ is orthogonal with $\mathbf{u}$)

What we know



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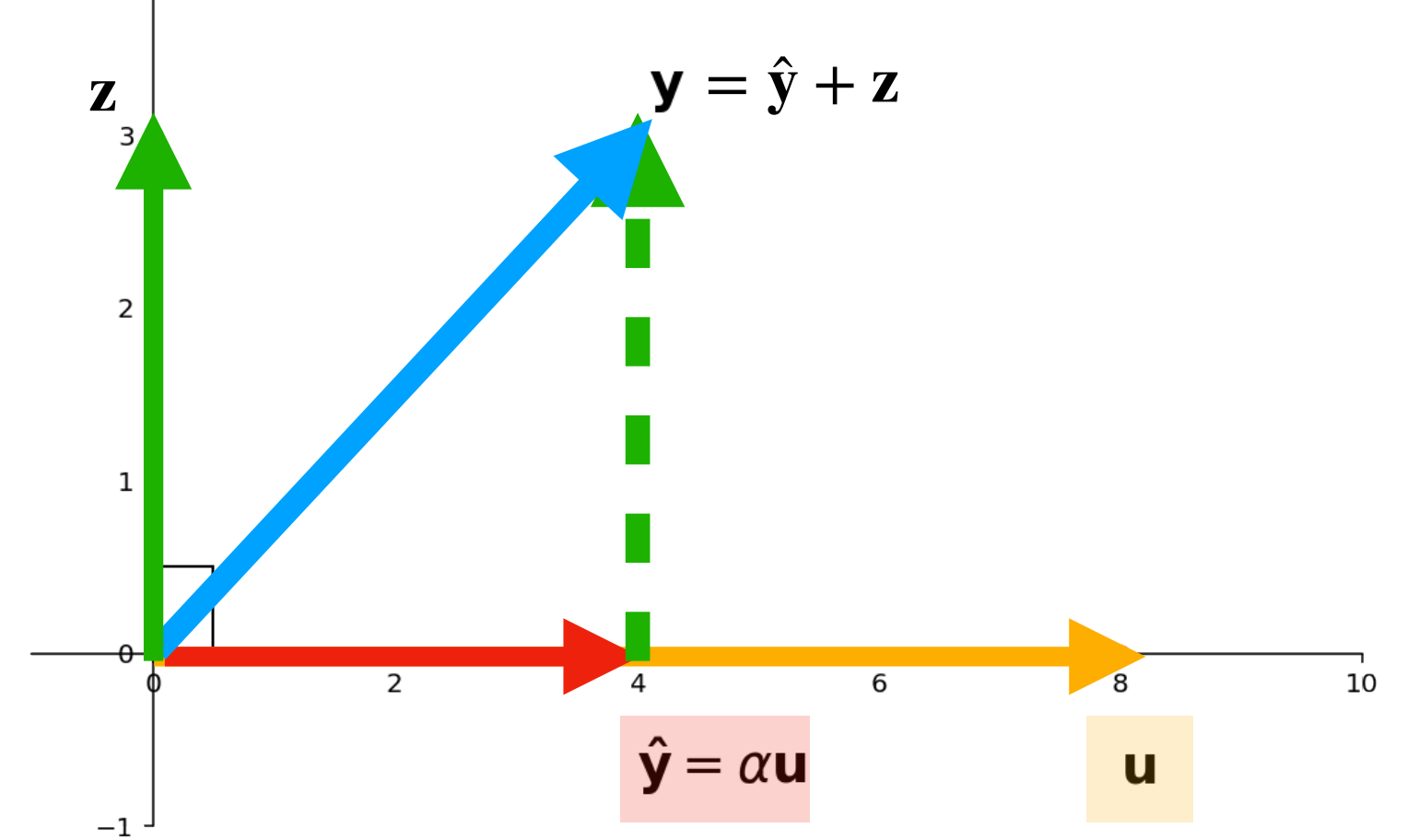
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Therefore:

$$\langle \mathbf{y} - \alpha \mathbf{u}, \mathbf{u} \rangle = 0$$

What we know



- » $\hat{\mathbf{y}} = \alpha \mathbf{u}$ for some scalar α (since $\hat{\mathbf{y}} \in \text{span}\{\mathbf{u}\}$)
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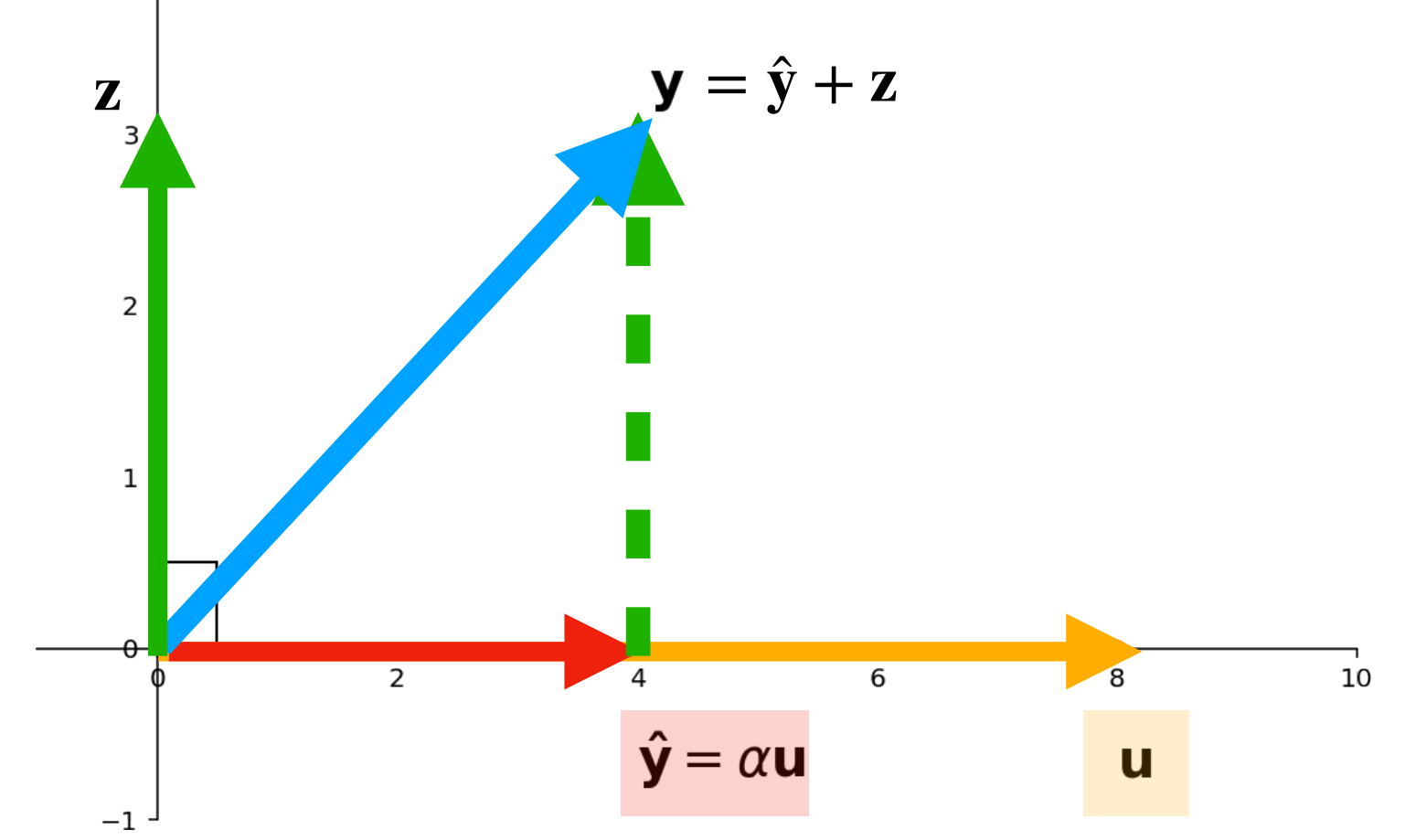
Therefore:

$$\langle \mathbf{y} - \alpha \mathbf{u}, \mathbf{u} \rangle = 0$$

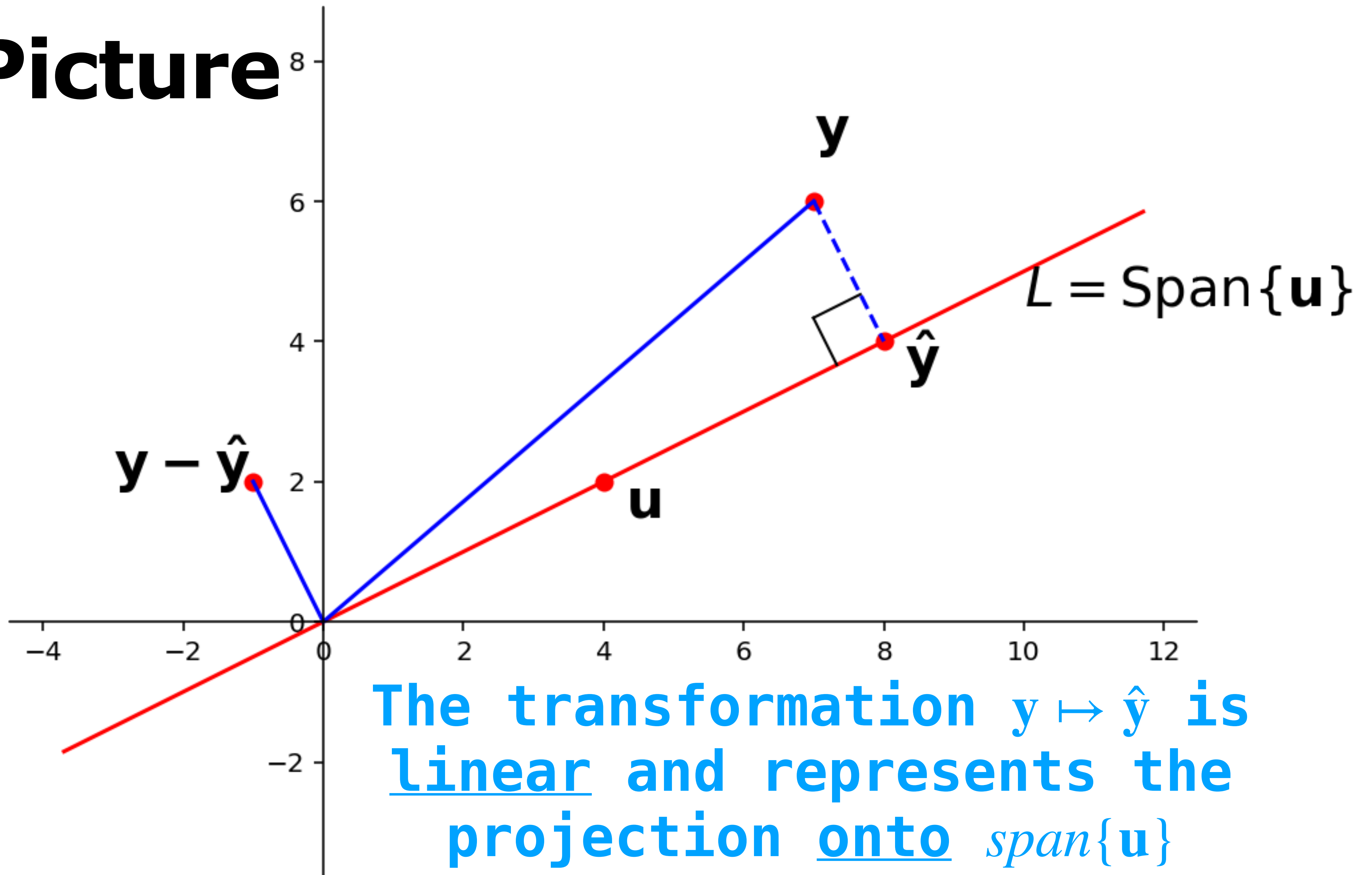
Once we have α , we can compute both $\hat{y}$ and z

Finding α

$$\langle \mathbf{y} - \alpha \mathbf{u}, \mathbf{u} \rangle = 0$$



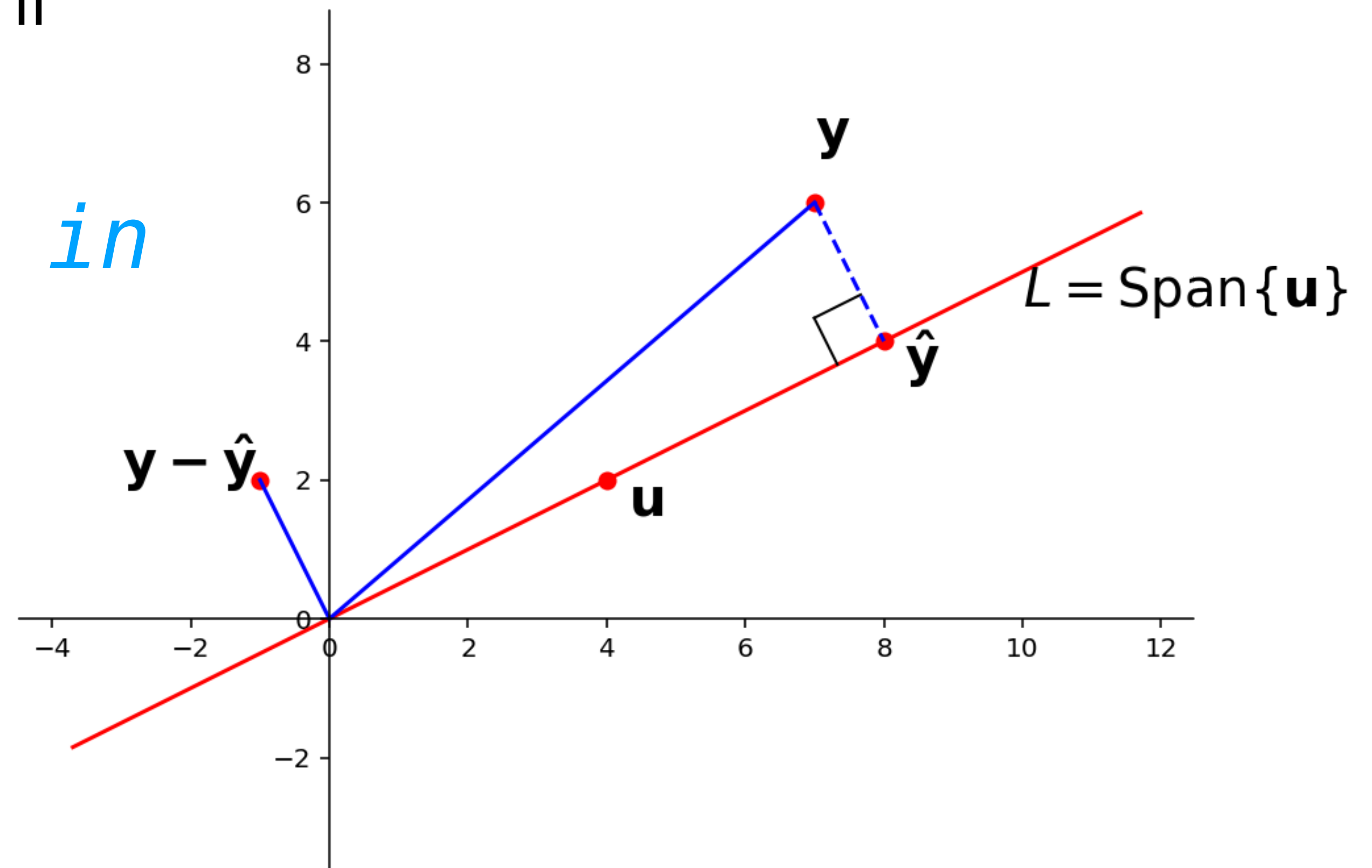
The Picture



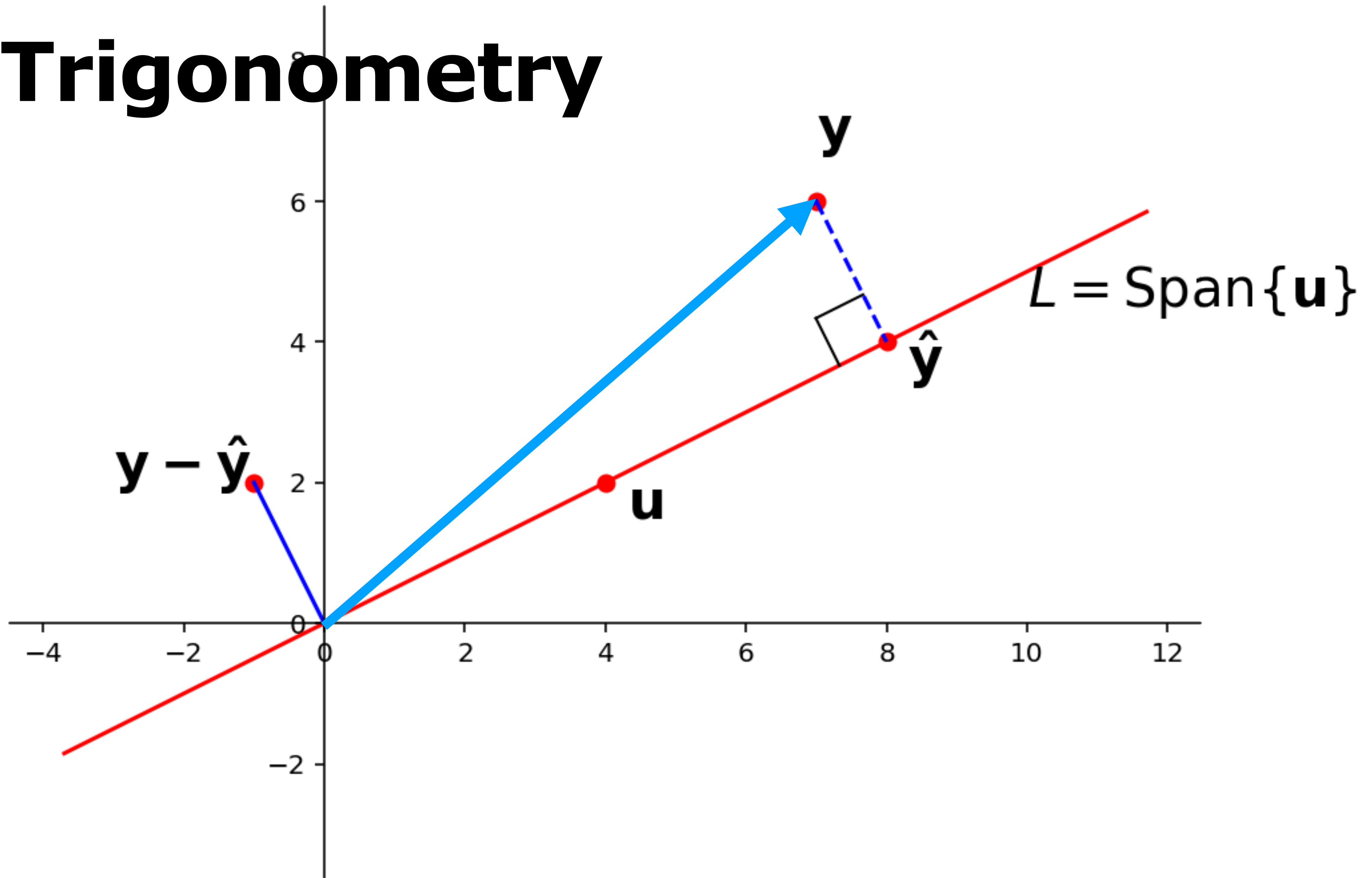
$\hat{\mathbf{y}}$ and Distance

Thm. $\|\hat{\mathbf{y}} - \mathbf{y}\| = \min_{\mathbf{w} \in \text{span}\{\mathbf{u}\}} \|\mathbf{w} - \mathbf{y}\|$

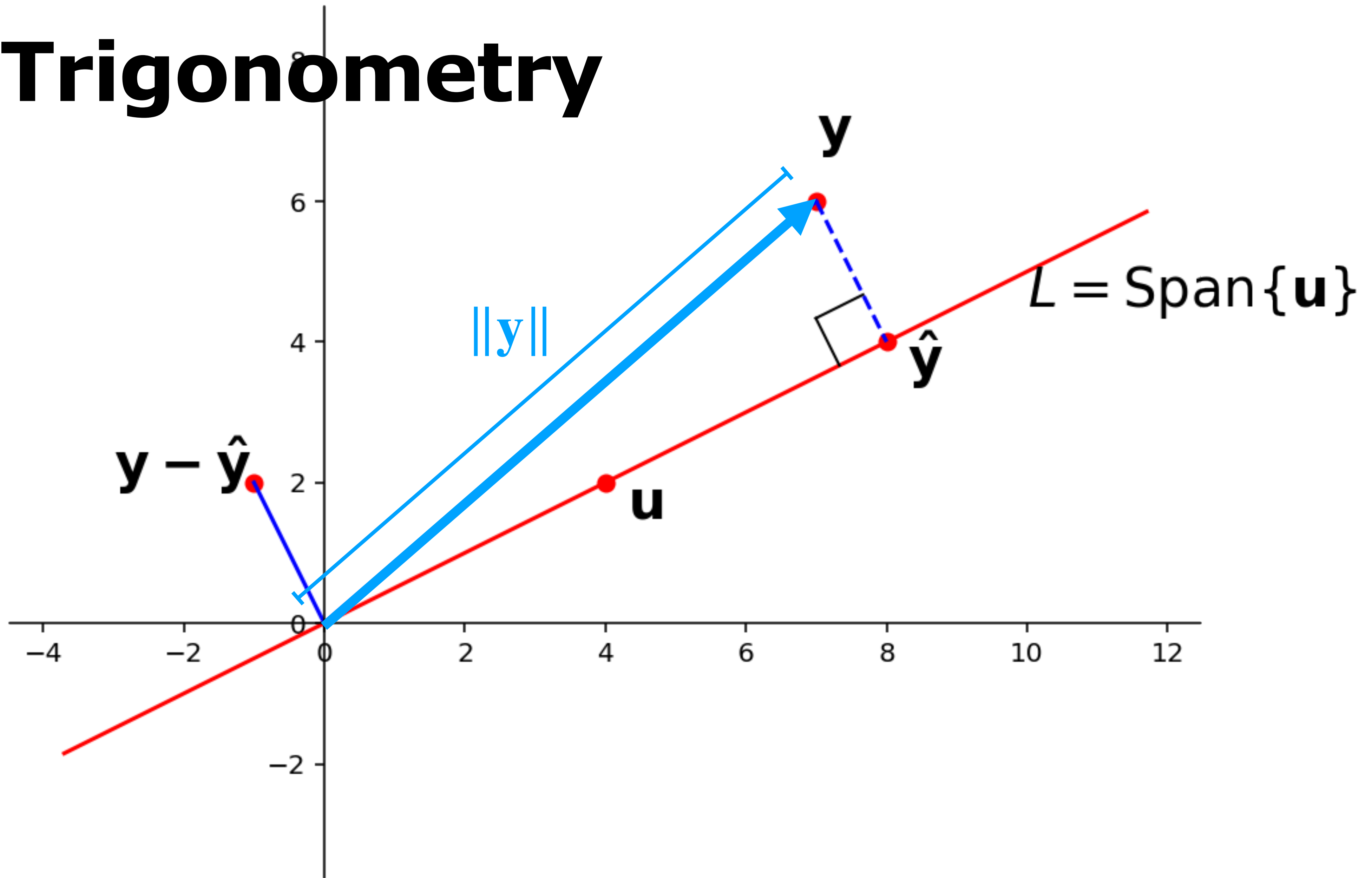
$\hat{\mathbf{y}}$ is the closest vector in $\text{span}\{\mathbf{u}\}$ to $\mathbf{y}$



The Trigonometry

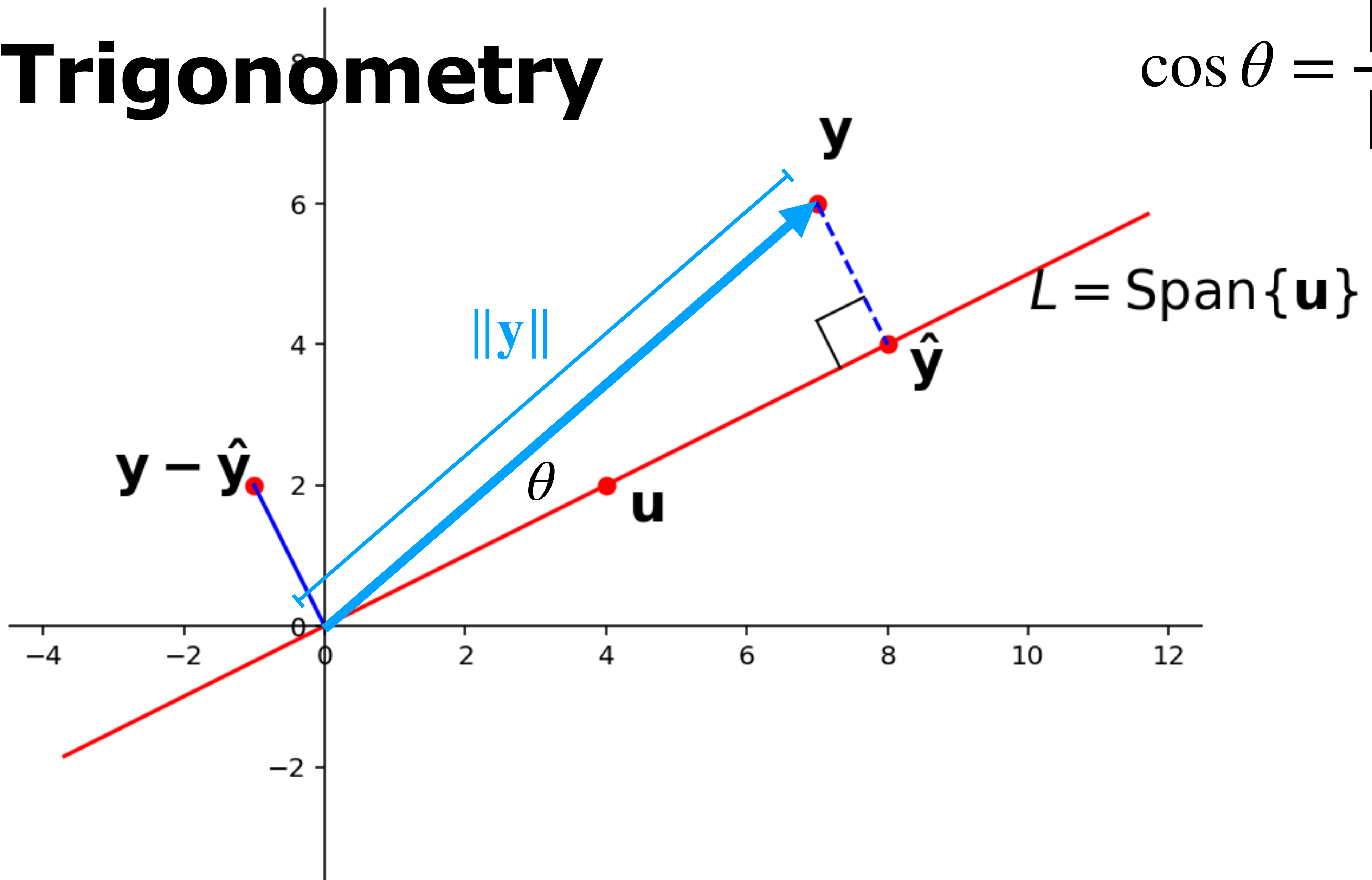


The Trigonometry



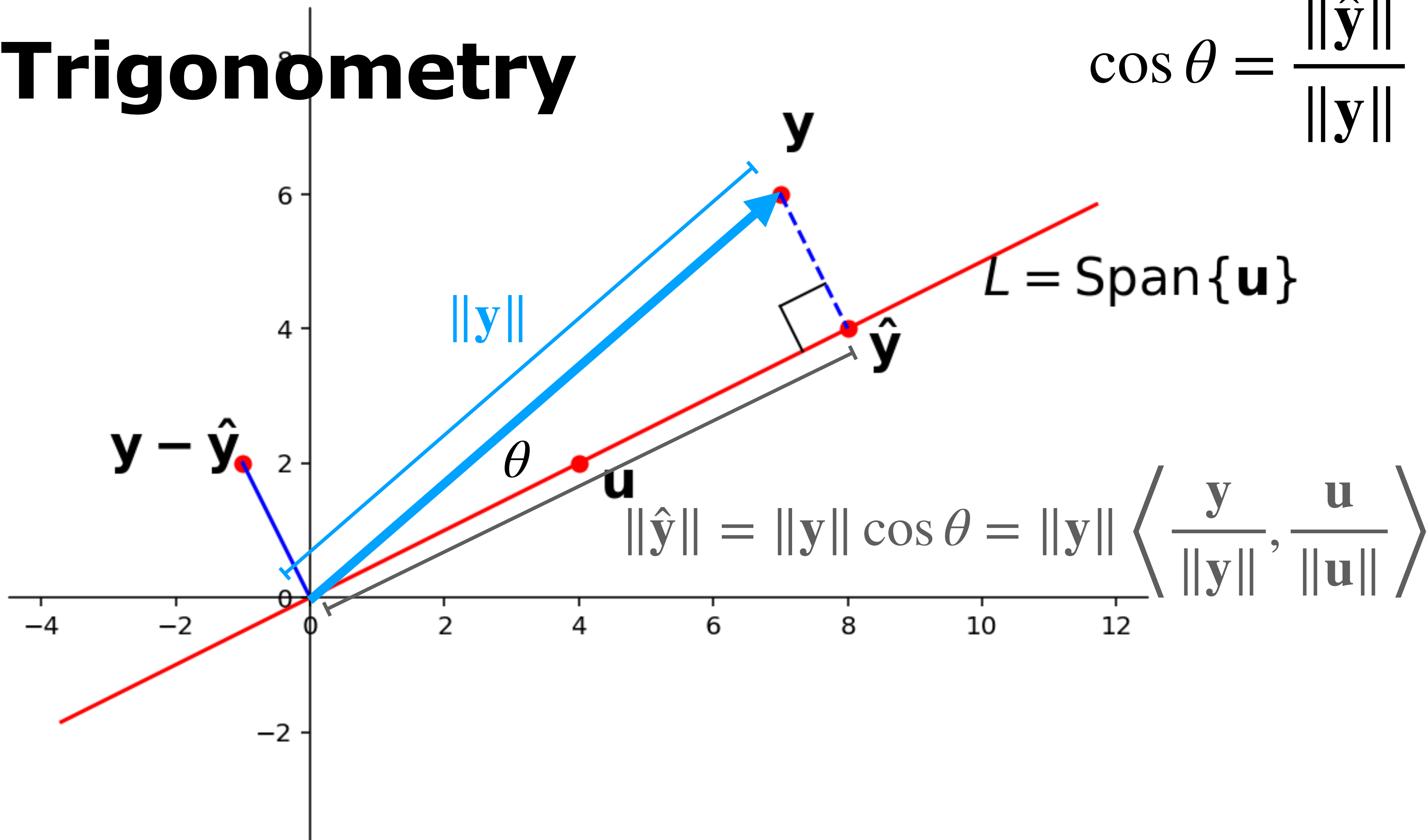
The Trigonometry

$$\cos \theta = \frac{\|\hat{\mathbf{y}}\|}{\|\mathbf{y}\|}$$



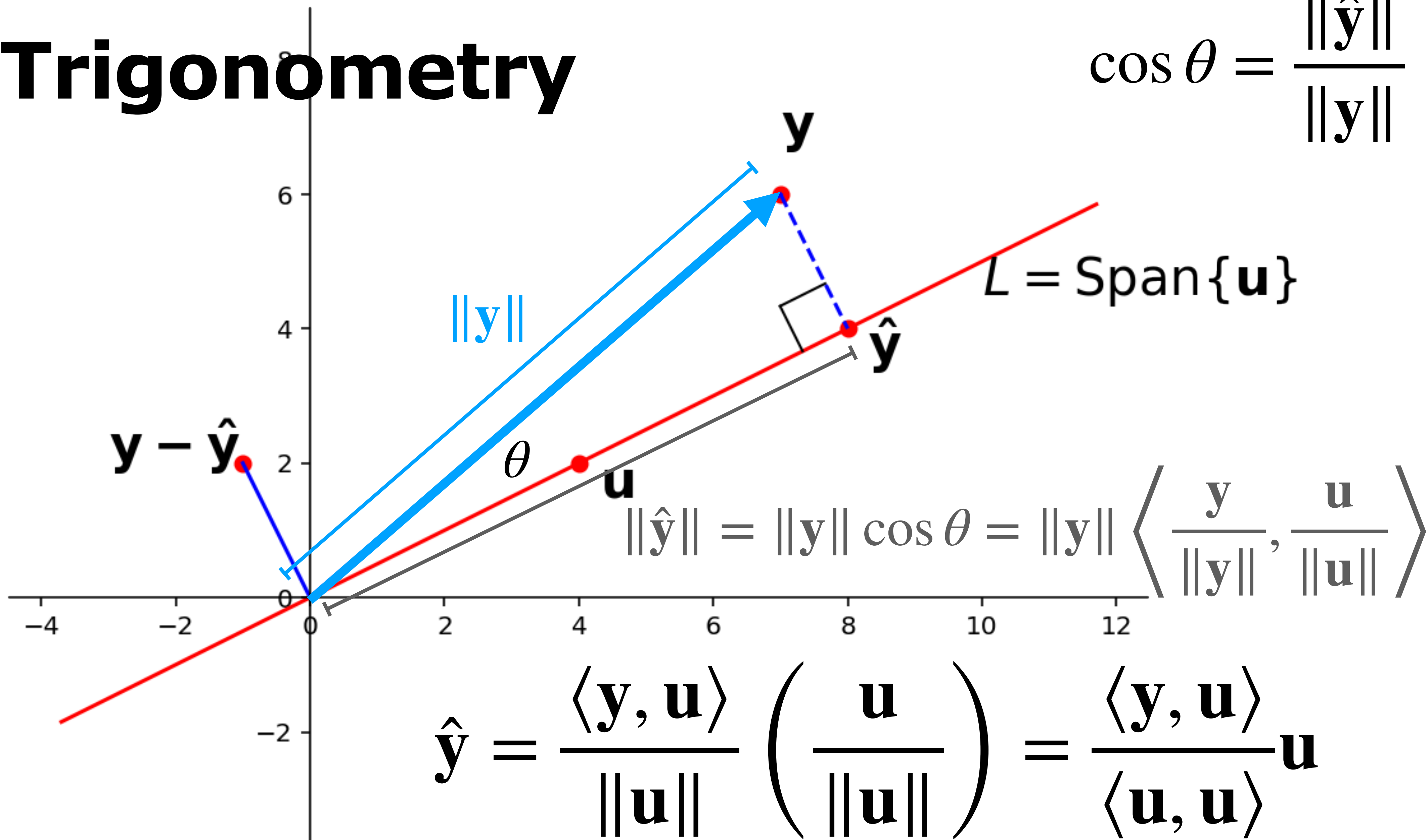
The Trigonometry

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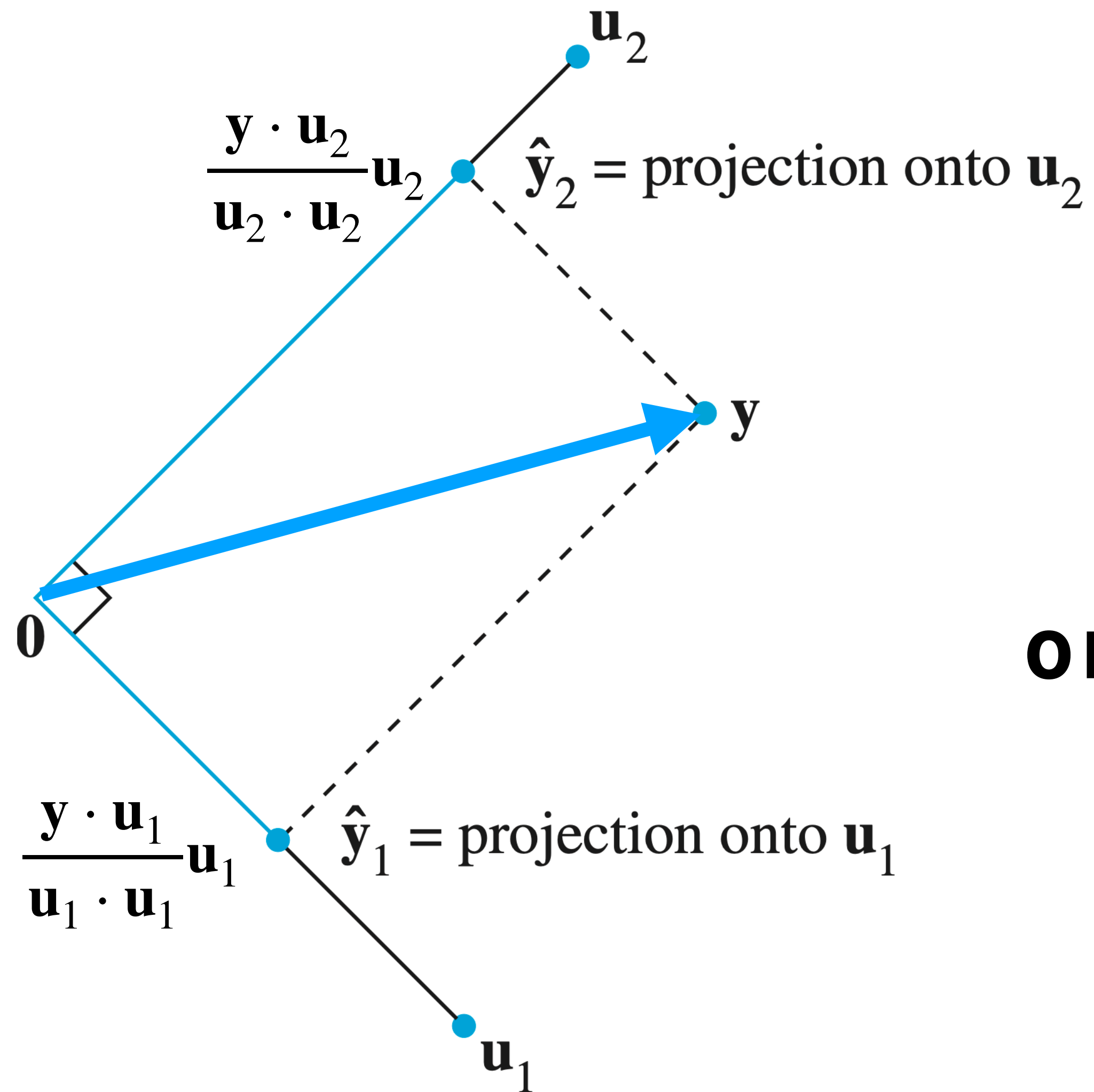


The Trigonometry

$$\cos \theta = \frac{\|\hat{\mathbf{y}}\|}{\|\mathbf{y}\|}$$



Orthogonal Projections and Orthogonal Bases



Each *component* of y
written in terms of an
orthogonal basis is an
**orthogonal projection onto
to a basis vector**

Question

$$\alpha = \frac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}}$$

Find the matrix which implements orthogonal projection onto the span of $\begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}$

Answer

$$\frac{1}{\sqrt{5}} \begin{bmatrix} 1 & -1 & 2 \\ -1 & 1 & -2 \\ 2 & -2 & 4 \end{bmatrix}$$

Orthonormal Sets

Orthogonal sets would be easier
to work with if every vector was
a unit vector

Orthonormality

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Def. A set $\{u_1, u_2, \dots, u_p\}$ is an **orthonormal set** if
of it an orthogonal set of *unit* vectors

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Orthonormal Matrices

Def. A matrix is **orthonormal** if its columns form an orthonormal set

This is sometimes called an **orthogonal** matrix

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And this is incredibly confusing, but we'll try to be consistent and clear

Orthonormal Matrices and Transposition

Thm. For an orthonormal matrix $U \in \mathbb{R}^{m \times n}$

$$U^T U = I_n$$

Inverses of Orthogonal Matrices

Thm. If a matrix $U \in \mathbb{R}^{n \times n}$ is orthogonal then it is invertible and

$$U^{-1} = U^T$$

Orthonormal Matrices and Inner Products

Thm. For an orthonormal matrix $U \in \mathbb{R}^{m \times n}$ and any vectors $\mathbf{x}$ and $\mathbf{y}$ in $\mathbb{R}^n$

$$\langle U\mathbf{x}, U\mathbf{y} \rangle = \langle \mathbf{x}, \mathbf{y} \rangle$$

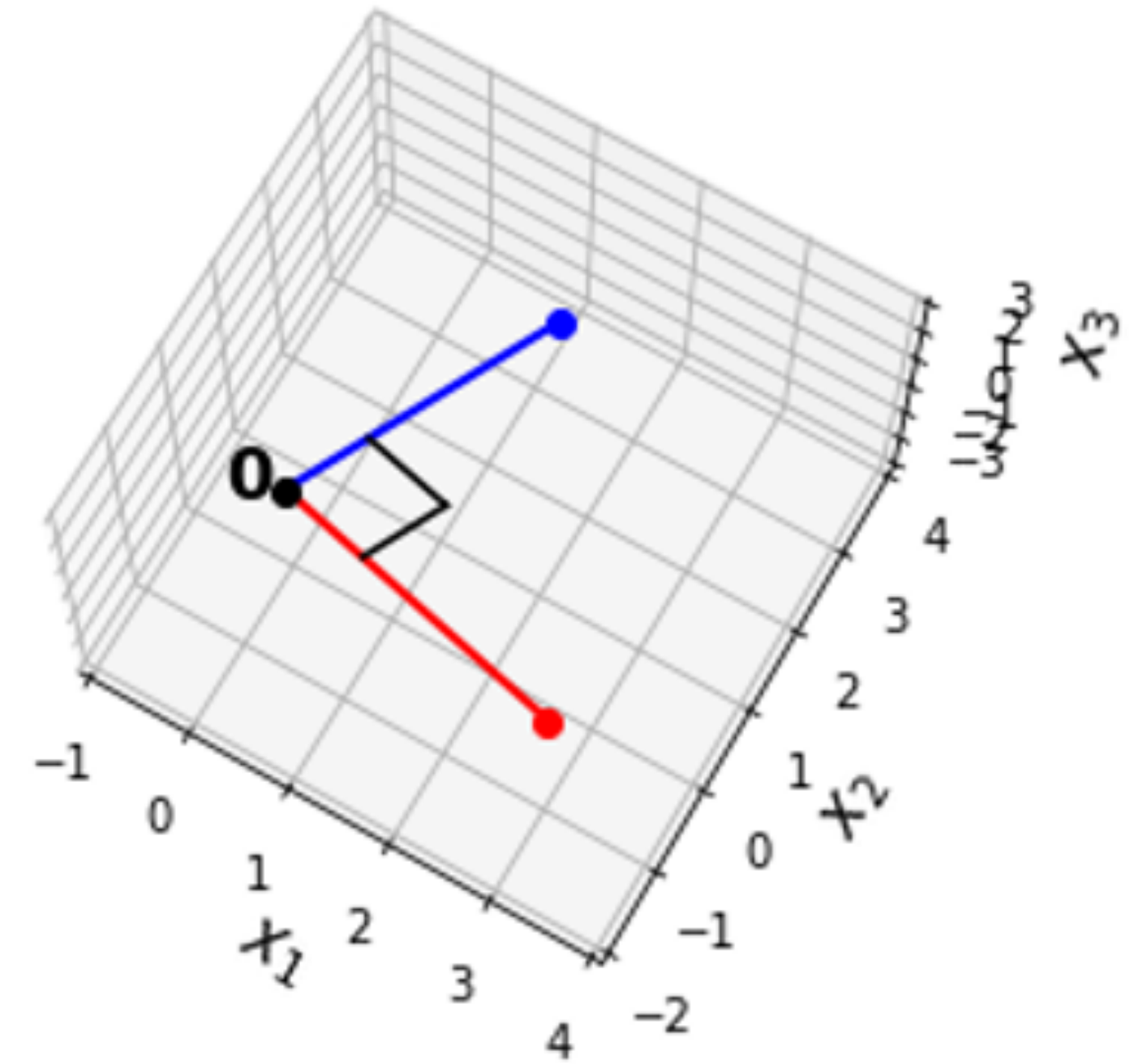
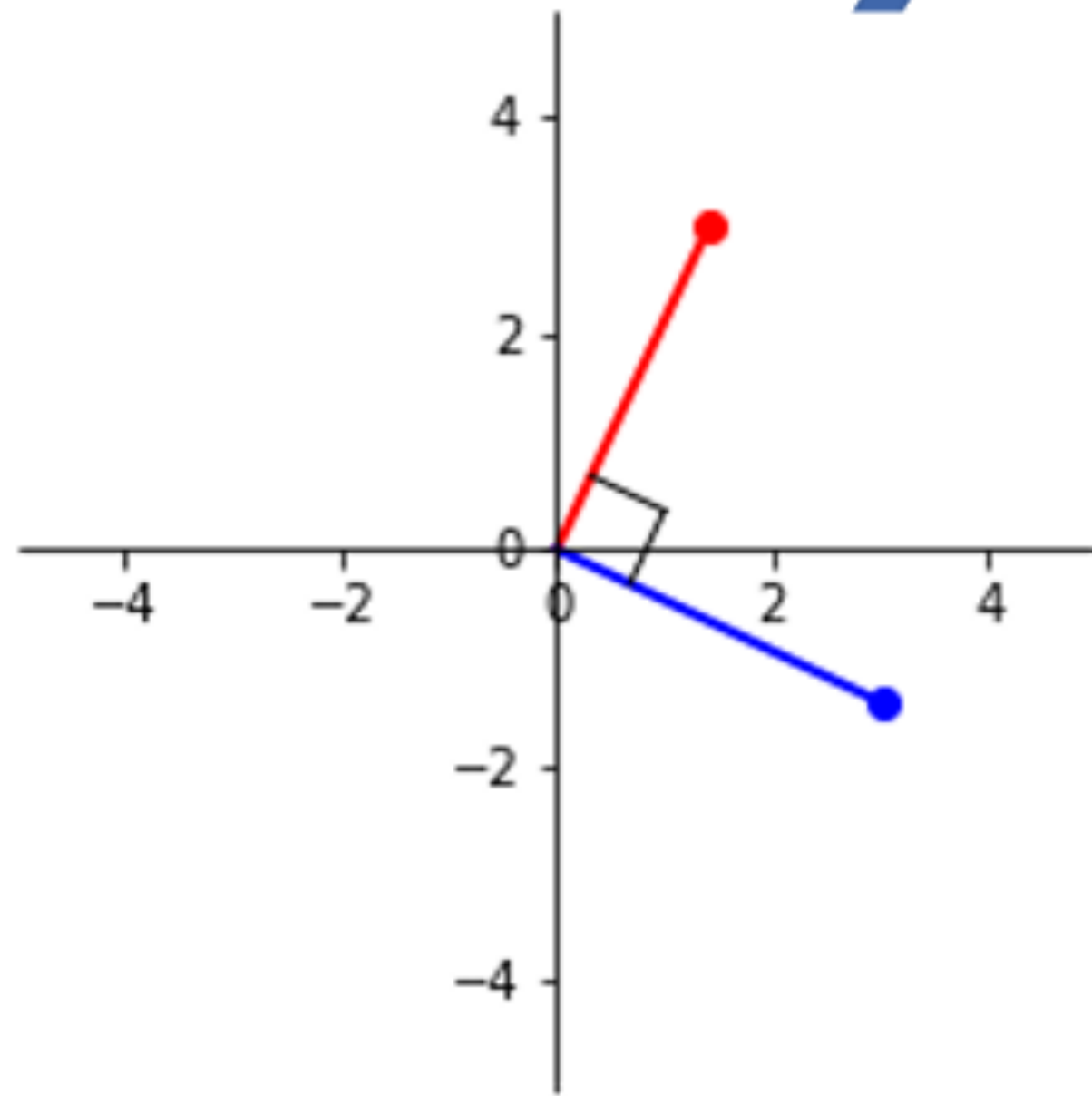
Orthonormal matrices preserve inner products

Length, Angle, Orthogonality Preservation

Since *lengths* and *angles* are defined in terms of inner products, they are also preserved by orthonormal matrices:

The Picture

Orthonormal U



Example

$$U = \begin{bmatrix} 1/\sqrt{2} & 2/3 \\ 1/\sqrt{2} & -2/3 \\ 0 & 1/3 \end{bmatrix}$$

$$x = \begin{bmatrix} \sqrt{2} \\ 3 \end{bmatrix}$$

Question

*Suppose A is an $m \times n$ matrix with orthogonal but **not** orthonormal columns. What is $A^T A$?*

Answer

Summary

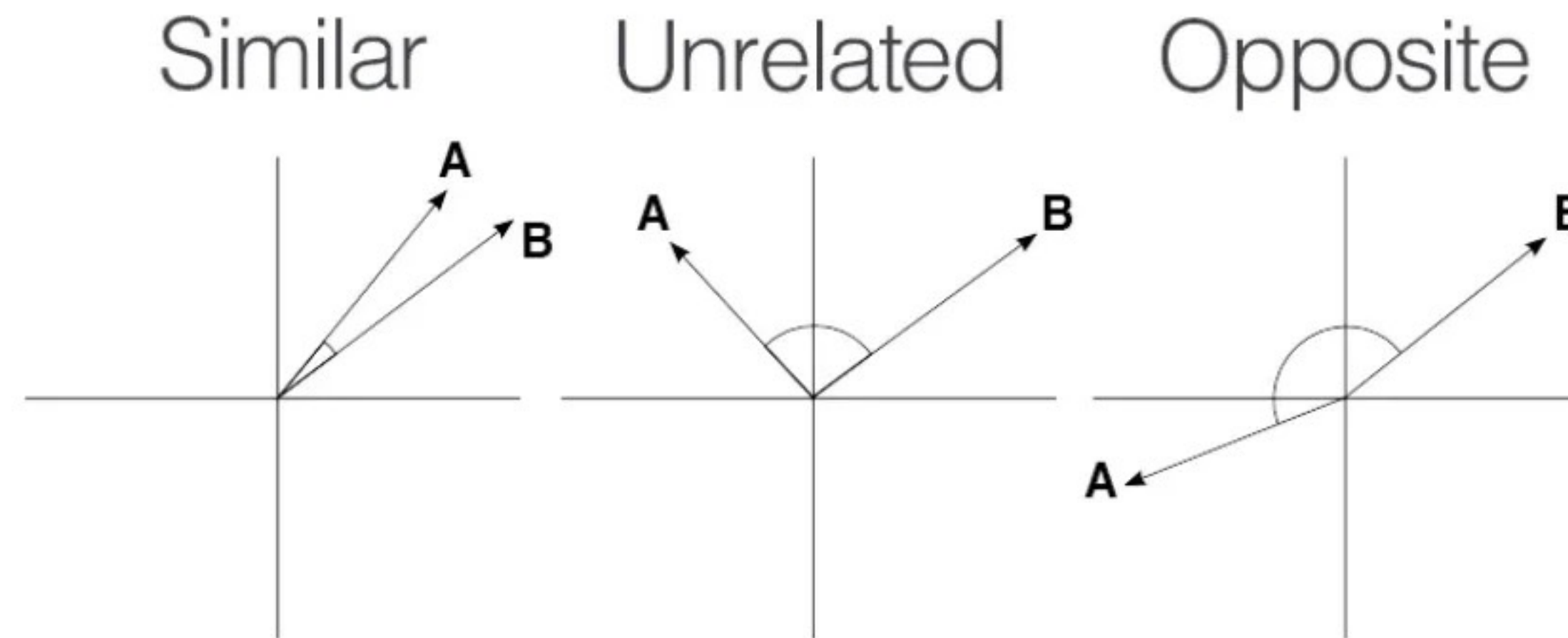
Orthogonal sets allow for *simpler calculations* of coordinates

Finding these coordinates is a really about find the **orthogonal projections** onto each vector in the orthogonal set

We can apply these ideas to matrices and describe a class of very well behaved transformations via **orthonormal matrices**

Application: Cosine Similarity

High Level



Data points are *very big vectors*

Similar vectors "point in nearly the same direction"

Example: Netflix Users

$$\text{user}_1 = \begin{bmatrix} 2 \\ 10 \\ 1 \\ 3 \end{bmatrix} \quad \text{user}_2 = \begin{bmatrix} 2 \\ 5 \\ 0 \\ 4 \end{bmatrix} \quad \text{user}_3 = \begin{bmatrix} 10 \\ 0 \\ 5 \\ 6 \end{bmatrix} \quad \begin{array}{l} \text{comedy} \\ \text{drama} \\ \text{horror} \\ \text{romance} \end{array}$$

A Netflix user might be represented as a vectors whose i th entry is the number of movies they've watched in a particular genre

Who are more likely to share similar interests in movies?

Cosine Similarity

Def. The **cosine similarity** of two vectors is the cosine of the angle between them

If its close to 0, then two Netflix users watch very different movies

If its close to 1, then two Netflix users watch very similar movies

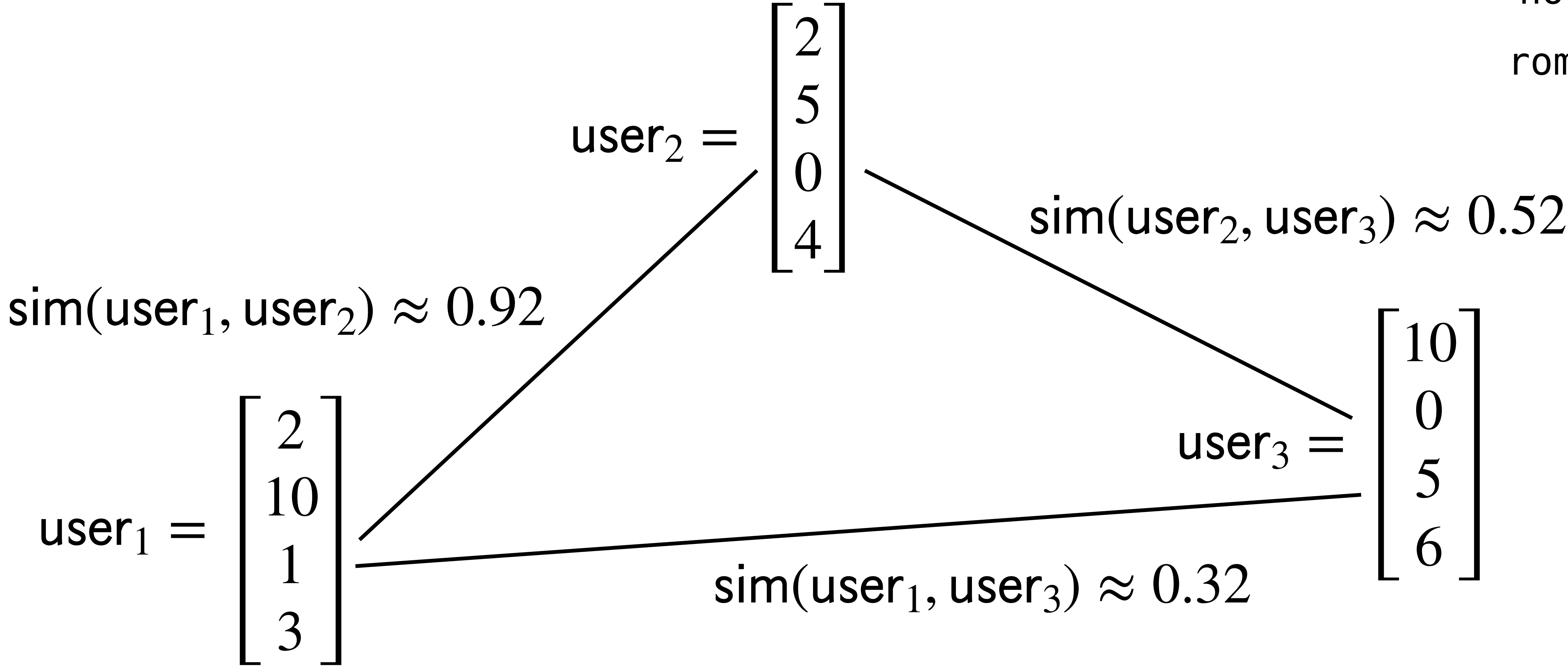
Example: Netflix Users

comedy

drama

horror

romance



Other Examples

- *Document similarity*
 - Documents $\mapsto$ word count vectors
 - Similar documents should use similar words
- *Word2Vec*
 - Words $\mapsto$ vector *somehow*
 - This underlies modern natural language processing (NLP)